

Lecture Notes on Quantum Field Theory II

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1 The Dirac Field

1.1 Lorentz Invariance

Readings

- Peskin & Schroeder Sec. 3.1
- Zee Chap. II

Relativity and Quantum Mechanics

In non-relativistic quantum mechanics, the dynamics of a particle is described by the Schrödinger equation

$$i\frac{\partial}{\partial t}\psi(t, \mathbf{x}) = -\frac{\nabla^2}{2m}\psi(t, \mathbf{x}). \quad (1.1)$$

However this equation is not compatible with special relativity, because time and space appear differently: the time derivative appears linearly while spatial derivatives appear quadratically.

In special relativity space and time must be treated on equal footing. The fundamental symmetry of spacetime is the **Lorentz symmetry**. **Lorentz invariance** means that if the original fields satisfy the equations of motion, the transformed fields must also satisfy the same equations.

Lorentz Transformations

In special relativity we combine time and space into spacetime coordinates

$$x^\mu = (t, \mathbf{x}) \quad (1.2)$$

where $\mu = 0, 1, 2, 3$. The invariant spacetime interval is

$$s^2 = -t^2 + \mathbf{x}^2. \quad (1.3)$$

Lorentz transformations are linear transformations

$$x'^\mu = \Lambda^\mu_\nu x^\nu \quad (1.4)$$

that leave the spacetime interval invariant

$$\eta_{\rho\sigma} \Lambda^\rho_\mu \Lambda^\sigma_\nu = \eta_{\mu\nu}, \quad (1.5)$$

where the Minkowski metric in our convention is

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.6)$$

For example, a boost in the x direction is

$$t' = \gamma(t - vx), \quad (1.7)$$

$$x' = \gamma(x - vt), \quad (1.8)$$

where $\gamma = 1/\sqrt{1 - v^2}$.

Relativistic Wave Equations

To construct relativistic wave equations we use Lorentz invariant operators. The derivative operator is

$$\partial_\mu = \left(\frac{\partial}{\partial t}, \nabla \right). \quad (1.9)$$

Using the Minkowski metric we define

$$\partial^\mu = \eta^{\mu\nu} \partial_\nu. \quad (1.10)$$

In our metric convention $(-, +, +, +)$ we obtain

$$\partial_\mu \partial^\mu = -\frac{\partial^2}{\partial t^2} + \nabla^2. \quad (1.11)$$

Using this operator we obtain the Klein-Gordon equation

$$(\partial_\mu \partial^\mu - m^2)\phi = 0. \quad (1.12)$$

This equation is Lorentz invariant and describes a relativistic scalar particle.

Limitations of Relativistic Quantum Mechanics

One may attempt to construct a relativistic quantum theory by writing wave equations for a single particle, such as the Klein-Gordon equation or the Dirac equation. However, this approach faces fundamental difficulties.

- First, in relativistic physics the number of particles is not conserved. Processes such as particle creation and annihilation can occur when the energy is sufficiently large. For example, an electron–positron pair can be created from a photon. A single-particle wave function $\psi(\mathbf{x}, t)$ cannot describe processes in which the number of particles changes.
- Second, the relativistic wave equations introduce negative-energy solutions. Interpreting these states within single-particle quantum mechanics leads to conceptual problems.

These difficulties indicate that relativistic quantum mechanics cannot be consistently formulated as a theory of a fixed number of particles. Instead, the correct framework is quantum field theory, in which fields are promoted to operators defined at each spacetime point and particles are interpreted as excitations of these fields.

We refer the reader to the discussion in Quantum Field Theory I for a more detailed explanation of these issues.

Transformation of Fields

The discussion above indicates that relativistic quantum theory must be formulated in terms of fields rather than single-particle wave functions. A field is a physical quantity defined at every spacetime point,

$$\phi(x), \quad x^\mu = (t, \mathbf{x}).$$

Since spacetime coordinates transform under Lorentz transformations, fields must transform in a consistent way so that the physical laws remain Lorentz invariant. In the following we discuss how different types of fields transform under Lorentz transformations.

Under a Lorentz transformation the coordinates change as

$$x \rightarrow x' = \Lambda x. \quad (1.13)$$

Fields transform according to their representation of the Lorentz group.

Scalar field (spin 0) A scalar field is defined as a field whose value is invariant under Lorentz transformations. This means that the value of the field at a given spacetime event does not depend on the choice of inertial frame. For a scalar field we require

$$\phi'(x') = \phi(x). \quad (1.14)$$

This equation states that the value of the field at the same spacetime event is the same in both coordinate systems. Since

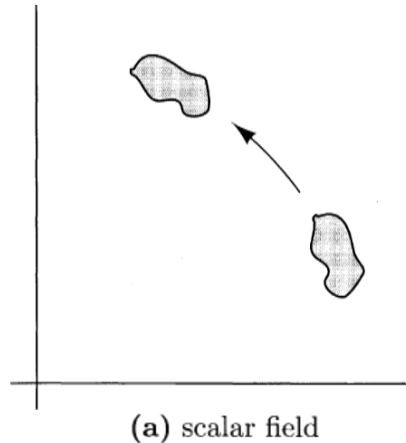
$$x = \Lambda^{-1}x',$$

we can rewrite the transformation law as

$$\phi'(x') = \phi(\Lambda^{-1}x'). \quad (1.15)$$

Finally, renaming the coordinate $x' \rightarrow x$, we obtain the commonly used form

$$\boxed{\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x)} \quad (1.16)$$



Example: Lorentz invariance of the Klein–Gordon Lagrangian Consider the Klein–Gordon Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}m^2\phi^2. \quad (1.17)$$

Using the scalar field transformation rule

$$\phi \rightarrow \phi'(x) = \phi(\Lambda^{-1}x), \quad (1.18)$$

we can determine how the derivatives transform. Using the chain rule we obtain

$$\partial_\mu\phi'(x) = \frac{\partial}{\partial x^\mu}\phi(\Lambda^{-1}x) = (\Lambda^{-1})^\nu{}_\mu\partial_\nu\phi(\Lambda^{-1}x). \quad (1.19)$$

Therefore

$$\partial_\mu\phi'(x)\partial^\mu\phi'(x) = (\Lambda^{-1})^\nu{}_\mu(\Lambda^{-1})^\rho{}_\sigma\eta^{\mu\sigma}\partial_\nu\phi(\Lambda^{-1}x)\partial_\rho\phi(\Lambda^{-1}x). \quad (1.20)$$

Using the property of Lorentz transformations

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}, \quad (1.21)$$

one finds

$$\partial_\mu\phi'(x)\partial^\mu\phi'(x) = \partial_\mu\phi(\Lambda^{-1}x)\partial^\mu\phi(\Lambda^{-1}x). \quad (1.22)$$

Similarly,

$$\phi'^2(x) = \phi^2(\Lambda^{-1}x). \quad (1.23)$$

Therefore the Lagrangian density transforms as

$$\boxed{\mathcal{L}(x) \rightarrow \mathcal{L}'(x) = \mathcal{L}(\Lambda^{-1}x)}. \quad (1.24)$$

This means that the Lagrangian density is a Lorentz scalar. The action is

$$S = \int d^4x \mathcal{L}(x). \quad (1.25)$$

Since Lorentz transformations satisfy

$$d^4x = d^4x', \quad (1.26)$$

the action is invariant,

$$S' = S. \quad (1.27)$$

Because the equations of motion follow from the stationary condition

$$\delta S = 0, \quad (1.28)$$

the resulting Euler–Lagrange equation

$$(\partial_\mu\partial^\mu - m^2)\phi = 0 \quad (1.29)$$

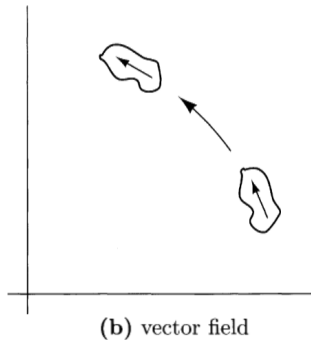
is automatically Lorentz invariant. Therefore, **if the Lagrangian density is a Lorentz scalar, the equations of motion derived from it are automatically Lorentz invariant.**

Vector field (spin 1) To understand how a vector field transforms, it is useful to first consider ordinary spatial rotations. Let $V^i(x)$ be a vector field in three-dimensional space. Under a rotation R , the coordinates transform as

$$x \rightarrow x' = Rx.$$

The vector field transforms as

$$V^i(x) \rightarrow V'^i(x) = R^i{}_j V^j(R^{-1}x). \quad (1.30)$$



(b) vector field

This transformation has two effects.

- The argument of the field changes from x to $R^{-1}x$ because the spacetime point itself is transformed.
- The components of the vector rotate according to the rotation matrix R^i_j .

In other words, a rotation affects both the position at which the field is evaluated and the orientation of the vector itself.

The relativistic generalization of this transformation is obtained by replacing rotations with Lorentz transformations. For a four-vector field $V^\mu(x)$ we therefore have

$$\boxed{V^\mu(x) \rightarrow V'^\mu(x) = \Lambda^\mu_\nu V^\nu(\Lambda^{-1}x)}. \quad (1.31)$$

Tensor field The transformation rule for vector fields can be generalized to tensor fields. Consider a rank-two tensor field $T^{\mu\nu}(x)$. Under a Lorentz transformation, the tensor field transforms as

$$\boxed{T^{\mu\nu}(x) \rightarrow T'^{\mu\nu}(x) = \Lambda^\mu_\rho \Lambda^\nu_\sigma T^{\rho\sigma}(\Lambda^{-1}x)}. \quad (1.32)$$

Each Lorentz index introduces one factor of the Lorentz transformation matrix Λ . More generally, for a tensor with multiple indices, each index transforms independently with its own Lorentz matrix.

Index matching and Lorentz covariance Lorentz symmetry imposes a powerful constraint on the form of relativistic equations. If every term in an equation carries the same set of free Lorentz indices, then each term transforms with exactly the same Lorentz matrices. Therefore the equation preserves its form under Lorentz transformations. In this sense, matching Lorentz indices acts as a “machine” that guarantees Lorentz covariance.

E.x. Consider the equation

$$A^\mu + B^\mu = 0. \quad (1.33)$$

Under a Lorentz transformation we obtain

$$A'^\mu = \Lambda^\mu_\nu A^\nu, \quad B'^\mu = \Lambda^\mu_\nu B^\nu.$$

Both terms acquire the same Lorentz matrix, so the equation becomes

$$\Lambda^\mu_\nu A^\nu + \Lambda^\mu_\nu B^\nu = 0, \quad (1.34)$$

which is equivalent to the original equation. Therefore the equation is Lorentz covariant.

However, if the indices do not match, the equation will generally not be Lorentz covariant. For example, consider

$$A^\mu + C = 0, \quad (1.35)$$

where C is a scalar. Under a Lorentz transformation we obtain

$$A'^\mu = \Lambda^\mu_\nu A^\nu, \quad C' = C.$$

The two terms now transform differently, so the transformed equation becomes

$$\Lambda^\mu_\nu A^\nu + C = 0, \quad (1.36)$$

which is not equivalent to the original equation. Therefore the equation is not Lorentz covariant.

Spinor field (spin- $\frac{1}{2}$) Scalar, vector and tensor fields transform with powers of the Lorentz matrix Λ . However, there exist other fields whose transformation properties cannot be written in this way. These are called **spinor fields**.

A spinor field transforms as

$$\boxed{\psi_a(x) \rightarrow \psi'_a(x) = S_{ab}(\Lambda) \psi_b(\Lambda^{-1}x)} \quad (1.37)$$

The matrix $S(\Lambda)$ is a representation of the Lorentz group acting on the spinor indices. Unlike vectors, this matrix is not equal to the Lorentz matrix Λ itself. Instead it provides a different representation of the same symmetry.

Continuous symmetry groups The Lorentz symmetry belongs to a class of symmetries called **Lie groups**. A Lie group is a group of transformations that depend smoothly on continuous parameters.

Examples include

- rotations (parameter: angle θ)
- Lorentz transformations (parameters: rotation angles and boosts)
- translations (parameter: displacement a)

Representations A representation of a symmetry group is a way to realize the symmetry transformation using matrices acting on vectors.

For example, a rotation in the plane can be represented by the matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (1.38)$$

Different physical objects transform according to different representations of the same symmetry group.

- scalar representation
- vector representation
- spinor representation

Generators Continuous symmetry transformations can be built from infinitesimal transformations. For example, a small rotation by an angle $d\theta$ can be written as

$$R(d\theta) = I - i d\theta J. \quad (1.39)$$

The matrix J is called the generator of the rotation.

Then, a finite transformation is obtained by composing infinitely many such infinitesimal transformations. This leads to the exponential form

$$R(\theta) = e^{-i\theta J}. \quad (1.40)$$

Thus, once the generator is known, the full continuous transformation is obtained by exponentiation.

Lie algebra The set of generators together with their commutation relations is called the **Lie algebra** of the group. The commutation relations tell us how different infinitesimal transformations combine.

For example, for spatial rotations the generators J_i satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk} J_k. \quad (1.41)$$

From algebra to representation To construct a representation of the group, we first find matrices that represent the generators and satisfy the same commutation relations. Once such matrices are known, the corresponding finite transformations are obtained by exponentiation.

For example, for spin- $\frac{1}{2}$ particles we may choose

$$J_i = \frac{\sigma_i}{2}, \quad (1.42)$$

where σ_i are the Pauli matrices. Since these matrices satisfy the rotation algebra, they provide a representation of the generators. The corresponding finite rotation is then

$$U(\theta) = e^{-i\theta_i J_i} = e^{-i\theta_i \sigma_i / 2}. \quad (1.43)$$

This gives a 2×2 representation of rotations acting on spinors.

Generators of the Lorentz group The Lorentz group is also a Lie group, so its finite transformations are obtained by exponentiating its generators.

To construct the generators, it is useful to first recall the generators of spatial rotations. In three dimensions, the generators of rotations can be written as

$$J_i = i\epsilon_{ijk}x^j\partial_k. \quad (1.44)$$

These generators satisfy the rotation algebra

$$[J_i, J_j] = i\epsilon_{ij}{}^k J_k. \quad (1.45)$$

To generalize this to relativistic spacetime, we introduce antisymmetric generators

$$J^{\mu\nu} = i(x^\mu\partial^\nu - x^\nu\partial^\mu). \quad (1.46)$$

These are the generators of Lorentz transformations. Because the indices are antisymmetric,

$$J^{\mu\nu} = -J^{\nu\mu}, \quad (1.47)$$

there are six independent generators. They correspond to

- three spatial rotations
- three Lorentz boosts.

Lorentz algebra The generators satisfy the Lorentz Lie algebra

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma} - g^{\nu\sigma}J^{\mu\rho} + g^{\mu\sigma}J^{\nu\rho}). \quad (1.48)$$

Any matrices that represent Lorentz transformations must satisfy the same commutation relations. Therefore, any representation $M(\Lambda)$ of the Lorentz group must be constructed from matrices that obey this algebra.

Example: vector representation For a four-vector V^μ , the generators are represented by 4×4 matrices

$$(J^{\mu\nu})^\alpha{}_\beta = i\left(\delta^\mu_\beta g^{\nu\alpha} - \delta^\nu_\beta g^{\mu\alpha}\right). \quad (1.49)$$

An infinitesimal Lorentz transformation can be written as

$$V^\alpha \rightarrow \left(\delta^\alpha_\beta - \frac{i}{2}\omega_{\mu\nu}(J^{\mu\nu})^\alpha{}_\beta\right)V^\beta, \quad (1.50)$$

where $\omega_{\mu\nu}$ are small antisymmetric parameters.

Examples Different choices of the parameters $\omega_{\mu\nu}$ correspond to different Lorentz transformations.

Rotation example For a rotation in the x - y plane we take

$$\omega_{12} = -\omega_{21} = \theta. \quad (1.51)$$

The corresponding transformation acts on a vector as

$$V \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\theta & 0 \\ 0 & \theta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V. \quad (1.52)$$

Boost example For a boost in the x -direction we take

$$\omega_{01} = -\omega_{10} = \beta. \quad (1.53)$$

The transformation becomes

$$V \rightarrow \begin{pmatrix} 1 & \beta & 0 & 0 \\ \beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V. \quad (1.54)$$

Example: spinor representation For a Dirac spinor $\psi_a(x)$, the generators of Lorentz transformations are represented by 4×4 matrices

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu], \quad (1.55)$$

where γ^μ are the Dirac gamma matrices. These matrices satisfy the same Lorentz algebra as the generators $J^{\mu\nu}$ and therefore provide a representation of the Lorentz group acting on spinors. We will introduce more details about the spinor representations in the Dirac fields.

1.2 The Dirac Equation

Readings

- Peskin & Schroeder Sec. 3.2
- Zee Chap. II

Historical motivation

Shortly after the development of quantum mechanics by Schrödinger and Heisenberg, physicists attempted to construct a relativistic wave equation for particles. The first successful relativistic equation was the Klein–Gordon equation,

$$(\partial_\mu \partial^\mu - m^2)\phi = 0, \quad (1.56)$$

which we discussed previously. Although this equation is Lorentz invariant, it is not satisfactory as a single-particle relativistic wave equation.

Two major difficulties arise if the Klein–Gordon equation is interpreted as a wave equation.

- The probability density obtained from the Klein–Gordon current is not positive definite.
- The equation admits negative-energy solutions.

At the time these problems were viewed as technical difficulties rather than indications of a deeper issue. In 1928 Dirac attempted to construct an alternative relativistic wave equation that would overcome these problems. Dirac’s idea was to construct an equation that is

- first order in time derivatives, like the Schrödinger equation,
- Lorentz covariant,
- and reproduces the relativistic dispersion relation

$$p^\mu p_\mu = -m^2. \quad (1.57)$$

This led Dirac to propose what is now known as the **Dirac equation**, which provides a relativistic description of spin- $\frac{1}{2}$ particles such as the electron.

Dirac’s construction

Dirac began with an equation similar in form to the Schrödinger equation

$$i\partial_t \psi = H\psi, \quad (1.58)$$

where ψ is now a multi-component wavefunction.

Lorentz covariance requires that time and space derivatives appear in a symmetric way. Therefore the Hamiltonian must be linear in spatial derivatives. The most general form is

$$H = \vec{\alpha} \cdot (-i\nabla) + m\beta, \quad (1.59)$$

where $\vec{\alpha}$ and β are constant matrices.

If $\vec{\alpha}$ and β were ordinary numbers, this equation would not be rotationally invariant¹. Dirac’s crucial insight was to interpret α_i and β as matrices acting on a multi-component wave function.

¹If α_i were numbers, the operator $\alpha_i \partial_i$ would contain a fixed spatial vector $\vec{\alpha}$, which selects a preferred direction and therefore breaks rotational invariance.

Thus the wave function must have several components,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_n \end{pmatrix}. \quad (1.60)$$

This object is called a **spinor**. Although ψ is written as a column vector, it is not a spacetime vector. The distinction lies in how the object transforms under Lorentz transformations. A vector transforms with the Lorentz matrix $\Lambda^\mu{}_\nu$, while a spinor transforms with a different matrix $S(\Lambda)$ constructed from the generators of the Lorentz group. Objects that transform in this way are called spinors.

A striking difference is that a 360° spatial rotation leaves vectors unchanged but changes the sign of a spinor. Only after a 720° rotation does a spinor return to its original value. This property is characteristic of particles with spin $1/2$.

Consistency with relativistic dispersion

To ensure that the equation describes relativistic particles, Dirac required that the equation admit plane-wave solutions with the relativistic dispersion relation

$$p^\mu p_\mu = -m^2. \quad (1.61)$$

A simple way to guarantee this property is to require that squaring the first-order equation reproduces the Klein–Gordon equation. Starting from

$$i\partial_t\psi = (\vec{\alpha} \cdot (-i\nabla) + m\beta)\psi, \quad (1.62)$$

we square both sides and obtain

$$-\frac{\partial^2}{\partial t^2}\psi = [\vec{\alpha} \cdot (-i\nabla) + m\beta]^2\psi. \quad (1.63)$$

Expanding the right-hand side gives

$$-\alpha^i\alpha^j\partial_i\partial_j\psi + m^2\beta^2\psi - im(\beta\alpha^i + \alpha^i\beta)\partial_i\psi. \quad (1.64)$$

For this expression to reduce to the Klein–Gordon equation

$$(-\partial_t^2 + \nabla^2 - m^2)\psi = 0, \quad (1.65)$$

the matrices α_i and β must satisfy the conditions

$$\alpha^i\alpha^j + \alpha^j\alpha^i = 0 \quad (i \neq j), \quad (1.66)$$

$$(\alpha^i)^2 = 1, \quad (1.67)$$

$$\beta^2 = 1, \quad (1.68)$$

$$\beta\alpha^i + \alpha^i\beta = 0. \quad (1.69)$$

In addition, since the Hamiltonian must be Hermitian, we require

$$(\alpha^i)^\dagger = \alpha^i, \quad \beta^\dagger = \beta. \quad (1.70)$$

If these relations are satisfied, equation (1.62) automatically admits plane-wave solutions with the relativistic dispersion relation.

We therefore need to find matrices that satisfy these algebraic conditions. It is easy to check that constant numbers cannot satisfy these relations. One can also verify that 2×2 matrices are not sufficient, and 3×3 matrices also fail to satisfy all the conditions simultaneously. The smallest matrices that work are 4×4 matrices.

Therefore the minimal dimension of the matrices is

$$n \geq 4. \quad (1.71)$$

In fact, there exist infinitely many matrices satisfying these conditions.

Example solutions One possible solution is

$$\beta = \begin{pmatrix} 1_{2 \times 2} & 0 \\ 0 & -1_{2 \times 2} \end{pmatrix}, \quad \alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}, \quad (1.72)$$

where σ^i are the Pauli matrices.
Another solution is

$$\beta = \begin{pmatrix} 0 & 1_{2 \times 2} \\ 1_{2 \times 2} & 0 \end{pmatrix}, \quad \alpha^i = \begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}. \quad (1.73)$$

One can verify directly that both of these choices satisfy the required relations.

Since α_i and β are 4×4 matrices, the wavefunction ψ must be a four-component object,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}. \quad (1.74)$$

The components ψ_α are complex functions. This new object is called a **spinor**. Later we will see that it describes particles with spin $\frac{1}{2}$.

Gamma matrices

For later convenience it is useful to rewrite the equation in a more symmetric form. Starting from

$$i\partial_t\psi = -i\vec{\alpha} \cdot \nabla\psi + m\beta\psi, \quad (1.75)$$

we multiply both sides by β ,

$$i\beta\partial_t\psi = -i\beta\alpha^i\partial_i\psi + m\psi. \quad (1.76)$$

We now introduce the matrices

$$\gamma^0 = i\beta, \quad \gamma^i = i\beta\alpha^i. \quad (1.77)$$

In terms of these matrices the equation becomes

$$(\gamma^\mu\partial_\mu - m)\psi = 0. \quad (1.78)$$

Because different authors use different conventions for the Minkowski metric, factors of i may appear in different places in the literature. In our convention the equation takes the form above.

Clifford algebra

The relations satisfied by α_i and β can be written more compactly in terms of the gamma matrices. They satisfy

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 0 \quad (\mu \neq \nu), \quad (1.79)$$

together with

$$(\gamma^0)^2 = -1, \quad (\gamma^i)^2 = 1. \quad (1.80)$$

These relations can be written in the compact form

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}}. \quad (1.81)$$

Here we use the anticommutator

$$\{A, B\} = AB + BA. \quad (1.82)$$

Equation (1.81) defines the **Clifford algebra**. Any set of matrices γ^μ satisfying this relation is called a representation of the Clifford algebra.

From the above properties of α_i and β we also obtain

$$(\gamma^0)^\dagger = -\gamma^0, \quad (\gamma^i)^\dagger = \gamma^i. \quad (1.83)$$

These relations can also be written in the compact form

$$\boxed{(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0}. \quad (1.84)$$

Explicit representations of the gamma matrices Using the solutions for α^i and β given earlier, we can construct explicit matrices γ^μ . For example, using the first solution we obtain

$$\gamma^0 = \begin{pmatrix} i\mathbf{1}_{2 \times 2} & 0 \\ 0 & -i\mathbf{1}_{2 \times 2} \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & i\sigma^i \\ -i\sigma^i & 0 \end{pmatrix}, \quad (1.85)$$

where σ^i are the Pauli matrices.

Using the second solution for α and β we obtain another set of gamma matrices

$$\gamma^0 = \begin{pmatrix} 0 & i\mathbf{1}_{2 \times 2} \\ i\mathbf{1}_{2 \times 2} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix}. \quad (1.86)$$

Both sets of matrices satisfy the Clifford algebra (1.81). Therefore they represent two different representations of the same algebra.

Remarks

• Massless case

Consider the case $m = 0$. In this case the dispersion relation becomes

$$p^2 = 0. \quad (1.87)$$

The Dirac equation reduces to

$$i\partial_t \psi = -i\alpha^i \partial_i \psi. \quad (1.88)$$

The matrices α_i must satisfy

$$\{\alpha_i, \alpha_j\} = 0, \quad \alpha_i^2 = 1. \quad (1.89)$$

These conditions can be satisfied by the Pauli matrices

$$\alpha_i = \sigma_i. \quad (1.90)$$

Therefore in the massless case it is sufficient to use 2×2 matrices, and the wavefunction can be written as a two-component spinor

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}. \quad (1.91)$$

This illustrates an important difference between massive and massless particles. Massive fermions require four components, while massless fermions can be described by two-component spinors.

We will discuss this point in more detail later.

• Conserved current

One of Dirac's original motivations was that the Dirac equation admits a conserved current, analogous to the probability current in the Schrödinger equation.

Starting from the Dirac equation

$$(\gamma^\mu \partial_\mu - m)\psi = 0, \quad (1.92)$$

The adjoint equation is

$$0 = (\partial_\mu \psi^\dagger) (\gamma^\mu)^\dagger - m\psi^\dagger = (\partial_\mu \psi^\dagger) \gamma^0 \gamma^\mu \gamma^0 - m\psi^\dagger. \quad (1.93)$$

where we use the relation

$$(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0. \quad (1.94)$$

Then, multiplying (1.93) from the right by γ^0 gives

$$0 = -(\partial_\mu \psi^\dagger) \gamma^0 \gamma^\mu - m\psi^\dagger \gamma^0, \quad (1.95)$$

we define the Dirac adjoint

$$\bar{\psi} \equiv \psi^\dagger \gamma^0. \quad (1.96)$$

In the end, one finds that the Hermitian conjugate of the Dirac equation implies the adjoint equation

$$(\gamma^\mu \partial_\mu + m)\bar{\psi} = 0. \quad (1.97)$$

Multiplying the Dirac equation from the left by $\bar{\psi}$ gives

$$\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi = 0, \quad (1.98)$$

while multiplying the adjoint equation from the right by ψ gives

$$(\partial_\mu\bar{\psi})\gamma^\mu\psi + m\bar{\psi}\psi = 0. \quad (1.99)$$

Adding these two equations, the mass terms cancel and we obtain

$$(\partial_\mu\bar{\psi})\gamma^\mu\psi + \bar{\psi}\gamma^\mu\partial_\mu\psi = 0. \quad (1.100)$$

This can be written as

$$\partial_\mu(\bar{\psi}\gamma^\mu\psi) = 0. \quad (1.101)$$

Therefore the conserved current is

$$j^\mu = \bar{\psi}\gamma^\mu\psi. \quad (1.102)$$

In our convention $(\gamma^0)^2 = -1$, so

$$j^0 = -\psi^\dagger\psi. \quad (1.103)$$

Equivalently, one may define

$$\tilde{j}^\mu = -\bar{\psi}\gamma^\mu\psi, \quad (1.104)$$

for which

$$\tilde{j}^0 = \psi^\dagger\psi \geq 0. \quad (1.105)$$

At the end of the day, we have the positive definite conserved current

$$\tilde{j}^\mu = -\bar{\psi}\gamma^\mu\psi \quad (1.106)$$

satisfying $\partial_\mu\tilde{j}^\mu = 0$ with positive definite classically $\tilde{j}^0 = \psi^\dagger\psi \geq 0$.

• Equivalence of different gamma matrices

There are infinitely many matrices satisfying the Clifford algebra. What is the meaning of these different solutions?

To understand this, consider the Dirac equation as a matrix equation acting on a four-component spinor ψ . Suppose we perform a change of basis in this spinor space

$$\psi \rightarrow \psi' = B\psi, \quad (1.107)$$

where B is an invertible constant matrix. Insert $\psi = B^{-1}\psi'$ into the original Dirac equation:

$$(\gamma^\mu\partial_\mu - m)B^{-1}\psi' = 0, \quad (1.108)$$

Multiplying equation (1.78) by B gives

$$(\gamma'^\mu\partial_\mu - m)\psi' = 0, \quad (1.109)$$

where

$$\gamma'^\mu = B\gamma^\mu B^{-1}. \quad (1.110)$$

Thus ψ' satisfies the same Dirac equation but with a different set of gamma matrices.

It is straightforward to check that the matrices γ'^μ also satisfy the Clifford algebra. Therefore any two sets of gamma matrices related by a similarity transformation describe the same physical theory. They correspond simply to different choices of basis for the spinor space.

Although these representations are mathematically equivalent, different choices of gamma matrices can be more convenient in different physical situations. For example, the representation introduced earlier

$$\gamma^0 = \begin{pmatrix} i\mathbf{1}_{2\times 2} & 0 \\ 0 & -i\mathbf{1}_{2\times 2} \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & i\sigma^i \\ -i\sigma^i & 0 \end{pmatrix} \quad (1.111)$$

is particularly useful for studying the non-relativistic limit, because it makes the connection with the Schrödinger equation more transparent.

On the other hand, another representation

$$\gamma^0 = \begin{pmatrix} 0 & i\mathbf{1}_{2\times 2} \\ i\mathbf{1}_{2\times 2} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix} \quad (1.112)$$

is often more convenient in the ultra-relativistic regime.

Therefore different representations of the gamma matrices are used in practice depending on the physical problem being studied. Since all such representations are related by similarity transformations, they lead to the same physical predictions.

1.3 Lorentz covariance of the Dirac equation

Readings

- Peskin & Schroeder Sec. 3.2 and 3.4
- Zee Chap. II

We now show that the Dirac equation is Lorentz covariant. Lorentz covariance means that the equation has the same form in every Lorentz frame. Under a Lorentz transformation,

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu, \quad (1.113)$$

the transformed field should satisfy the same Dirac equation. Before discussing the Dirac field, let us briefly recall simpler examples.

For a scalar field, the active transformation law is

$$\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x). \quad (1.114)$$

Similarly, for a vector field such as the Maxwell potential,

$$A_\mu(x) \rightarrow A'_\mu(x) = \Lambda_\mu{}^\nu A_\nu(\Lambda^{-1}x). \quad (1.115)$$

In this case the spacetime argument changes and the vector index also transforms with a Lorentz matrix. Maxwell's equations then keep the same form in all frames. For example,

$$\partial_\mu F^{\mu\nu} = 0 \quad \Longleftrightarrow \quad \partial'_\mu F'^{\mu\nu} = 0. \quad (1.116)$$

For the Dirac equation, the field ψ is neither a scalar nor a spacetime vector. It is a new object, a spinor. Therefore we must determine how ψ transforms under Lorentz transformations.

Lorentz transformation of the Dirac field

We postulate that under a Lorentz transformation Λ , the Dirac field transforms as

$$\psi(x) \rightarrow \psi'(x) = S(\Lambda)\psi(\Lambda^{-1}x), \quad (1.117)$$

where $S(\Lambda)$ is a 4×4 matrix acting on the spinor indices.

This is the active point of view: we keep the coordinate x fixed, but the transformed field at x is expressed in terms of the old field evaluated at the transformed point $\Lambda^{-1}x$.

Lorentz covariance of the Dirac equation means that whenever $\psi(x)$ satisfies

$$(\gamma^\mu \partial_\mu - m)\psi(x) = 0, \quad (1.118)$$

then the transformed field $\psi'(x)$ must also satisfy

$$(\gamma^\mu \partial_\mu - m)\psi'(x) = 0. \quad (1.119)$$

Here the matrices γ^μ themselves do not transform as spacetime vectors. They are fixed matrices acting in spinor space. The Lorentz transformation acts on the field ψ , not on the gamma matrices as dynamical variables.

Condition for Lorentz covariance

To determine $S(\Lambda)$, we compute the derivative of the transformed field:

$$\partial_\mu \psi'(x) = \partial_\mu [S(\Lambda)\psi(\Lambda^{-1}x)] = S(\Lambda)(\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi(\Lambda^{-1}x), \quad (1.120)$$

since $S(\Lambda)$ is a constant matrix for a global Lorentz transformation. Therefore

$$\gamma^\mu \partial_\mu \psi'(x) = \gamma^\mu S(\Lambda)(\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi(\Lambda^{-1}x). \quad (1.121)$$

For (1.119) to follow from the original Dirac equation, it is sufficient that

$$\gamma^\mu S(\Lambda)(\Lambda^{-1})^\nu{}_\mu = \gamma^\nu. \quad (1.122)$$

Multiplying this equation from the left by $S^{-1}(\Lambda)$ gives the standard condition

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu. \quad (1.123)$$

Equivalently,

$$\boxed{S(\Lambda)\gamma^\mu S^{-1}(\Lambda) = (\Lambda^{-1})^\mu{}_\nu \gamma^\nu}. \quad (1.124)$$

So the problem reduces to finding a matrix $S(\Lambda)$ satisfying (1.123).

Infinitesimal Lorentz transformations

As usual, it is much easier to first solve the problem for infinitesimal Lorentz transformations. Let

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad \omega_{\mu\nu} = -\omega_{\nu\mu}, \quad (1.125)$$

where $\omega^\mu{}_\nu$ is infinitesimal. Then

$$(\Lambda^{-1})^\mu{}_\nu = \delta^\mu{}_\nu - \omega^\mu{}_\nu \quad (1.126)$$

to first order in ω . Since Lorentz transformations form a continuous group, the corresponding spinor transformation must also depend smoothly on the parameters $\omega_{\mu\nu}$. Near the identity transformation we can therefore expand $S(\Lambda)$ linearly in $\omega_{\mu\nu}$,

$$S(\Lambda) = 1 - \frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu} + O(\omega^2),$$

where the matrices $\Sigma^{\mu\nu}$ act in spinor space and are called the generators of Lorentz transformations. Similarly,

$$S^{-1}(\Lambda) = 1 + \frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu} \quad (1.127)$$

to first order in ω . Substituting these expressions into (1.123) and keeping only terms linear in ω , we obtain

$$i[\Sigma^{\lambda\rho}, \gamma^\mu] = \eta^{\lambda\mu} \gamma^\rho - \eta^{\rho\mu} \gamma^\lambda. \quad (1.128)$$

Thus the problem is reduced to finding matrices $\Sigma^{\mu\nu}$ satisfying (1.128).

Generators in spinor space

A solution of (1.128) is

$$\Sigma^{\lambda\rho} = \frac{i}{4} [\gamma^\lambda, \gamma^\rho]. \quad (1.129)$$

One can verify this directly by substituting into (1.128) and repeatedly using the Clifford algebra (1.81). The matrices $\Sigma^{\mu\nu}$ are the generators of Lorentz transformations in spinor space. For a finite Lorentz transformation, the corresponding spinor transformation is obtained by exponentiation:

$$S(\Lambda) = \exp \left[-\frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu} \right]. \quad (1.130)$$

Remarks on the generators

- The matrices Σ^{ij} generate spatial rotations in spinor space:

$$\Sigma^{ij} = \frac{i}{4} [\gamma^i, \gamma^j]. \quad (1.131)$$

Since γ^i and γ^j are Hermitian, we find

$$(\Sigma^{ij})^\dagger = \Sigma^{ij}. \quad (1.132)$$

Therefore the spinor transformation corresponding to a pure rotation,

$$S = \exp \left[-\frac{i}{2} \omega_{ij} \Sigma^{ij} \right], \quad (1.133)$$

is unitary.

- The matrices Σ^{0i} generate boosts in spinor space:

$$\Sigma^{0i} = \frac{i}{4} [\gamma^0, \gamma^i]. \quad (1.134)$$

Since $(\gamma^0)^\dagger = -\gamma^0$ and $(\gamma^i)^\dagger = \gamma^i$, we obtain

$$(\Sigma^{0i})^\dagger = -\Sigma^{0i}. \quad (1.135)$$

So Σ^{0i} is anti-Hermitian, and the corresponding boost transformation

$$S = \exp \left[-\frac{i}{2} \omega_{0i} \Sigma^{0i} \right] \quad (1.136)$$

is not unitary.

- For a general Lorentz transformation containing both rotations and boosts,

$$SS^\dagger \neq 1. \quad (1.137)$$

This fact will be important when constructing Lorentz invariant quantities.

Lorentz invariant bilinears

Since S is not unitary in general, the quantity $\psi^\dagger \psi$ is not Lorentz invariant. Indeed,

$$\psi^\dagger \psi \rightarrow \psi'^\dagger \psi' = \psi^\dagger (\Lambda^{-1} x) S^\dagger S \psi (\Lambda^{-1} x), \quad (1.138)$$

which is not equal to $\psi^\dagger \psi$ unless $S^\dagger S = 1$. So we need to find another scalar built from ψ .

From (1.129), one can show that

$$(\Sigma^{\mu\nu})^\dagger = -\gamma^0 \Sigma^{\mu\nu} \gamma^0. \quad (1.139)$$

Therefore

$$S^\dagger = \exp \left[\frac{i}{2} \omega_{\mu\nu} (\Sigma^{\mu\nu})^\dagger \right] = \exp \left[-\frac{i}{2} \omega_{\mu\nu} \gamma^0 \Sigma^{\mu\nu} \gamma^0 \right] = -\gamma^0 \exp \left[\frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu} \right] \gamma^0. \quad (1.140)$$

Using $(\gamma^0)^2 = -1$, one finds

$$S^\dagger = -\gamma^0 S^{-1} \gamma^0. \quad (1.141)$$

This suggests defining the Dirac adjoint

$$\boxed{\bar{\psi} \equiv \psi^\dagger \gamma^0}. \quad (1.142)$$

Then

$$\bar{\psi} \psi \quad (1.143)$$

is a Lorentz scalar. Indeed,

$$\bar{\psi}'(x) \psi'(x) = \psi'^\dagger(x) \gamma^0 \psi'(x) \quad (1.144)$$

$$= \psi^\dagger (\Lambda^{-1} x) S^\dagger \gamma^0 S \psi (\Lambda^{-1} x) \quad (1.145)$$

$$= \psi^\dagger (\Lambda^{-1} x) \gamma^0 \psi (\Lambda^{-1} x) \quad (1.146)$$

$$= \bar{\psi} (\Lambda^{-1} x) \psi (\Lambda^{-1} x). \quad (1.147)$$

It is also useful to write the transformation of $\bar{\psi}$ itself:

$$\bar{\psi}(x) \rightarrow \bar{\psi}'(x) = \bar{\psi}(\Lambda^{-1} x) S^{-1}. \quad (1.148)$$

Similarly, from the transformation law of ψ one finds

$$\begin{aligned} \gamma^\mu \partial_\mu \psi'(x) &= \gamma^\mu S (\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi (\Lambda^{-1} x) = S (S^{-1} \gamma^\mu S) (\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi (\Lambda^{-1} x) \\ &= S \Lambda^\mu{}_\rho \gamma^\rho (\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi (\Lambda^{-1} x) = S \gamma^\mu \partial_\mu \psi (\Lambda^{-1} x). \end{aligned} \quad (1.149)$$

It is convenient to introduce the slash notation

$$\not{\partial} \equiv \gamma^\mu \partial_\mu. \quad (1.150)$$

Then (1.149) becomes

$$\not{\partial} \psi'(x) = S \not{\partial} \psi (\Lambda^{-1} x). \quad (1.151)$$

One can also show that

$$\bar{\psi} \gamma^\mu \psi \quad (1.152)$$

transforms as a Lorentz vector.

1.4 Classical solutions of the Dirac Equation

Readings

- Peskin & Schroeder Sec. 3.3
- Zee Chap. II

Dirac action

Using the Lorentz invariant quantities constructed above, we can write the Dirac action as

$$S = -i \int d^4x \bar{\psi} (\gamma^\mu \partial_\mu - m) \psi. \quad (1.153)$$

This action is Lorentz invariant because $\bar{\psi}\psi$ is a scalar and $\bar{\psi}\gamma^\mu\partial_\mu\psi$ is also a scalar. The overall factor of i is included so that the action is real. Indeed,

$$(\bar{\psi}\psi)^\dagger = -\bar{\psi}\psi \quad (1.154)$$

and

$$(\bar{\psi}\gamma^\mu\partial_\mu\psi)^\dagger = -\bar{\psi}\gamma^\mu\partial_\mu\psi + \text{total derivative}. \quad (1.155)$$

The total derivative contributes only a boundary term to the action, which is assumed to vanish at infinity.

The action (1.153) reproduces the Dirac equation. Treating ψ and $\bar{\psi}$ as independent variables, variation with respect to $\bar{\psi}$ gives

$$(\gamma^\mu \partial_\mu - m)\psi = 0, \quad (1.156)$$

while variation with respect to ψ gives the adjoint equation for $\bar{\psi}$:

$$(\gamma^\mu \partial_\mu + m)\bar{\psi} = 0. \quad (1.157)$$

Classical solutions of the Dirac equation

We have now derived the Dirac equation, determined its Lorentz transformation properties, and constructed the corresponding action. The next step is to study its classical solutions. This is the classical starting point for quantizing the Dirac field.

By construction, the Dirac equation implies the Klein–Gordon equation, so its solutions must be built from plane waves with

$$k^2 = -m^2, \quad k^\mu = (\omega_{\vec{k}}, \vec{k}). \quad (1.158)$$

However, unlike the scalar field case, this does not completely determine the solution, because the Dirac field has four components.

We therefore separate the solutions into two types:

$$\psi_k^{(+)}(x) = u_k e^{ik \cdot x}, \quad \psi_k^{(-)}(x) = v_k e^{-ik \cdot x}, \quad (1.159)$$

where u_k and v_k are four-component complex vectors.

The first type is usually called the positive-energy solution, and the second the negative-energy solution. As in the Klein–Gordon case, this terminology refers to the classical plane-wave form. After quantization, the physical excitations all have positive energy.

Substituting these ansätze into the Dirac equation

$$(\gamma^\mu \partial_\mu - m)\psi = 0 \quad (1.160)$$

gives

$$(i\not{k} - m)u_k = 0, \quad (i\not{k} + m)v_k = 0, \quad (1.161)$$

where

$$\not{k} \equiv k_\mu \gamma^\mu. \quad (1.162)$$

Taking the Dirac adjoint gives

$$\bar{u}_k(i\not{k} - m) = 0, \quad \bar{v}_k(i\not{k} + m) = 0. \quad (1.163)$$

So the problem reduces to solving these algebraic equations for the spinors u_k and v_k .

Solutions in the rest frame

A convenient shortcut is to first consider the rest frame, where

$$\omega = m, \quad \vec{k} = 0. \quad (1.164)$$

Then the equations become

$$(-im\gamma^0 - m)u = 0, \quad (-im\gamma^0 + m)v = 0, \quad (1.165)$$

or equivalently

$$i\gamma^0 u = -u, \quad i\gamma^0 v = v. \quad (1.166)$$

Thus u and v are eigenvectors of γ^0 .

Using the explicit representation

$$\gamma^0 = \begin{pmatrix} iI & 0 \\ 0 & -iI \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & i\sigma^i \\ -i\sigma^i & 0 \end{pmatrix}, \quad (1.167)$$

equation (1.166) implies

$$u^{(0)} = \begin{pmatrix} \xi \\ 0 \end{pmatrix}, \quad v^{(0)} = \begin{pmatrix} 0 \\ \eta \end{pmatrix}, \quad (1.168)$$

where ξ and η are arbitrary two-component complex vectors.

Choosing a basis, we may take

$$u_1^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u_2^{(0)} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad (1.169)$$

and

$$v_1^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_2^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (1.170)$$

Once the rest-frame solutions are known, one can obtain the solutions for general momentum by performing a Lorentz boost:

$$u_k = S u^{(0)}, \quad v_k = S v^{(0)}. \quad (1.171)$$

This is already simpler than solving the original equations directly, although there is an even more efficient method, which we will discuss next.

Constructing solutions for general momentum

Instead of explicitly constructing the Lorentz boost, there is a more efficient method based on a useful algebraic identity. Recall that

$$\not{k} = \gamma^\mu k_\mu. \quad (1.172)$$

Using the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \quad (1.173)$$

one finds

$$\not{k}^2 = \gamma^\mu k_\mu \gamma^\nu k_\nu = k^\mu k_\mu = -m^2, \quad (1.174)$$

where we used the mass-shell condition. From this it follows that

$$(i\not{k} + m)(i\not{k} - m) = -\not{k}^2 - m^2 = 0. \quad (1.175)$$

Using this identity we can construct solutions of the Dirac equation

$$(i\not{k} - m)u = 0, \quad (i\not{k} + m)v = 0. \quad (1.176)$$

Indeed, if we define ²

$$u = N(i\not{k} + m)\tilde{u}, \quad v = N(-i\not{k} + m)\tilde{v}, \quad (1.177)$$

then equation (1.175) guarantees that the Dirac equations are automatically satisfied. Here N is a normalization constant and $\tilde{u}, \tilde{v}$ are arbitrary spinors.

Since we know that there are only two independent solutions for each of u and v , it is convenient to choose the rest-frame solutions as a basis. Thus we take

$$u_s(k) = N(i\not{k} + m)u_s^{(0)}, \quad v_s(k) = N(-i\not{k} + m)v_s^{(0)}, \quad s = 1, 2. \quad (1.178)$$

These spinors form a complete basis of solutions to the Dirac equation.

²The construction uses a simple algebraic trick. Suppose we want to solve an operator equation $Au = 0$. If there exists another operator B such that $AB = 0$, then any vector of the form

$$u = B\tilde{u}$$

automatically satisfies the equation, since

$$Au = A(B\tilde{u}) = (AB)\tilde{u}.$$

If $AB = 0$, the equation holds for any $\tilde{u}$. In the present case the operators are $A = (i\not{k} - m)$ and $B = (i\not{k} + m)$, and the identity

$$(i\not{k} + m)(i\not{k} - m) = 0$$

plays a role analogous to the elementary factorization $(x + m)(x - m) = 0$. Thus $(i\not{k} + m)$ maps arbitrary spinors into the kernel of $(i\not{k} - m)$, providing solutions of the Dirac equation.

Normalization of the spinors

Taking the Dirac conjugate

$$\bar{u} = u^\dagger \gamma^0, \quad \bar{v} = v^\dagger \gamma^0, \quad (1.179)$$

one finds³

$$\bar{u}_s(k) = N \bar{u}_s^{(0)}(i\not{k} + m), \quad \bar{v}_s(k) = N \bar{v}_s^{(0)}(-i\not{k} + m). \quad (1.180)$$

Using the identity (1.175), one immediately obtains

$$\bar{u}_r(k)v_s(k) = 0, \quad \bar{v}_r(k)u_s(k) = 0. \quad (1.181)$$

We now fix the normalization by requiring

$$\bar{u}_r(k)u_s(k) = i 2m \delta_{rs}, \quad \bar{v}_r(k)v_s(k) = -i 2m \delta_{rs}. \quad (1.182)$$

To determine N , consider

$$\bar{u}_r(k)u_s(k) = N^2 \bar{u}_r^{(0)}(i\not{k} + m)^2 u_s^{(0)}. \quad (1.183)$$

Using the on-shell relation $\not{k}^2 = -m^2$, one finds

$$(i\not{k} + m)^2 = -\not{k}^2 + 2im\not{k} + m^2 = 2m(i\not{k} + m), \quad (1.184)$$

so that

$$\bar{u}_r(k)u_s(k) = 2mN^2 \bar{u}_r^{(0)}(i\not{k} + m)u_s^{(0)}. \quad (1.185)$$

Now write

$$\not{k} = \gamma^\mu k_\mu = -E\gamma^0 + k_i \gamma^i, \quad E = \omega_{\vec{k}}. \quad (1.186)$$

For the rest-frame spinors $u_s^{(0)}$, one has

$$\bar{u}_r^{(0)}u_s^{(0)} = i \delta_{rs}, \quad \bar{u}_r^{(0)}\gamma^0 u_s^{(0)} = -\delta_{rs}, \quad \bar{u}_r^{(0)}\gamma^i u_s^{(0)} = 0. \quad (1.187)$$

Hence

$$\bar{u}_r^{(0)}(i\not{k} + m)u_s^{(0)} = \bar{u}_r^{(0)}(-iE\gamma^0 + ik_i \gamma^i + m)u_s^{(0)} \quad (1.188)$$

$$= i(E + m)\delta_{rs}. \quad (1.189)$$

Therefore

$$\bar{u}_r(k)u_s(k) = i 2m N^2 (E + m) \delta_{rs}. \quad (1.190)$$

Comparing with the chosen normalization,

$$\bar{u}_r(k)u_s(k) = i 2m \delta_{rs}, \quad (1.191)$$

we obtain

$$N^2(E + m) = 1. \quad (1.192)$$

Thus the normalization constant is

$$\boxed{N = \frac{1}{\sqrt{E + m}}}. \quad (1.193)$$

³Starting from $u_s(k) = N(i\not{k} + m)u_s^{(0)}$, taking the Hermitian conjugate and multiplying by γ^0 gives

$$\bar{u}_s(k) = u_s^\dagger(k)\gamma^0 = N(u_s^{(0)})^\dagger(i\not{k} + m)^\dagger\gamma^0.$$

Using $(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$, one obtains

$$(i\not{k} + m)^\dagger \gamma^0 = \gamma^0 (i\not{k} + m),$$

and therefore

$$\bar{u}_s(k) = N \bar{u}_s^{(0)}(i\not{k} + m).$$

Similarly,

$$\bar{v}_s(k) = N \bar{v}_s^{(0)}(-i\not{k} + m).$$

Explicit form of the spinors

Using the explicit representation of the gamma matrices and the rest frame basis spinors, one obtains

$$u_s(k) = \begin{pmatrix} \sqrt{E+m} \xi_s \\ \frac{\vec{\sigma} \cdot \vec{k}}{\sqrt{E+m}} \xi_s \end{pmatrix}, \quad v_s(k) = \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{k}}{\sqrt{E+m}} \xi_s \\ \sqrt{E+m} \xi_s \end{pmatrix}. \quad (1.194)$$

The two-component spinors are

$$\xi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \xi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (1.195)$$

Non-relativistic limit

In the non-relativistic limit

$$\frac{|\vec{k}|}{m} \ll 1 \iff E \simeq m, \quad (1.196)$$

the solutions reduce to

$$u_s(k) = \sqrt{2m} \begin{pmatrix} \xi_s \\ 0 \end{pmatrix}, \quad v_s(k) = \sqrt{2m} \begin{pmatrix} 0 \\ \xi_s \end{pmatrix}. \quad (1.197)$$

Thus the four-component Dirac spinor effectively reduces to the two-component Pauli spinor ξ_s describing a spin- $\frac{1}{2}$ particle in non-relativistic quantum mechanics.

Ultra-relativistic limit

In the ultra-relativistic limit

$$E \gg m, \quad E \simeq |\vec{k}|, \quad \hat{k} = \frac{\vec{k}}{|\vec{k}|}, \quad (1.198)$$

the spinors become

$$u_s(k) = \sqrt{E} \begin{pmatrix} \xi_s \\ \vec{\sigma} \cdot \hat{k} \xi_s \end{pmatrix}, \quad v_s(k) = \sqrt{E} \begin{pmatrix} \vec{\sigma} \cdot \hat{k} \xi_s \\ \xi_s \end{pmatrix}. \quad (1.199)$$

Using the identity⁴

$$(\vec{\sigma} \cdot \hat{k})^2 = \mathbf{1}_{2 \times 2}. \quad (1.200)$$

Since $(\vec{\sigma} \cdot \hat{k})^2 = 1$, defining $\eta_s = (\vec{\sigma} \cdot \hat{k}) \xi_s$ implies $\xi_s = (\vec{\sigma} \cdot \hat{k}) \eta_s$. Hence

$$v_s(k) = \sqrt{E} \begin{pmatrix} \eta_s \\ \vec{\sigma} \cdot \hat{k} \eta_s \end{pmatrix},$$

which has the same form as $u_s(k)$ with ξ_s replaced by η_s . Therefore the v_s spinors lie in the span of u_1, u_2 , so only two independent solutions remain in the massless limit. This is consistent with the massless Dirac equation that only requires 2 components.

Useful relations among the spinors

Besides the normalization relations with the Dirac adjoint, the spinors also satisfy the following identities:

$$u_r^\dagger(\vec{k}) u_s(\vec{k}) = 2E \delta_{rs}, \quad v_r^\dagger(\vec{k}) v_s(\vec{k}) = 2E \delta_{rs}. \quad (1.201)$$

Moreover,

$$u_r^\dagger(\vec{k}) v_s(-\vec{k}) = 0, \quad v_r^\dagger(\vec{k}) u_s(-\vec{k}) = 0. \quad (1.202)$$

⁴Using $(\vec{\sigma} \cdot \hat{k}) = \sigma^i \hat{k}_i$, we obtain

$$(\vec{\sigma} \cdot \hat{k})^2 = \hat{k}_i \hat{k}_j \sigma^i \sigma^j.$$

Using the Pauli matrix identity

$$\sigma^i \sigma^j = \delta^{ij} I + i \epsilon^{ijk} \sigma^k,$$

this becomes

$$(\vec{\sigma} \cdot \hat{k})^2 = \hat{k}_i \hat{k}_j \delta^{ij} I + i \hat{k}_i \hat{k}_j \epsilon^{ijk} \sigma^k.$$

The first term gives $\hat{k}_i \hat{k}_i = |\hat{k}|^2 = 1$. The second term vanishes because $\hat{k}_i \hat{k}_j$ is symmetric in (i, j) while ϵ^{ijk} is antisymmetric. Hence $(\vec{\sigma} \cdot \hat{k})^2 = I$.

On the other hand,

$$u_r^\dagger(\vec{k})v_s(\vec{k}) \neq 0, \quad (1.203)$$

so the orthogonality here only holds after reversing the spatial momentum.

These relations can be verified directly from the explicit expressions for the spinors. Although the explicit form of $u_s(k)$ and $v_s(k)$ depends on the chosen representation of the gamma matrices, the relations (1.201) and (1.202) are in fact representation-independent. They follow from the Dirac equation and the general algebraic properties of the gamma matrices, and therefore hold in any representation.

Projectors onto positive and negative energy states

One can construct projection operators onto the subspaces spanned by u_s and v_s . The projector onto positive-energy solutions is

$$P_+ = \frac{1}{2im} \sum_{s=1}^2 u_s(k) \bar{u}_s(k). \quad (1.204)$$

Using the normalization relations one finds

$$P_+^2 = P_+, \quad (1.205)$$

and explicitly

$$P_+ = \frac{i\not{k} + m}{2m}. \quad (1.206)$$

Similarly the projector onto the negative-energy solutions is

$$P_- = \frac{1}{2im} \sum_{s=1}^2 v_s(k) \bar{v}_s(k) = -\frac{i\not{k} + m}{2m}. \quad (1.207)$$

These projectors satisfy

$$P_-^2 = P_-, \quad P_+ P_- = 0, \quad P_+ + P_- = 1. \quad (1.208)$$

The operators P_+ and P_- act as projectors onto the subspaces of positive-energy and negative-energy solutions respectively. In particular, one finds

$$P_+ u_s(k) = u_s(k), \quad P_+ v_s(k) = 0, \quad (1.209)$$

and

$$P_- v_s(k) = v_s(k), \quad P_- u_s(k) = 0. \quad (1.210)$$

Thus P_+ projects onto the space spanned by the spinors $u_s(k)$, while P_- projects onto the space spanned by the spinors $v_s(k)$. In this sense the projectors filter out the positive-energy and negative-energy components of a Dirac spinor.

1.5 Quantization of the Dirac field

Readings

- Peskin & Schroeder Sec. 3.5
- Zee Chap. II

Canonical structure

Having obtained a complete set of classical solutions, we now proceed to quantize the Dirac theory. The Dirac action is

$$S = -i \int d^4x \bar{\psi}(\gamma^\mu \partial_\mu - m)\psi. \quad (1.211)$$

To perform canonical quantization we first compute the canonical momentum conjugate to ψ . Treating ψ and $\psi^\dagger$ as independent fields, the Lagrangian density can be written as

$$\mathcal{L} = i\psi^\dagger \partial_0 \psi - i\bar{\psi}(\gamma^i \partial_i - m)\psi. \quad (1.212)$$

The canonical momentum is therefore

$$\pi_\psi = \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi)} = i\psi^\dagger. \quad (1.213)$$

However,

$$\pi_{\psi^\dagger} = 0, \quad (1.214)$$

which indicates that the Dirac system is a constrained system. The vanishing of $\pi_{\psi^\dagger}$ indicates that the Dirac theory is a constrained system. In general, constrained systems require a more elaborate formalism (such as Dirac's constrained quantization) in order to perform canonical quantization. Fortunately, in the present case there is a simple way to proceed which leads to the correct result.

Since the canonical momentum conjugate to ψ is

$$\pi_\psi = i\psi^\dagger, \quad (1.215)$$

we may interpret ψ as the configuration variable and $\psi^\dagger$ as its canonical momentum. With this interpretation the Lagrangian density can be written in canonical form

$$\mathcal{L} = \pi_\psi \partial_0 \psi - \mathcal{H}, \quad (1.216)$$

where the Hamiltonian density is

$$\mathcal{H} = i\bar{\psi}(\gamma^i \partial_i - m)\psi. \quad (1.217)$$

Thus the phase space variables are $(\psi, \psi^\dagger)$. We now impose the canonical equal-time commutation relations

$$[\psi_\alpha(t, \vec{x}), \pi_{\psi_\beta}(t, \vec{x}')] = \delta_{\alpha\beta} \delta^{(3)}(\vec{x} - \vec{x}'), \quad \alpha, \beta = 1, 2, 3, 4. \quad (1.218)$$

The next step is to expand the field operator in terms of the complete set of classical solutions,

$$\psi(x) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \sum_{s=1,2} \left[a_{\vec{k}}^{(s)} u_s(k) e^{ik \cdot x} + b_{\vec{k}}^{(s)} v_s(k) e^{-ik \cdot x} \right]. \quad (1.219)$$

where $u_s(k)e^{-ik \cdot x}$ and $v_s(k)e^{ik \cdot x}$ form a complete set of plane-wave solutions of the Dirac equation. The coefficients $a_{\vec{k}}^{(s)}$ and $b_{\vec{k}}^{(s)}$ are promoted to operators upon quantization. Substituting this expansion into the commutation relations (1.218) leads to

$$[a_{\vec{k}}^{(s)}, (a_{\vec{k}'}^{(r)})^\dagger] = [b_{\vec{k}}^{(s)}, (b_{\vec{k}'}^{(r)})^\dagger] = (2\pi)^3 \delta_{rs} \delta^{(3)}(\vec{k} - \vec{k}'), \quad (1.220)$$

with all other commutators vanishing. One may then define the vacuum state $|0\rangle$ by

$$a_{\vec{k}}^{(s)} |0\rangle = 0, \quad b_{\vec{k}}^{(s)} |0\rangle = 0, \quad (1.221)$$

and build the Hilbert space by acting with the creation operators $(a_{\vec{k}}^{(s)})^\dagger$ and $(b_{\vec{k}}^{(s)})^\dagger$.

Why commutators fail

At first sight this procedure appears to work, but it leads to two serious problems.

- First, the commutation relations imply that the quanta created by $a^\dagger$ and $b^\dagger$ obey bosonic statistics, whereas the Dirac field should describe spin- $\frac{1}{2}$ particles which obey fermionic statistics.
- Second, computing the Hamiltonian

$$H = \int d^3 x \mathcal{H} \quad (1.222)$$

and substituting the expansion (1.219) gives

$$H = \int \frac{d^3 \vec{k}}{(2\pi)^3} \omega_{\vec{k}} \sum_{s=1,2} \left[(a_{\vec{k}}^{(s)})^\dagger a_{\vec{k}}^{(s)} - (b_{\vec{k}}^{(s)})^\dagger b_{\vec{k}}^{(s)} \right]. \quad (1.223)$$

The minus sign implies that states with many b -particles have arbitrarily negative energy, so the Hamiltonian is not bounded from below and the vacuum would be unstable.

Canonical anticommutation relations

Both problems are resolved by replacing the commutation relations with *anticommutation* relations. Instead of the usual commutator, we impose

$$\left\{ \psi_\alpha(t, \vec{x}), \pi_\beta(t, \vec{x}') \right\} = i\delta_{\alpha\beta}\delta^{(3)}(\vec{x} - \vec{x}'), \quad (1.224)$$

or equivalently

$$\left\{ \psi_\alpha(t, \vec{x}), \psi_\beta^\dagger(t, \vec{x}') \right\} = \delta_{\alpha\beta}\delta^{(3)}(\vec{x} - \vec{x}'). \quad (1.225)$$

These relations are symmetric between ψ and $\psi^\dagger$ because the anticommutator involves a plus sign rather than a minus sign.

This choice is also self-consistent. Evaluated at $\vec{x} = \vec{x}'$, the left-hand side is positive definite and matches the sign of the right-hand side. The sign ultimately traces back to the relation $\pi_\psi = i\psi^\dagger$ appearing in the Lagrangian.

Mode expansion of the Dirac field

As in the scalar case, we expand the field operator in terms of a complete set of classical solutions,

$$\psi(x) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \sum_{s=1,2} \left(a_{\vec{k}}^{(s)} u_s e^{ik \cdot x} + \left(b_{\vec{k}}^{(s)} \right)^\dagger v_s e^{-ik \cdot x} \right). \quad (1.226)$$

Here $a_{\vec{k}}^{(s)}$ and $b_{\vec{k}}^{(s)}$ are operators which annihilate particles and antiparticles respectively.

The Dirac spinor has four complex components. Correspondingly, the Dirac equation admits four independent plane-wave solutions: $u_s(k)$ and $v_s(k)$ with $s = 1, 2$. The u_s describe particles, the v_s describe antiparticles, and the index s labels the two spin polarizations. Thus the four solutions correspond to particle/antiparticle together with spin up/down.

Anticommutation relations for creation operators

Substituting the expansion (1.226) into the canonical anticommutation relations leads to

$$\left\{ a_{\vec{k}}^{(s)}, \left(a_{\vec{k}'}^{(r)} \right)^\dagger \right\} = \left\{ b_{\vec{k}}^{(s)}, \left(b_{\vec{k}'}^{(r)} \right)^\dagger \right\} = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}') \delta_{sr}, \quad (1.227)$$

with all other anticommutators vanishing. The vacuum state is defined by

$$a_{\vec{k}}^{(s)} |0\rangle = b_{\vec{k}}^{(s)} |0\rangle = 0. \quad (1.228)$$

Hamiltonian and vacuum energy

Substituting the mode expansion into the Hamiltonian gives

$$H = \int \frac{d^3\vec{k}}{(2\pi)^3} \omega_{\vec{k}} \sum_{s=1,2} \left[\left(a_{\vec{k}}^{(s)} \right)^\dagger a_{\vec{k}}^{(s)} + \left(b_{\vec{k}}^{(s)} \right)^\dagger b_{\vec{k}}^{(s)} \right] + E_0. \quad (1.229)$$

The constant E_0 represents the vacuum energy. The Hamiltonian is now positive definite, so it is natural to define the vacuum state $|0\rangle$ as the state of lowest energy,

$$a_{\vec{k}}^{(s)} |0\rangle = b_{\vec{k}}^{(s)} |0\rangle = 0. \quad (1.230)$$

The vacuum energy E_0 is formally divergent, as in the scalar field case, but it can be removed by normal ordering or by redefining the zero of energy.

Fermionic harmonic oscillators

By analogy with the ordinary harmonic oscillator,

$$H = \omega \left(a^\dagger a + \frac{1}{2} \right),$$

a fermionic oscillator is defined by operators satisfying anti-commutation relations

$$\{a, a^\dagger\} = 1, \quad \{a, a\} = \{a^\dagger, a^\dagger\} = 0. \quad (1.231)$$

Using $\{a, a^\dagger\} = 1$, one can write the Hamiltonian in a symmetric form as

$$H = \omega \left(a^\dagger a - \frac{1}{2} \right).$$

However, in quantum field theory the constant term is usually dropped, and one writes

$$H = \omega a^\dagger a,$$

since the zero-point energy is physically irrelevant.

Pauli exclusion and spectrum A key consequence of the fermionic algebra is that

$$(a^\dagger)^2 = 0,$$

so the oscillator can only be excited once. Acting on the vacuum,

$$a^\dagger |0\rangle$$

creates a single excitation, while

$$(a^\dagger)^2 |0\rangle = 0.$$

Thus the only allowed states are

$$|0\rangle, \quad a^\dagger |0\rangle.$$

Equivalently, the number operator $N = a^\dagger a$ has eigenvalues 0 and 1. This is the quantum mechanical origin of the Pauli exclusion principle.

Application to the Dirac field For the Dirac field, the Hamiltonian decomposes into independent modes labeled by $(\vec{k}, s)$:

$$H = \sum_{\vec{k}, s} \omega_k \left(a_{\vec{k}}^{(s)\dagger} a_{\vec{k}}^{(s)} + b_{\vec{k}}^{(s)\dagger} b_{\vec{k}}^{(s)} \right).$$

For each $(\vec{k}, s)$ there are therefore two independent fermionic oscillators, associated with particles $(a_{\vec{k}}^{(s)})$ and antiparticles $(b_{\vec{k}}^{(s)})$.

Single-particle states

We can now define the one-particle states created from the vacuum. The particle states are

$$|\vec{k}, s\rangle = \sqrt{2\omega_{\vec{k}}} \left(a_{\vec{k}}^{(s)} \right)^\dagger |0\rangle, \quad (1.232)$$

while the antiparticle states are

$$|\vec{k}, \bar{s}\rangle = \sqrt{2\omega_{\vec{k}}} \left(b_{\vec{k}}^{(s)} \right)^\dagger |0\rangle. \quad (1.233)$$

Each particle has two spin polarizations labeled by $s = 1, 2$, and the four-momentum satisfies the mass-shell condition

$$k^2 = -m^2. \quad (1.234)$$

Momentum operator

The Noether charge associated with spatial translations is the momentum operator (See Appendix B)

$$\vec{P} = \int d^3\vec{x} \psi^\dagger (-i\nabla) \psi. \quad (1.235)$$

Using the mode expansion one finds

$$\vec{P} = \int \frac{d^3\vec{k}}{(2\pi)^3} \vec{k} \sum_{s=1,2} (a^\dagger a + b^\dagger b). \quad (1.236)$$

Therefore the states (1.232) and (1.233) are eigenstates of the momentum operator with eigenvalue $\vec{k}$.

Normalization of one-particle states

The normalization of the particle states is

$$\langle \vec{k}, s | \vec{k}', r \rangle = 2\omega_{\vec{k}} \delta_{rs} \delta^{(3)}(\vec{k} - \vec{k}'). \quad (1.237)$$

Similarly for antiparticles,

$$\langle \vec{k}, \bar{s} | \vec{k}', \bar{r} \rangle = 2\omega_{\vec{k}} \delta_{\bar{s}\bar{r}} \delta^{(3)}(\vec{k} - \vec{k}'). \quad (1.238)$$

Particle and antiparticle states are orthogonal,

$$\langle \vec{k}, s | \vec{k}', \bar{r} \rangle = 0, \quad (1.239)$$

since the operators a and b anticommute.

Spin and statistics

Each excitation of the Dirac field has two independent spin polarizations. In quantum mechanics the number of spin states of a particle with spin s is $2s + 1$. Since the Dirac particle has two spin states, it follows that

$$2s + 1 = 2 \quad \Rightarrow \quad s = \frac{1}{2}.$$

Thus the Dirac field describes spin- $\frac{1}{2}$ particles.

One can verify this explicitly by constructing the angular momentum operator from the Noether charge associated with Lorentz symmetry, see Appendix A.

More generally, the spin-statistics theorem states that fields with half-integer spin must obey anticommutation relations, while fields with integer spin obey commutation relations. Thus fermionic fields obey Fermi–Dirac statistics and satisfy the Pauli exclusion principle, while bosonic fields obey Bose–Einstein statistics.

Charge

Global $U(1)$ symmetry and conserved current

In the Dirac theory, the action

$$S = -i \int d^4x \bar{\psi} (\gamma^\mu \partial_\mu - m) \psi \quad (1.240)$$

is invariant under a global phase rotation

$$\psi \rightarrow e^{i\alpha} \psi, \quad \bar{\psi} \rightarrow e^{-i\alpha} \bar{\psi}, \quad (1.241)$$

where α is a constant. This defines a global $U(1)$ symmetry.

By Noether's theorem, this symmetry implies the existence of a conserved current

$$\partial_\mu j^\mu = 0. \quad (1.242)$$

The associated current is

$$j^\mu = -\bar{\psi} \gamma^\mu \psi, \quad j^0 = \psi^\dagger \psi. \quad (1.243)$$

In the original interpretation of the Dirac equation as a wave equation, j^0 was interpreted as a probability density and is manifestly positive definite. However, in quantum field theory, ψ is an operator-valued field, and j^μ is interpreted instead as a conserved charge current.

Total charge operator

The conserved charge is defined as

$$Q = \int d^3x j^0 = \int d^3x \psi^\dagger \psi. \quad (1.244)$$

Substituting the mode expansion (1.226), one finds

$$Q = \int \frac{d^3\vec{k}}{(2\pi)^3} \sum_{s=1,2} \left[\left(a_{\vec{k}}^{(s)} \right)^\dagger a_{\vec{k}}^{(s)} - \left(b_{\vec{k}}^{(s)} \right)^\dagger b_{\vec{k}}^{(s)} \right] + Q_0, \quad (1.245)$$

where Q_0 is an infinite constant arising from the vacuum contribution. As in the Hamiltonian, this constant is removed by normal ordering or by defining the vacuum charge to vanish:

$$Q|0\rangle = 0. \quad (1.246)$$

Particles and antiparticles

After removing the vacuum contribution, the charge operator becomes

$$Q = \int \frac{d^3\vec{k}}{(2\pi)^3} \sum_{s=1,2} \left[\left(a_{\vec{k}}^{(s)} \right)^\dagger a_{\vec{k}}^{(s)} - \left(b_{\vec{k}}^{(s)} \right)^\dagger b_{\vec{k}}^{(s)} \right]. \quad (1.247)$$

This shows that:

- The operators $a^\dagger$ create states with charge $+1$
- The operators $b^\dagger$ create states with charge -1

Indeed, acting on one-particle states,

$$Q|\vec{k}, s\rangle = +|\vec{k}, s\rangle, \quad Q|\vec{k}, \bar{s}\rangle = -|\vec{k}, \bar{s}\rangle. \quad (1.248)$$

Thus, the two types of excitations created by $a^\dagger$ and $b^\dagger$ have:

- the same mass
- the same spin ($\frac{1}{2}$)
- opposite charge

This is precisely the definition of particle and antiparticle.

Physical interpretation

The conserved charge Q is an abstract quantum number associated with the global $U(1)$ symmetry. Its physical meaning depends on how the Dirac field is coupled to other fields. When coupled to electromagnetism, this $U(1)$ symmetry becomes the electric charge symmetry, and Q is identified (up to a normalization factor) with electric charge.

Historical remark

One of the most remarkable predictions of the Dirac equation is the existence of antiparticles. Although originally motivated by the search for a relativistic wave equation, the theory naturally predicts states with the same mass and spin as the electron but opposite charge.

Dirac initially speculated that these states might correspond to protons, but this interpretation was quickly ruled out due to the mass difference. The correct interpretation is that they represent a new particle, the positron.

This prediction was experimentally confirmed in 1932 by Carl Anderson, who observed a particle in cosmic rays with the same mass as the electron but opposite charge.

Correlation Functions

Motivation

In quantum field theory, the fundamental observables are not the fields themselves but their correlation functions. While in classical field theory one studies solutions of equations of motion, in quantum theory the fields become operators and cannot be measured directly.

Instead, physical information is encoded in expectation values such as

$$\langle 0 | \psi(x_1) \bar{\psi}(x_2) \cdots | 0 \rangle. \quad (1.249)$$

These correlation functions describe how excitations created at one spacetime point propagate to another.

In particular:

- Two-point functions (propagators) describe the propagation of a single particle.
- Time-ordered correlation functions are the basic building blocks of perturbation theory and Feynman diagrams.
- All scattering amplitudes can ultimately be expressed in terms of correlation functions.

Therefore, instead of solving the Dirac equation directly, quantum field theory focuses on computing correlation functions.

Correlation functions are the basic observables of quantum field theory.

The Wightman function

We first define the Wightman function, which does not involve time ordering:

$$D_+^{\alpha\beta}(x-y) \equiv \langle 0 | \psi_\alpha(x) \bar{\psi}_\beta(y) | 0 \rangle. \quad (1.250)$$

Substituting the mode expansion (1.226), only the particle part contributes, and we obtain

$$D_+^{\alpha\beta}(x-y) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} \sum_{s=1,2} u_s^\alpha(k) \bar{u}_s^\beta(k) e^{ik \cdot (x-y)}. \quad (1.251)$$

Using the completeness relation for Dirac spinors,

$$\sum_{s=1,2} u_s(k) \bar{u}_s(k) = (i\not{k} + m), \quad (1.252)$$

we find

$$D_+^{\alpha\beta}(x-y) = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} i(i\not{k} + m)_{\alpha\beta} e^{ik \cdot (x-y)}. \quad (1.253)$$

Now we use the fact that

$$k_\mu \rightarrow i\partial_\mu^x$$

when acting on the exponential. Therefore,

$$D_+^{\alpha\beta}(x-y) = i(\not{\partial}_x + m)_{\alpha\beta} \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{ik \cdot (x-y)}. \quad (1.254)$$

Defining the scalar Wightman function

$$G_+(x-y) = \langle 0 | \phi(x) \phi^\dagger(y) | 0 \rangle = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{2\omega_{\vec{k}}} e^{ik \cdot (x-y)}, \quad (1.255)$$

we obtain the important relation

$$\boxed{D_+^{\alpha\beta}(x-y) = i(\not{\partial}_x + m)_{\alpha\beta} G_+(x-y)}. \quad (1.256)$$

This shows that fermionic correlation functions are obtained from scalar ones by acting with a Dirac operator.

Retarded function

The retarded Green's function is defined using the anticommutator:

$$D_R^{\alpha\beta}(x-y) = \theta(x^0 - y^0) \langle 0 | \{ \psi_\alpha(x), \bar{\psi}_\beta(y) \} | 0 \rangle. \quad (1.257)$$

It satisfies

$$D_R^{\alpha\beta}(x-y) = i(\not{\partial}_x + m)_{\alpha\beta} G_R(x-y), \quad (1.258)$$

where $G_R(x-y)$ is the retarded Green's function for the scalar field.

Feynman propagator

The most important object for perturbation theory is the time-ordered (Feynman) propagator:

$$D_F^{\alpha\beta}(x-y) = \langle 0 | T \psi_\alpha(x) \bar{\psi}_\beta(y) | 0 \rangle. \quad (1.259)$$

For fermions, time ordering includes an extra minus sign:

$$D_F^{\alpha\beta}(x-y) = \begin{cases} \langle 0 | \psi_\alpha(x) \bar{\psi}_\beta(y) | 0 \rangle, & x^0 > y^0, \\ -\langle 0 | \bar{\psi}_\beta(y) \psi_\alpha(x) | 0 \rangle, & x^0 < y^0. \end{cases} \quad (1.260)$$

This minus sign is a direct consequence of the anticommutation relations of fermionic fields.

As before, one finds

$$\boxed{D_F^{\alpha\beta}(x-y) = i(\not{\partial}_x + m)_{\alpha\beta} G_F(x-y)} \quad (1.261)$$

where $G_F(x-y)$ is the scalar Feynman propagator.

Momentum space representation

Taking the Fourier transform, we obtain the propagator in momentum space:

$$D_F(k) = i(\not{k} + m) \frac{-i}{k^2 + m^2 + i\epsilon}. \quad (1.262)$$

This can be understood from the identity

$$(i\not{k} - m)(i\not{k} + m) = (-k^2 - m^2)\mathbf{1}_{4 \times 4}. \quad (1.263)$$

Thus the Dirac propagator can be written compactly as

$$D_F(k) = \frac{1}{-i\not{k} + m - i\epsilon} \quad (1.264)$$

which is a 4×4 matrix in spinor space.

Summary

We see that all fermionic propagators can be obtained from scalar propagators via

$$D(x - y) = i(\not{\partial} + m) G(x - y). \quad (1.265)$$

This relation reflects the fact that the Dirac equation is essentially a “square root” of the Klein–Gordon equation.

1.6 Chiral Fermions

Readings

- Peskin & Schroeder Sec. 3.2
- Zee Chap. II

We will start a new topic: **chiral fermions**. Recall that under a Lorentz transformation Λ ,

$$x' = \Lambda x, \quad (1.266)$$

the Dirac spinor transforms as

$$\psi(x) \rightarrow \psi'(x') = S(\Lambda) \psi(x), \quad (1.267)$$

where

$$S(\Lambda) = \exp\left(-\frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu}\right), \quad \Sigma^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu]. \quad (1.268)$$

Thus, Lorentz transformations are generated by the commutators of gamma matrices. This structure ensures that the Dirac equation is consistent with Lorentz symmetry.

Motivation: why study chiral fermions? In the construction of the Dirac equation, we introduced a four-component spinor ψ in order to write a relativistic wave equation that is linear in derivatives and consistent with Lorentz symmetry. This naturally leads to the question:

Is the four-component structure required by Lorentz symmetry itself, or is it a consequence of additional physical requirements?

To answer this, we analyze how spinors transform under Lorentz transformations. We will find that the Dirac spinor representation is **reducible**: it can be decomposed into two independent two-component pieces. Each of these pieces already furnishes a valid representation of the Lorentz group.

This observation leads to the notion of **chiral (Weyl) fermions**, which describe two-component spinors transforming irreducibly under Lorentz transformations.

The four-component Dirac spinor then arises when we combine the two chiral components, which becomes necessary when additional structures such as a mass term or parity symmetry are imposed. We first study the chiral decomposition of the Dirac spinor.

Choice of gamma matrices To make the reduction explicit, we choose a specific representation - the chiral representation:

$$\gamma^0 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix}, \quad (1.269)$$

where σ^i are the Pauli matrices. In this representation, the Lorentz generators take a particularly simple form. Using

$$\Sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu], \quad (1.270)$$

we obtain the **boost generators**

$$\Sigma^{0i} = -\frac{i}{2} \begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}. \quad (1.271)$$

Similarly, the rotation generator is

$$\Sigma^{ij} = -\frac{1}{2} \epsilon^{ijk} \begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}. \quad (1.272)$$

These together form the Lorentz generators. One finds that both boost and rotation generators have the form

$$\Sigma^{\mu\nu} = \begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix}, \quad (1.273)$$

which is **block diagonal**. Since

$$S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right), \quad (1.274)$$

it also follows the form of the block matrix

$$S(\Lambda) = \begin{pmatrix} S_L & 0 \\ 0 & S_R \end{pmatrix}. \quad (1.275)$$

Therefore, Lorentz transformations do **not mix** the upper and lower components.

Decomposition of the Dirac spinor We write

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}. \quad (1.276)$$

Then:

- ψ_L transforms with S_L ,
- ψ_R transforms with S_R ,
- they transform independently.

In summary, each of ψ_L and ψ_R :

- forms a valid representation of the Lorentz group,
- can be treated independently.

Thus,

$$\text{we can consistently reduce from 4 components to 2 components.} \quad (1.277)$$

They are:

- ψ_L : left-handed (chiral) fermion
- ψ_R : right-handed (chiral) fermion

This leads to the notion of **Weyl fermions**.

The **key reason** this works is that the representation is **reducible**: the block diagonal structure of $\Sigma^{\mu\nu}$ implies that the Dirac representation decomposes into two independent pieces.

Why two components are enough From the block diagonal form of $S(\Lambda)$, we see that:

- the upper two components and lower two components do not mix,
- each transforms independently under Lorentz transformations.

Therefore, each of ψ_L and ψ_R already has a well-defined Lorentz transformation. This shows that:

Lorentz covariance itself only requires two-component spinors.

The four-component Dirac spinor arises when we combine the two chiralities, which becomes necessary when introducing a mass term or imposing additional symmetries such as parity.

Relation to the massless case In fact, we already encountered this earlier. For a massless fermion ($m = 0$), the Dirac equation reduces effectively to two components, and remains Lorentz covariant. Thus:

- Massless fermions can be described by two-component spinors.
- Lorentz symmetry is compatible with two components.

This suggests that:

It is the mass term that requires the full four-component Dirac spinor.

General gamma matrices: need for a systematic construction So far, the decomposition $\psi = (\psi_L, \psi_R)$ relied on a special representation (1.269), where the generators are manifestly block diagonal. However, for a **general choice of gamma matrices**, this block structure is not obvious.

Since all representations of gamma matrices are equivalent, the decomposition into two components must exist in any representation. The question is how to construct it in general.

The key object is the matrix

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3. \quad (1.278)$$

The factor of i is chosen such that γ^5 is Hermitian:

$$(\gamma^5)^\dagger = \gamma^5. \quad (1.279)$$

One can verify the following important properties:

- Square:

$$(\gamma^5)^2 = 1 \quad (1.280)$$

- Anticommutation with gamma matrices:

$$\{\gamma^5, \gamma^\mu\} = 0 \quad (1.281)$$

- Traceless:

$$\text{Tr}(\gamma^5) = 0 \quad (1.282)$$

Since γ^5 is Hermitian and satisfies $(\gamma^5)^2 = 1$:

- its eigenvalues are ± 1 ,
- the trace condition implies equal numbers of $+1$ and -1 eigenvalues.

Therefore, the Dirac spinor space decomposes into two two-dimensional eigenspaces ⁵.

⁵To clarify the meaning of a “two-dimensional eigenspace,” consider

$$\gamma^5 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Its eigenvalues are $\{-1, -1, +1, +1\}$. A convenient choice of corresponding eigenvectors is:

- For eigenvalue $+1$:

$$v_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

Projection operators Thus, we define the projection operators:

$$P_L = \frac{1 - \gamma^5}{2}, \quad P_R = \frac{1 + \gamma^5}{2}. \quad (1.283)$$

They satisfy:

$$P_L^2 = P_L, \quad P_R^2 = P_R, \quad P_L P_R = 0, \quad P_L + P_R = 1. \quad (1.284)$$

The projectors $P_L = \frac{1 - \gamma^5}{2}$ and $P_R = \frac{1 + \gamma^5}{2}$ project onto the eigenspaces of γ^5 with eigenvalues -1 and $+1$, respectively, i.e.,

$$P_L \gamma^5 = -P_L, \quad P_R \gamma^5 = +P_R.$$

Definition of chiral components We define:

$$\psi_L = P_L \psi, \quad \psi_R = P_R \psi. \quad (1.285)$$

These satisfy:

$$\gamma^5 \psi_L = -\psi_L, \quad \gamma^5 \psi_R = +\psi_R. \quad (1.286)$$

Thus, ψ_L and ψ_R are eigenstates of γ^5 with eigenvalues -1 and $+1$ respectively. From above, we can see that even for general gamma matrices:

- the Dirac spinor decomposes into two independent components,
- the decomposition is defined by γ^5 ,
- each component transforms independently under Lorentz symmetry.

This provides a representation-independent definition of **chirality**.

Lorentz invariance of chirality We now show an important property:

$$[\gamma^5, \Sigma^{\mu\nu}] = 0. \quad (1.287)$$

Recall that

$$\Sigma^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu]. \quad (1.288)$$

Using the anticommutation relation

$$\{\gamma^5, \gamma^\mu\} = 0, \quad (1.289)$$

we compute:

$$\gamma^5 \Sigma^{\mu\nu} = \frac{i}{4} \gamma^5 (\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) \quad (1.290)$$

$$= \frac{i}{4} (-\gamma^\mu \gamma^5 \gamma^\nu + \gamma^\nu \gamma^5 \gamma^\mu) \quad (1.291)$$

$$= \frac{i}{4} (\gamma^\mu \gamma^\nu \gamma^5 - \gamma^\nu \gamma^\mu \gamma^5) \quad (1.292)$$

$$= \Sigma^{\mu\nu} \gamma^5. \quad (1.293)$$

Thus,

$$[\gamma^5, \Sigma^{\mu\nu}] = 0. \quad (1.294)$$

Since

$$S(\Lambda) = \exp\left(-\frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu}\right), \quad (1.295)$$

-
- For eigenvalue -1 :

$$w_1 = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}, \quad w_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

The eigenspace corresponding to eigenvalue $+1$ is defined as all linear combinations of v_1 and v_2 :

$$V_+ = \text{span}\{v_1, v_2\}.$$

Similarly, the eigenspace for eigenvalue -1 is

$$V_- = \text{span}\{w_1, w_2\}.$$

Since each eigenspace is generated by two linearly independent vectors, it has dimension 2. In general, the dimension of an eigenspace is the number of independent eigenvectors spanning it.

and γ^5 commutes with all $\Sigma^{\mu\nu}$, it follows that

$$[\gamma^5, S(\Lambda)] = 0. \quad (1.296)$$

Now consider the transformed spinor:

$$\psi'(x') = S(\Lambda)\psi(x). \quad (1.297)$$

Acting with γ^5 , we obtain:

$$\gamma^5\psi' = \gamma^5 S(\Lambda)\psi = S(\Lambda)\gamma^5\psi. \quad (1.298)$$

If ψ is an eigenstate of γ^5 , for example

$$\gamma^5\psi_L = -\psi_L, \quad (1.299)$$

then

$$\gamma^5\psi'_L = -\psi'_L. \quad (1.300)$$

Similarly,

$$\gamma^5\psi'_R = +\psi'_R. \quad (1.301)$$

Therefore:

- ψ_L transforms into ψ_L ,
- ψ_R transforms into ψ_R ,
- they do **not mix under Lorentz transformations**.

In Summary, we have shown that:

$$[\gamma^5, S(\Lambda)] = 0 \quad \Rightarrow \quad \text{chirality is preserved under Lorentz transformations.} \quad (1.302)$$

Thus, the decomposition

$$\psi = \psi_L + \psi_R \quad (1.303)$$

is Lorentz invariant, and each component forms a consistent representation of the Lorentz group. This confirms that the notion of **chiral fermions** is well-defined in any representation.

Dirac equation in terms of ψ_L and ψ_R Let us now rewrite the Dirac theory in terms of the chiral components

$$\psi_L = P_L\psi, \quad \psi_R = P_R\psi. \quad (1.304)$$

In the chiral representation (1.269), the Dirac spinor takes the form

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad (1.305)$$

where ψ_L and ψ_R are two-component spinors. Substituting this into the Dirac Lagrangian,

$$\mathcal{L} = -i\bar{\psi}(\gamma^\mu\partial_\mu - m)\psi, \quad (1.306)$$

one can rewrite it in terms of ψ_L and ψ_R . In the chiral representation, the kinetic terms become diagonal, and one finds schematically:

$$\mathcal{L}_{\text{kin}} = i\psi_L^\dagger(\partial_0 + i\sigma^i\partial_i)\psi_L + i\psi_R^\dagger(\partial_0 - i\sigma^i\partial_i)\psi_R. \quad (1.307)$$

Important observation:

- There are **no cross terms** between ψ_L and ψ_R in the kinetic term.

The mass term, however, couples them:

$$\mathcal{L}_{\text{mass}} = -m \left(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L \right). \quad (1.308)$$

Now consider the case $m = 0$. Then:

- the mass term vanishes,
- ψ_L and ψ_R completely decouple.

Thus, the theory splits into two independent parts:

$$\mathcal{L} = \mathcal{L}_L + \mathcal{L}_R. \quad (1.309)$$

This confirms again that:

a massless fermion can be described entirely by a two-component spinor.

This is consistent with our earlier observation that Lorentz symmetry only requires two components.

Chiral symmetry In the massive Dirac theory, we have a global $U(1)$ symmetry:

$$\psi \rightarrow e^{i\alpha}\psi. \quad (1.310)$$

In the massless case ($m = 0$), the left- and right-handed components decouple, and can be transformed independently:

$$\psi_L \rightarrow e^{i\alpha_L}\psi_L, \quad \psi_R \rightarrow e^{i\alpha_R}\psi_R. \quad (1.311)$$

Therefore, the symmetry is enhanced to:

$$U(1)_L \times U(1)_R. \quad (1.312)$$

This is called a **chiral symmetry**.

Alternative form of chiral symmetry: It is often useful to rewrite the transformation as:

$$\psi \rightarrow e^{i\alpha}\psi, \quad \psi \rightarrow e^{i\beta\gamma^5}\psi. \quad (1.313)$$

The second transformation distinguishes ψ_L and ψ_R , since:

$$\gamma^5\psi_L = -\psi_L, \quad \gamma^5\psi_R = +\psi_R. \quad (1.314)$$

The transformation $e^{i\beta\gamma^5}$ is called an **axial (chiral) symmetry**. It is a symmetry of the theory only in the massless case. When a mass term $m\bar{\psi}\psi$ is present, this symmetry is broken because the mass term mixes ψ_L and ψ_R .

Physical implications Chiral symmetry plays a very important role in physics:

- It is crucial in particle physics (for example, in the physics of pions).
- It can be present at the classical level but may be broken at the quantum level (anomalies).

In particular:

a classical symmetry may not survive quantization, leading to important physical effects.

Remarks Massless vs massive fermions: the difference between massless and massive fermions is profound:

- Massive fermion:
 - ψ_L and ψ_R are coupled,
 - cannot be separated dynamically.
- Massless fermion:
 - ψ_L and ψ_R are independent,
 - each describes a separate degree of freedom.

Thus:

$$\text{mass is what couples left- and right-handed components.} \quad (1.315)$$

1.7 Majorana Fermions

We now turn to another possible reduction of the Dirac spinor: the **Majorana fermion**.

Motivation: reducing complex degrees of freedom Let us compare different types of fermions:

- Dirac fermion: 4 complex components $\Rightarrow$ 8 real components
- Weyl fermion ψ_L or ψ_R : 2 complex components $\Rightarrow$ 4 real components
- Majorana fermion: 4 real components

Thus, Majorana fermions reduce the degrees of freedom by imposing a **reality condition**.

Basic idea: can a spinor be real? A natural question is:

Can we impose a reality condition $\psi^* = \psi$ on a Dirac spinor?

At first sight, this seems impossible, because the Dirac equation involves γ^μ , which are generally complex matrices. Let us examine this more carefully.

Compatibility with the Dirac equation The Dirac equation is

$$(\gamma^\mu \partial_\mu - m)\psi = 0. \quad (1.316)$$

Taking complex conjugation gives:

$$(\gamma^{\mu*} \partial_\mu - m)\psi^* = 0. \quad (1.317)$$

If we want $\psi^* = \psi$, then consistency requires:

$$\gamma^{\mu*} = \gamma^\mu. \quad (1.318)$$

Conclusion:

- A real spinor is possible only if we can choose a representation where γ^μ are real matrices.

Majorana representation It turns out that such a representation **does exist**. It is

$$\gamma_M^0 = \begin{pmatrix} 0 & i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix}, \quad \gamma_M^1 = \begin{pmatrix} \sigma^1 & 0 \\ 0 & \sigma^1 \end{pmatrix}, \quad \gamma_M^2 = \begin{pmatrix} 0 & -i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix}, \quad \gamma_M^3 = \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \quad (1.319)$$

In this representation:

$$\gamma_M^\mu = \text{real matrices}, \quad (\gamma_M^\mu)^\dagger = \gamma_M^\mu, \quad (\gamma_M^\mu)^2 = 1 \quad (1.320)$$

Then:

- the Dirac equation is real,
- a real spinor $\psi^* = \psi$ is consistent.

Such a spinor is called a **Majorana spinor**.

Compatibility with Lorentz transformations We must check that the Majorana condition is preserved under Lorentz transformations. Recall that a Dirac spinor transforms as

$$\psi \rightarrow \psi' = S(\Lambda)\psi, \quad S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right). \quad (1.321)$$

In the Majorana representation, the gamma matrices γ_M^μ are real, so their commutators $[\gamma_M^\mu, \gamma_M^\nu]$ are also real. It follows that

$$\Sigma^{\mu\nu} = \frac{i}{4}[\gamma_M^\mu, \gamma_M^\nu] \quad (1.322)$$

are purely imaginary, and therefore $S(\Lambda)$ is a real matrix. Now suppose that ψ satisfies the Majorana condition

$$\psi^* = \psi. \quad (1.323)$$

After a Lorentz transformation,

$$\psi' = S(\Lambda)\psi, \quad (1.324)$$

we take the complex conjugate:

$$\psi'^* = S(\Lambda)^*\psi^*. \quad (1.325)$$

Since $S(\Lambda)$ is real, $S(\Lambda)^* = S(\Lambda)$, and using $\psi^* = \psi$, we obtain

$$\psi'^* = S(\Lambda)\psi = \psi'. \quad (1.326)$$

Conclusion:

The Majorana condition is preserved under Lorentz transformations. Therefore, the space of Majorana spinors is closed under Lorentz symmetry.

Relation between different gamma matrix representations So far, the Majorana condition $\psi^* = \psi$ relied on a special representation where γ_M^μ are real. However, in a **general representation**, the gamma matrices are complex, and the condition $\psi^* = \psi$ no longer makes sense. We therefore need a **representation-independent** definition of the Majorana condition.

All sets of gamma matrices satisfying the Clifford algebra are equivalent up to a similarity transformation. In particular, let us denote:

- γ^μ : a general representation,
- γ_M^μ : the Majorana representation (where all γ_M^μ are real).

Then there exists an invertible matrix C such that

$$C\gamma^\mu C^{-1} = \gamma_M^\mu. \quad (1.327)$$

This matrix C implements a **change of basis** from the general representation to the Majorana representation.

Under this change of basis, the spinor transforms accordingly:

$$\psi_M = C\psi. \quad (1.328)$$

In the Majorana representation, we can impose the reality condition:

$$\psi_M^* = \psi_M. \quad (1.329)$$

Substituting $\psi_M = C\psi$, we obtain:

$$(C\psi)^* = C\psi. \quad (1.330)$$

This gives

$$C^* \psi^* = C\psi. \quad (1.331)$$

Multiplying both sides by C^{*-1} , we find:

$$\boxed{\psi^* = B\psi, \quad B \equiv C^{*-1}C} \quad (1.332)$$

which is the correct generalization of the Majorana condition.

Properties of B We now derive an important property of B . Starting from

$$C\gamma^\mu C^{-1} = \gamma_M^\mu, \quad (1.333)$$

and using that γ_M^μ are real:

$$(\gamma_M^\mu)^* = \gamma_M^\mu, \quad (1.334)$$

we take the complex conjugate:

$$C^* \gamma^{\mu*} (C^{-1})^* = \gamma_M^\mu. \quad (1.335)$$

Comparing with the original equation, we obtain:

$$C^* \gamma^{\mu*} (C^{-1})^* = C\gamma^\mu C^{-1}. \quad (1.336)$$

Rearranging, we find:

$$\gamma^{\mu*} = B\gamma^\mu B^{-1}, \quad B = C^{*-1}C. \quad (1.337)$$

Thus, B relates γ^μ to its complex conjugate.

Compatibility with Lorentz transformations We now check that the Majorana condition

$$\psi^* = B\psi \quad (1.338)$$

is preserved under Lorentz transformations. Recall that under a Lorentz transformation:

$$\psi \rightarrow \psi' = S(\Lambda)\psi, \quad S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right). \quad (1.339)$$

The condition is compatible with Lorentz symmetry if:

$$\boxed{\psi'^* = B\psi'} \quad (1.340)$$

whenever $\psi^* = B\psi$.

Step 1: action of B on $\Sigma^{\mu\nu}$

From

$$\gamma^{\mu*} = B\gamma^\mu B^{-1}, \quad (1.341)$$

we derive how B acts on the Lorentz generators:

$$\Sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]. \quad (1.342)$$

Taking complex conjugation:

$$\Sigma^{\mu\nu*} = \frac{-i}{4}[\gamma^{\mu*}, \gamma^{\nu*}] \quad (1.343)$$

$$= \frac{-i}{4}[B\gamma^\mu B^{-1}, B\gamma^\nu B^{-1}] \quad (1.344)$$

$$= \frac{-i}{4}B[\gamma^\mu, \gamma^\nu]B^{-1} \quad (1.345)$$

$$= -B\Sigma^{\mu\nu}B^{-1}. \quad (1.346)$$

Thus:

$$\boxed{\Sigma^{\mu\nu*} = -B\Sigma^{\mu\nu}B^{-1}} \quad (1.347)$$

The minus sign comes from complex conjugation of i .

Step 2: relation for $S(\Lambda)^*$

Now consider the Lorentz transformation matrix:

$$S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right). \quad (1.348)$$

Taking complex conjugation:

$$S(\Lambda)^* = \exp\left(+\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu*}\right) \quad (1.349)$$

$$= \exp\left(+\frac{i}{2}\omega_{\mu\nu}(-B\Sigma^{\mu\nu}B^{-1})\right) \quad (1.350)$$

$$= \exp\left(-\frac{i}{2}\omega_{\mu\nu}B\Sigma^{\mu\nu}B^{-1}\right). \quad (1.351)$$

Using the identity

$$\exp(BAB^{-1}) = B\exp(A)B^{-1}, \quad (1.352)$$

we obtain:

$$\boxed{S(\Lambda)^* = B S(\Lambda) B^{-1}} \quad (1.353)$$

This is a crucial relation.

Step 3: check the Majorana condition

Now compute:

$$\psi'^* = (S(\Lambda)\psi)^* = S(\Lambda)^*\psi^*. \quad (1.354)$$

Using:

$$S(\Lambda)^* = BS(\Lambda)B^{-1}, \quad \psi^* = B\psi, \quad (1.355)$$

we find:

$$\psi'^* = (BS(\Lambda)B^{-1})(B\psi) \quad (1.356)$$

$$= BS(\Lambda)\psi \quad (1.357)$$

$$= B\psi'. \quad (1.358)$$

Thus:

$$\boxed{\psi'^* = B\psi'} \quad (1.359)$$

We have shown that:

- if ψ satisfies the Majorana condition $\psi^* = B\psi$,
- then the transformed spinor $\psi' = S(\Lambda)\psi$ also satisfies it.

Therefore:

$$\boxed{\text{The Majorana condition is compatible with Lorentz symmetry.}} \quad (1.360)$$

Majorana spinor in the chiral basis Let us now work out the Majorana condition explicitly in the chiral basis. In the Majorana representation, we had simply

$$B = 1. \quad (1.361)$$

In a general basis, however, the Majorana condition takes the form

$$\psi^* = B\psi. \quad (1.362)$$

We now determine B in the chiral representation.

Step 1: reality properties of the gamma matrices. Recall that in the chiral basis,

$$\gamma^0 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix}. \quad (1.363)$$

From these explicit expressions, one finds:

- γ^0, γ^1 , and γ^3 are purely imaginary,
- γ^2 is real.

Thus,

$$(\gamma^0)^* = -\gamma^0, \quad (\gamma^1)^* = -\gamma^1, \quad (\gamma^3)^* = -\gamma^3, \quad (1.364)$$

while

$$(\gamma^2)^* = \gamma^2. \quad (1.365)$$

Step 2: determine B . Recall that B is defined by

$$\gamma^{\mu*} = B\gamma^\mu B^{-1}. \quad (1.366)$$

For those gamma matrices that are purely imaginary, this becomes

$$-\gamma^\mu = B\gamma^\mu B^{-1}, \quad (1.367)$$

which means that B must **anticommute** with them. For the gamma matrix that is real, namely γ^2 , we have

$$\gamma^2 = B\gamma^2 B^{-1}, \quad (1.368)$$

which means that B must **commute** with γ^2 . Therefore, in the chiral basis, B must:

- anticommute with $\gamma^0, \gamma^1, \gamma^3$,
- commute with γ^2 .

A matrix with precisely these properties is

$$B = \gamma^2. \quad (1.369)$$

Hence, in the chiral basis, the Majorana condition becomes

$$\boxed{\psi^* = \gamma^2 \psi}. \quad (1.370)$$

This is exactly the explicit form of the Majorana condition in the chiral basis.

Step 3: write the condition in terms of ψ_L and ψ_R . Now decompose the Dirac spinor as

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad (1.371)$$

where ψ_L and ψ_R are two-component spinors. In the chiral basis,

$$\gamma^2 = \begin{pmatrix} 0 & -i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix}. \quad (1.372)$$

Thus,

$$\psi^* = \begin{pmatrix} \psi_L^* \\ \psi_R^* \end{pmatrix} = \gamma^2 \psi = \begin{pmatrix} 0 & -i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} -i\sigma^2 \psi_R \\ i\sigma^2 \psi_L \end{pmatrix}. \quad (1.373)$$

Therefore, the Majorana condition implies

$$\psi_L^* = -i\sigma^2 \psi_R, \quad \psi_R^* = i\sigma^2 \psi_L. \quad (1.374)$$

Equivalently, we can write

$$\psi_R = i\sigma^2 \psi_L^*, \quad \psi_L = -i\sigma^2 \psi_R^*. \quad (1.375)$$

So in the Majorana case, ψ_L and ψ_R are no longer independent. Once one of them is given, the other is completely determined.

Step 4: explicit form of the Majorana spinor. Since ψ_R is determined by ψ_L , a Majorana spinor in the chiral basis can be written as

$$\psi = \begin{pmatrix} \psi_L \\ i\sigma^2 \psi_L^* \end{pmatrix}. \quad (1.376)$$

Thus the whole four-component spinor is determined by a single two-component spinor ψ_L . This makes the counting of degrees of freedom very clear:

- ψ_L has two complex components,
- therefore it contains four real components,
- so the Majorana spinor has four independent real degrees of freedom.

This agrees with our earlier counting:

$$\text{Majorana fermion} \iff 4 \text{ real components.} \quad (1.377)$$

Interpretation. In the chiral basis, a Majorana fermion contains both left- and right-handed pieces, but they are related by complex conjugation rather than being independent fields. In this sense, a Majorana spinor is different from a Weyl spinor:

- a Weyl spinor keeps only ψ_L or only ψ_R ,
- a Majorana spinor contains both ψ_L and ψ_R , but ties them together by the Majorana condition.

Therefore, the Majorana condition does **not** mean that one simply sets $\psi_R = 0$ or $\psi_L = 0$. Instead, it means that the two chiral components are related to each other.

Remark on the massless case. This condition is independent of whether the fermion is massive or massless. However, in the massless case one may work directly with a single chiral component, and then the description reduces to the same number of degrees of freedom as for a massless Weyl fermion.

1.8 Discrete Symmetries

So far, we have mainly discussed **continuous symmetries**, i.e. transformations depending on continuous parameters. We now turn to **discrete symmetries**.

Definition of discrete symmetries A discrete symmetry is a transformation that:

- does not depend on any continuous parameter,
- cannot be generated infinitesimally.

In contrast:

- continuous symmetries are generated by infinitesimal transformations,
- discrete symmetries are **finite transformations** only.

Example: $\mathbb{Z}_2$ symmetry Consider a real scalar field with Lagrangian:

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4. \quad (1.378)$$

This theory is invariant under the transformation:

$$\phi \rightarrow -\phi. \quad (1.379)$$

This is because:

- the kinetic term is quadratic in ϕ ,
- the mass term is quadratic,
- the interaction term involves ϕ^4 ,

so all terms are even in ϕ . Applying the transformation twice gives:

$$(\phi \rightarrow -\phi)^2 = 1, \quad (1.380)$$

so this defines a $\mathbb{Z}_2$ **symmetry**.

Internal vs spacetime discrete symmetries Discrete symmetries can be classified into two types:

- **Internal symmetries:** act on fields but not spacetime
- **Spacetime symmetries:** act on spacetime coordinates

The example $\phi \rightarrow -\phi$ is an **internal symmetry**.

Complex scalar field

Now consider a complex scalar field with Lagrangian:

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi - \lambda (\phi^* \phi)^2. \quad (1.381)$$

This theory has a continuous global $U(1)$ symmetry:

$$\phi \rightarrow e^{i\alpha} \phi. \quad (1.382)$$

Remark: $\phi \rightarrow -\phi$ is no longer independent In this case,

$$\phi \rightarrow -\phi \quad (1.383)$$

is not an independent discrete symmetry, since it can be obtained as a special case of the continuous symmetry:

$$\phi \rightarrow e^{i\pi} \phi = -\phi. \quad (1.384)$$

Therefore, it does not define a new discrete symmetry.

Charge conjugation symmetry However, the transformation

$$\phi \rightarrow \phi^* \quad (1.385)$$

is a genuine internal discrete symmetry. Under this transformation:

- $\phi^* \phi \rightarrow \phi \phi^* = \phi^* \phi$,
- $\partial_\mu \phi^* \partial^\mu \phi \rightarrow \partial_\mu \phi \partial^\mu \phi^*$,

so the Lagrangian is invariant. This symmetry is called **charge conjugation**:

$$C : \quad \phi \rightarrow \phi^*. \quad (1.386)$$

Physically, it exchanges particles with antiparticles.

Discrete symmetries We now introduce discrete symmetries acting on spacetime.

Parity $\mathcal{P}$: Parity transformation flips the spatial coordinates:

$$\mathcal{P} : \quad x^\mu = (t, \vec{x}) \rightarrow (t, -\vec{x}). \quad (1.387)$$

The field transforms accordingly (details depend on the theory, but for scalar fields it is typically)

$$\phi(t, \vec{x}) \rightarrow \phi(t, -\vec{x}). \quad (1.388)$$

Remark: One may ask why we reverse all three spatial directions rather than just one or two.

- Reversing two directions is equivalent to a 180° rotation,
- which is part of a continuous symmetry,
- therefore it does not define an independent discrete symmetry.

Thus, reversing all three spatial directions gives the only independent spatial reflection symmetry.

Time reversal $\mathcal{T}$: Time reversal flips the time coordinate:

$$\mathcal{T} : \quad x^\mu = (t, \vec{x}) \rightarrow (-t, \vec{x}). \quad (1.389)$$

The internal discrete symmetry is

Charge conjugation $\mathcal{C}$: Charge conjugation acts on the field itself:

$$\mathcal{C}: \phi(x) \rightarrow \phi^*(x). \quad (1.390)$$

For scalar fields:

$$\phi \rightarrow \phi^*. \quad (1.391)$$

For fermions, this will be generalized using the charge conjugation operator:

$$\psi \rightarrow \psi^c. \quad (1.392)$$

Symmetries at the quantum level Our goal is to understand how these symmetries are implemented in **quantum theory**. In quantum mechanics, symmetry transformations are represented by operators acting on the Hilbert space:

$$\text{symmetry} \longleftrightarrow \text{operator on states}. \quad (1.393)$$

For **continuous symmetries**, we have already seen that they are implemented by **unitary operators**.

The same idea applies to discrete symmetries, with one important subtlety:

- **Parity (P)** and **Charge conjugation (C)** are implemented by **unitary operators**
- **Time reversal (T)** is implemented by an **anti-unitary operator**

We now explain why time reversal must be anti-unitary.

Time reversal in quantum mechanics Let us first discuss time reversal in ordinary quantum mechanics. Denote the time reversal unitary operator by U_T . We assume the system is **time-reversal invariant**, meaning that the transformation of the symmetry should commute with the Hamiltonian:

$$[U_T, H] = 0, \quad (1.394)$$

where H is the Hamiltonian. Let $\psi(t)$ satisfy the Schrödinger equation:

$$i \frac{\partial}{\partial t} \psi(t) = H \psi(t). \quad (1.395)$$

Define the transformed state:

$$\psi'(t) = U_T \psi(t). \quad (1.396)$$

Physically, time reversal means that $\psi'(t)$ should satisfy the Schrödinger equation with $t \rightarrow -t$:

$$i \frac{\partial}{\partial t} \psi'(t) = -H \psi'(t). \quad (1.397)$$

Why unitary is not enough Let us act with U_T on the Schrödinger equation:

$$U_T \left(i \frac{\partial}{\partial t} \psi \right) = U_T (H \psi). \quad (1.398)$$

Using $[U_T, H] = 0$, the right-hand side becomes:

$$U_T (H \psi) = H (U_T \psi) = H \psi'. \quad (1.399)$$

So we obtain:

$$U_T \left(i \frac{\partial}{\partial t} \psi \right) = H \psi'. \quad (1.400)$$

Now comes the key point: If U_T were an ordinary linear (unitary) operator, we would have:

$$U_T (i \partial_t \psi) = i \partial_t (U_T \psi), \quad (1.401)$$

which would give the same Schrödinger equation, *without* the required minus sign. This is inconsistent with time reversal.

Anti-linearity and anti-unitarity To obtain the correct transformation, we must require:

$$U_T i = -i U_T. \quad (1.402)$$

This means that U_T must take:

$$i \rightarrow -i. \quad (1.403)$$

This property defines an **anti-linear operator**.

Definition (anti-linear operator) An operator A is called anti-linear if for any complex number c :

$$A(c\psi) = c^* A\psi. \quad (1.404)$$

In particular:

$$A(i\psi) = (-i)A\psi. \quad (1.405)$$

Thus, the time reversal operator must be anti-linear.

Adjoint of anti-linear operators We have seen that the time reversal operator U_T must be **anti-linear**. Because of this unusual property, the definition of the **adjoint** must also be modified.

Ordinary (linear) operators For a linear operator A , the adjoint $A^\dagger$ is defined by:

$$\langle \psi, A^\dagger \chi \rangle = \langle A\psi, \chi \rangle. \quad (1.406)$$

Anti-linear operators For an anti-linear operator A , the definition must be modified to:

$$\langle \psi, A^\dagger \chi \rangle = \langle A\psi, \chi \rangle^*. \quad (1.407)$$

The additional complex conjugation is required for consistency with anti-linearity.

Remark This definition ensures that complex numbers can be consistently moved in and out of inner products, given that anti-linear operators conjugate coefficients:

$$A(c\psi) = c^* A\psi. \quad (1.408)$$

Anti-unitary operators We now define **anti-unitary operators**.

Unitary case For a unitary operator U , we have:

$$\langle U\psi, U\chi \rangle = \langle \psi, \chi \rangle. \quad (1.409)$$

Anti-unitary case For an anti-unitary operator U , we instead have:

$$\langle U\psi, U\chi \rangle = \langle \psi, \chi \rangle^*. \quad (1.410)$$

Thus:

- Anti-unitary operators preserve norms
- But they involve complex conjugation
- They are necessarily anti-linear

Conclusion The time reversal operator U_T must be **anti-unitary** in order to be a symmetry of quantum mechanics.

Discrete symmetries in quantum field theory We now move from quantum mechanics to **quantum field theory**. Before studying Dirac fields, we consider a simpler example: a **complex scalar field**.

Classical symmetry of the scalar field Consider a complex scalar field $\phi(x)$ with a standard relativistic Lagrangian. This theory is invariant under the discrete transformations:

- **Parity (P):**

$$\phi(x) \rightarrow \phi'(x) = \eta_P \phi(Px), \quad (1.411)$$

- **Time reversal (T):**

$$\phi(x) \rightarrow \phi'(x) = \eta_T \phi(Tx), \quad (1.412)$$

- **Charge conjugation (C):**

$$\phi(x) \rightarrow \phi'(x) = \eta_C \phi^*(Cx). \quad (1.413)$$

Here:

$$Px = (t, -\vec{x}), \quad Tx = (-t, \vec{x}), \quad (1.414)$$

and η_P, η_T, η_C are phase factors.

Quantum implementation At the quantum level, these transformations are implemented by operators acting on the Hilbert space. For example:

$$U_P \phi(x) U_P^\dagger = \phi'(x) = \eta_P \phi(Px), \quad (1.415)$$

$$U_T \phi(x) U_T^\dagger = \phi'(x) = \eta_T \phi(Tx), \quad (1.416)$$

$$U_C \phi(x) U_C^\dagger = \phi'(x) = \eta_C \phi^*(Cx). \quad (1.417)$$

Here:

- U_P, U_C are unitary
- U_T is anti-unitary

At first sight, the phases η_P, η_T, η_C appear arbitrary. However, they are constrained by consistency conditions.

Example: parity squared Applying parity twice:

$$\mathcal{P}^2 = 1. \quad (1.418)$$

Thus, acting twice on the field:

$$\phi(x) \xrightarrow{\mathcal{P}} \eta_P \phi(Px) \xrightarrow{\mathcal{P}} \eta_P^2 \phi(x). \quad (1.419)$$

Consistency requires:

$$\eta_P^2 = 1, \quad (1.420)$$

so that:

$$\eta_P = \pm 1. \quad (1.421)$$

This leads to an important distinction:

- $\eta_P = +1$:

$$\text{scalar field} \quad (1.422)$$

- $\eta_P = -1$:

$$\text{pseudoscalar field} \quad (1.423)$$

A pseudoscalar changes sign under parity.

Similar constraints The same reasoning applies to:

- time reversal: $\eta_T^2 = 1$
- charge conjugation: $\eta_C^2 = 1$

Thus, all these discrete symmetry phases are restricted to:

$$\eta = \pm 1. \quad (1.424)$$

Action on creation and annihilation operators To fully understand these symmetries, we must examine how they act on:

- creation and annihilation operators
- particle states

This will allow us to determine their action in interacting quantum field theory.

We now determine how discrete symmetries act on the creation and annihilation operators. Recall that the scalar field can be expanded as:

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_{\vec{k}} e^{ik \cdot x} + b_{\vec{k}}^\dagger e^{-ik \cdot x} \right). \quad (1.425)$$

By requiring

$$U_P \phi(x) U_P^\dagger = \eta_P \phi(Px), \quad (1.426)$$

we can deduce the transformation of the operators.

Parity One finds:

$$U_P a_{\vec{k}} U_P^\dagger = \eta_P a_{-\vec{k}}, \quad U_P b_{\vec{k}} U_P^\dagger = \eta_P b_{-\vec{k}}. \quad (1.427)$$

This reflects the fact that under spatial inversion:

$$\vec{k} \rightarrow -\vec{k}. \quad (1.428)$$

Time reversal Similarly:

$$U_T a_{\vec{k}} U_T^\dagger = \eta_T a_{-\vec{k}}, \quad U_T b_{\vec{k}} U_T^\dagger = \eta_T b_{-\vec{k}}. \quad (1.429)$$

This is consistent with the fact that time reversal flips momentum.

Charge conjugation For charge conjugation:

$$U_C a_{\vec{k}} U_C^\dagger = \eta_C b_{\vec{k}}, \quad U_C b_{\vec{k}} U_C^\dagger = \eta_C a_{\vec{k}}. \quad (1.430)$$

Thus:

$$\text{charge conjugation exchanges particles and antiparticles.} \quad (1.431)$$

CPT symmetries in Dirac field

We now turn to the **Dirac field**, where the situation becomes more subtle. The Dirac action is:

$$S = -i \int d^4x \bar{\psi} (\gamma^\mu \partial_\mu - m) \psi, \quad (1.432)$$

with equation of motion:

$$(\gamma^\mu \partial_\mu - m) \psi = 0. \quad (1.433)$$

Parity transformation for Dirac fields We now ask:

How should parity act on a Dirac spinor?

Naive guess By analogy with the scalar field, one might try:

$$\psi(x) \rightarrow \eta_P \psi(Px). \quad (1.434)$$

However, this is **not sufficient**.

Key point The Dirac field carries internal spinor indices. Under Lorentz transformations, spinors transform non-trivially in this internal space. Therefore, under parity, we must allow:

$$\psi(x) \rightarrow \psi'(x') = D \psi(x), \quad x' = Px, \quad (1.435)$$

where D is a **matrix acting on spinor space**. We say that the Dirac theory is parity invariant if:

Whenever $\psi(x)$ satisfies the Dirac equation, the transformed field $\psi'(x')$ also satisfies the same equation in the transformed coordinates.

This is completely analogous to Lorentz covariance.

Deriving the condition on D Start with the Dirac equation:

$$(\gamma^\mu \partial_\mu - m)\psi(x) = 0. \quad (1.436)$$

Under parity:

$$x^\mu = (t, \vec{x}) \rightarrow (t, -\vec{x}), \quad (1.437)$$

so derivatives transform as:

$$\partial_0 \rightarrow \partial_0, \quad \partial_i \rightarrow -\partial_i. \quad (1.438)$$

Thus the transformed equation becomes:

$$(\gamma^0 \partial_0 - \gamma^i \partial_i - m) \psi'(x') = 0. \quad (1.439)$$

Using $\psi'(x') = D\psi(x)$:

$$(\gamma^0 \partial_0 - \gamma^i \partial_i - m) D\psi(x) = 0. \quad (1.440)$$

On the other hand, multiplying the original equation by D :

$$D(\gamma^0 \partial_0 + \gamma^i \partial_i - m)\psi(x) = 0. \quad (1.441)$$

For these two equations to be equivalent, we require:

$$D\gamma^0 = \gamma^0 D, \quad D\gamma^i = -\gamma^i D. \quad (1.442)$$

Therefore, D commutes with γ^0 and anti-commutes with γ^i . These conditions uniquely determine D (up to a phase):

$$\boxed{D = \eta_P \gamma^0}. \quad (1.443)$$

Thus, the parity transformation of the Dirac field is:

$$\psi(t, \vec{x}) \rightarrow \psi'(t, -\vec{x}) = \eta_P \gamma^0 \psi(t, \vec{x}). \quad (1.444)$$

Remark on the phase η_P The phase η_P is constrained by consistency conditions, but the situation is more subtle than for scalar fields. For fermions:

- A 2π rotation can produce a minus sign
- Therefore, η_P is not restricted in the same naive way

We will not explore these subtleties in detail here, but keep in mind that fermions behave differently from scalars.

Charge conjugation for Dirac fields We now construct the charge conjugation transformation for the Dirac field.

Strategy For scalar fields, charge conjugation acts as:

$$\phi \rightarrow \phi^*. \quad (1.445)$$

For Dirac fields, we want a transformation that maps:

$$\psi \rightarrow (\text{something related to}) \psi^*. \quad (1.446)$$

while remaining consistent with Lorentz structure.

Construction A useful object is:

$$\bar{\psi} = \psi^\dagger \gamma^0. \quad (1.447)$$

Taking transpose:

$$\bar{\psi}^T = (\psi^\dagger \gamma^0)^T, \quad (1.448)$$

which is closely related to ψ^* . We therefore define:

$$\psi'(x) = C \bar{\psi}^T(x), \quad (1.449)$$

where C is a matrix acting on spinor indices.

Condition on the charge conjugation matrix We require that if ψ satisfies the Dirac equation:

$$(\gamma^\mu \partial_\mu - m)\psi = 0, \quad (1.450)$$

then ψ' also satisfies the same equation. Taking Hermitian conjugate and transpose of the Dirac equation, one finds:

$$(\gamma^\mu)^T \partial_\mu + m \quad (1.451)$$

acting on $\bar{\psi}^T$. To recover the original Dirac equation, we require:

$$C (\gamma^\mu)^T C^{-1} = -\gamma^\mu. \quad (1.452)$$

Such a matrix C always exists. The reason is that:

- γ^μ satisfy the Clifford algebra
- $-(\gamma^\mu)^T$ also satisfies the same algebra

Since all representations of the Clifford algebra are equivalent, there must exist a similarity transformation relating them:

$$\gamma^\mu \sim -(\gamma^\mu)^T. \quad (1.453)$$

This guarantees the existence of C .

Remark The explicit form of C depends on the representation of gamma matrices. For example, in the Dirac basis:

$$C = i\gamma^2\gamma^0. \quad (1.454)$$

Physical interpretation Charge conjugation acts as:

$$\text{particle} \longleftrightarrow \text{antiparticle}. \quad (1.455)$$

Time reversal for Dirac fields We now consider time reversal.

Ansatz We take:

$$\psi(x) \rightarrow \psi'(x') = D_T \psi(x), \quad x' = Tx = (-t, \vec{x}), \quad (1.456)$$

where D_T is a matrix acting on spinor space.

Definition of symmetry Time reversal invariance means that if ψ satisfies the Dirac equation, then ψ' satisfies:

$$(\gamma^\mu \partial'_\mu - m)\psi'(x') = 0. \quad (1.457)$$

Condition on D_T Following a derivation similar to parity, one finds that D_T must satisfy:

$$(\gamma^\mu)^T = D_T^T \gamma^\mu (D_T^{-1})^T. \quad (1.458)$$

Relation to charge conjugation Comparing this condition with that for C :

$$C(\gamma^\mu)^T C^{-1} = -\gamma^\mu, \quad (1.459)$$

we see that they differ by a minus sign. This minus sign can be generated by γ^5 , since:

$$\gamma^5 \gamma^\mu = -\gamma^\mu \gamma^5. \quad (1.460)$$

Thus, one can choose:

$$D_T^T = C^{-1} \gamma^5. \quad (1.461)$$

Conclusion Once C is known, the time reversal matrix D_T can be constructed.

Remarks on time reversal

- The operator U_T is anti-unitary
- However, the matrix D_T acting on spinor indices is an ordinary (linear) matrix
- Anti-linearity comes from the operator U_T , not from D_T

Summary of C, P, T for Dirac fields

- **Parity:**

$$\psi(t, \vec{x}) \rightarrow \eta_P \gamma^0 \psi(t, -\vec{x}) \quad (1.462)$$

- **Charge conjugation:**

$$\psi(x) \rightarrow C \bar{\psi}^T(x) \quad (1.463)$$

- **Time reversal:**

$$\psi(t, \vec{x}) \rightarrow D_T \psi(-t, \vec{x}) \quad (1.464)$$

2 Path Integrals for Fermions

We now turn to the path integral formulation for fermionic fields. This will provide a powerful and elegant framework for treating interacting fermionic quantum field theories, in complete analogy with the bosonic path integral.

Motivation: Why Ordinary Path Integrals Fail for Fermions

Let us first recall the structure of the path integral in ordinary quantum mechanics. For a particle with position operator $\hat{x}$ and momentum operator $\hat{p}$, the canonical commutation relation is

$$[\hat{x}, \hat{p}] = i. \quad (2.1)$$

The quantum mechanical path integral is constructed by integrating over all classical trajectories $x(t)$:

$$\int \mathcal{D}x(t) e^{iS[x]}. \quad (2.2)$$

Here $x(t)$ is an ordinary commuting function.

Similarly, in bosonic quantum field theory, for a scalar field $\hat{\phi}(t, \mathbf{x})$ and its conjugate momentum $\hat{\pi}_\phi(t, \mathbf{x})$, the canonical commutation relation is

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}_\phi(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad (2.3)$$

The corresponding path integral is obtained by integrating over all classical field configurations:

$$\int \mathcal{D}\phi e^{iS[\phi]}. \quad (2.4)$$

Again, $\phi(x)$ is an ordinary commuting classical field.

However, fermionic fields satisfy **anti-commutation relations** rather than commutation relations. For the Dirac field,

$$\{\hat{\psi}_\alpha(t, \mathbf{x}), \hat{\pi}_{\psi, \beta}(t, \mathbf{y})\} = i\delta_{\alpha\beta}\delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad (2.5)$$

This raises a fundamental question:

What kind of “classical” object should appear in the path integral for fermions?

If one naively integrates over ordinary commuting functions, the resulting path integral fails to reproduce the correct fermionic quantum theory.

2.1 Grassmann Variables

The solution was introduced by **Felix Berezin** in the early 1960s. Berezin proposed that the classical variables appearing in the fermionic path integral should themselves be **anti-commuting objects**. These are now called:

$$\boxed{\text{Grassmann variables}} \quad (2.6)$$

Thus, in the fermionic path integral, the classical fermion field is replaced by an anti-commuting Grassmann-valued field:

$$\psi(x) \rightarrow \text{Grassmann-valued field}. \quad (2.7)$$

The fermionic path integral then takes the schematic form

$$\int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{iS[\psi, \bar{\psi}]}. \quad (2.8)$$

Remarkably, this prescription reproduces the correct anti-commutation relations and fermionic propagators.

Grassmann Numbers

A Grassmann variable θ is defined by the property that it anti-commutes with itself:

$$\theta\theta = -\theta\theta. \quad (2.9)$$

Therefore,

$$\boxed{\theta^2 = 0.} \quad (2.10)$$

Similarly, for two Grassmann variables θ, η ,

$$\boxed{\theta\eta = -\eta\theta.} \quad (2.11)$$

Thus Grassmann variables anti-commute with one another.

Nilpotency The condition

$$\theta^2 = 0 \quad (2.12)$$

is called **nilpotency**. It implies that all powers higher than one vanish:

$$\theta^n = 0, \quad n \geq 2. \quad (2.13)$$

Functions of Grassmann Variables

Because Grassmann variables square to zero, functions of Grassmann variables have finite Taylor expansions. For a single Grassmann variable θ , any function can be expanded as

$$f(\theta) = f_0 + f_1\theta. \quad (2.14)$$

All higher powers vanish because $\theta^2 = 0$. Thus:

$$\boxed{\text{Any function of one Grassmann variable is linear.}} \quad (2.15)$$

Example The exponential truncates:

$$e^\theta = 1 + \theta. \quad (2.16)$$

More generally,

$$e^{a\theta} = 1 + a\theta. \quad (2.17)$$

Moreover,

$$\frac{1}{1 + \theta} = 1 - \theta. \quad (2.18)$$

Functions of Multiple Grassmann Variables

For two Grassmann variables θ_1, θ_2 , the most general function is

$$f(\theta_1, \theta_2) = f_0 + f_1\theta_1 + f_2\theta_2 + f_{12}\theta_1\theta_2. \quad (2.19)$$

No higher-order terms appear because any term containing

$$\theta_1^2 \quad \text{or} \quad \theta_2^2$$

vanishes. Thus:

$$\boxed{\text{The algebra of Grassmann variables is finite-dimensional.}} \quad (2.20)$$

Differentiation with Respect to Grassmann Variables

Differentiation is defined analogously to ordinary calculus. For

$$f(\theta) = f_0 + f_1\theta, \quad (2.21)$$

we define

$$\frac{\partial f}{\partial \theta} = f_1. \quad (2.22)$$

Important Subtlety: Left vs Right Derivatives For functions of several Grassmann variables, the order of differentiation matters because Grassmann variables anti-commute. Consider a function of two Grassmann variables:

$$f(\theta_1, \theta_2) = f_0 + f_1\theta_1 + f_2\theta_2 + f_{12}\theta_1\theta_2. \quad (2.23)$$

Left derivative Taking the derivative with respect to θ_1 from the left gives

$$\frac{\partial}{\partial\theta_1}f(\theta_1, \theta_2) = f_1 + f_{12}\theta_2. \quad (2.24)$$

Right derivative If instead we differentiate from the right,

$$f(\theta_1, \theta_2) \frac{\overleftarrow{\partial}}{\partial\theta_1}, \quad (2.25)$$

then the term $\theta_1\theta_2$ must first be reordered so that θ_1 is adjacent to the derivative operator:

$$\theta_1\theta_2 = -\theta_2\theta_1. \quad (2.26)$$

Thus,

$$f(\theta_1, \theta_2) \frac{\overleftarrow{\partial}}{\partial\theta_1} = f_1 - f_{12}\theta_2. \quad (2.27)$$

Hence:

Left and right Grassmann derivatives differ by signs.

(2.28)

Convention Unless stated otherwise, we will always use **left derivatives** throughout these notes.

Berezin Integration

To formulate fermionic path integrals, we must define integration over Grassmann variables. Unlike ordinary integration, Berezin integration is not geometrically intuitive and must be defined axiomatically. We require the Grassmann integral to satisfy the following properties:

(1) Linearity For Grassmann-valued functions $f(\theta), g(\theta)$ and constants a, b ,

$$\int d\theta [af(\theta) + bg(\theta)] = a \int d\theta f(\theta) + b \int d\theta g(\theta). \quad (2.29)$$

(2) Integral of a total derivative vanishes We impose

$\int d\theta \frac{\partial f(\theta)}{\partial\theta} = 0.$

(2.30)

This is the Grassmann analog of the statement that total derivatives integrate to boundary terms.

Determining the Integration Rules

Let

$$f(\theta) = f_0 + f_1\theta. \quad (2.31)$$

Then

$$\frac{\partial f}{\partial\theta} = f_1. \quad (2.32)$$

Using the requirement

$$\int d\theta \frac{\partial f}{\partial\theta} = 0,$$

we obtain

$$\int d\theta f_1 = 0. \quad (2.33)$$

Since f_1 is an ordinary number, it factors out of the integral:

$$f_1 \int d\theta = 0. \quad (2.34)$$

Thus we conclude

$$\boxed{\int d\theta \, 1 = 0.} \quad (2.35)$$

Now the most general integral is

$$\int d\theta \, f(\theta) = \int d\theta \, (f_0 + f_1 \theta) = f_0 \int d\theta \, 1 + f_1 \int d\theta \, \theta. \quad (2.36)$$

The first term vanishes, so

$$\int d\theta \, f(\theta) = f_1 \int d\theta \, \theta. \quad (2.37)$$

We fix the normalization by definition:

$$\boxed{\int d\theta \, \theta = 1.} \quad (2.38)$$

Therefore,

$$\boxed{\int d\theta \, f(\theta) = f_1.} \quad (2.39)$$

Integration Equals Differentiation

Comparing with

$$\frac{\partial f}{\partial \theta} = f_1,$$

we obtain the remarkable result:

$$\boxed{\int d\theta \, f(\theta) = \frac{\partial f}{\partial \theta}.} \quad (2.40)$$

Thus, for a single Grassmann variable:

Berezin integration is identical to differentiation.

This is one of the most unusual features of Grassmann calculus.

Grassmann Constants and Sign Rules

Suppose η is another Grassmann variable, independent of θ . Then when moving η past the integration measure $d\theta$, one picks up a minus sign:

$$\boxed{\int d\theta \, \eta f(\theta) = -\eta \int d\theta \, f(\theta) = -f_1 \eta.} \quad (2.41)$$

This reflects the fact that the measure $d\theta$ is itself Grassmann-odd.

Change of Variables in Grassmann Integration

Grassmann integration behaves differently from ordinary integration under a change of variables. Consider

$$\int d\theta \, f(a\theta), \quad (2.42)$$

where a is an ordinary number. Writing

$$f(\theta) = f_0 + f_1 \theta, \quad (2.43)$$

we have

$$f(a\theta) = f_0 + f_1 a\theta. \quad (2.44)$$

Therefore,

$$\int d\theta \, f(a\theta) = a f_1. \quad (2.45)$$

Now define

$$\theta' = a\theta. \quad (2.46)$$

Since θ' is a dummy integration variable,

$$\int d\theta' \, f(\theta') = f_1. \quad (2.47)$$

Comparing the two expressions gives

$$\boxed{d\theta' = \frac{1}{a}d\theta.} \quad (2.48)$$

Thus Grassmann integration transforms oppositely to ordinary integration:

$$\boxed{\theta' = a\theta \quad \Rightarrow \quad d\theta' = \frac{1}{a}d\theta.} \quad (2.49)$$

Integration over Multiple Grassmann Variables

For functions of several Grassmann variables, the order of integration matters because Grassmann variables anti-commute. In particular, the integration measures themselves also anti-commute:

$$\boxed{d\theta_1 d\theta_2 = -d\theta_2 d\theta_1.} \quad (2.50)$$

Consider a function of two Grassmann variables:

$$f(\theta_1, \theta_2) = f_0 + f_1\theta_1 + f_2\theta_2 + f_{12}\theta_1\theta_2. \quad (2.51)$$

Then

$$\int d\theta_1 d\theta_2 f(\theta_1, \theta_2) = f_{12} \int d\theta_1 d\theta_2 \theta_1\theta_2. \quad (2.52)$$

To evaluate the integral, we reorder the Grassmann variables:

$$\theta_1\theta_2 = -\theta_2\theta_1. \quad (2.53)$$

Hence

$$\int d\theta_1 d\theta_2 \theta_1\theta_2 = -1. \quad (2.54)$$

Therefore,

$$\boxed{\int d\theta_1 d\theta_2 f(\theta_1, \theta_2) = -f_{12}.} \quad (2.55)$$

More generally, for multiple Grassmann integrals one must carefully keep track of the ordering of both the integration measures and the Grassmann variables.

A useful convention to remember is

$$\boxed{\int d\theta_1 d\theta_2 \theta_2\theta_1 = 1.} \quad (2.56)$$

Grassmann Gaussian Integrals

Consider the Gaussian integral for two Grassmann variables:

$$\int d\theta_1 d\theta_2 \exp(-\theta_1 A_{12}\theta_2). \quad (2.57)$$

Since Grassmann variables are nilpotent, the exponential truncates:

$$\exp(-\theta_1 A_{12}\theta_2) = 1 - \theta_1 A_{12}\theta_2. \quad (2.58)$$

Thus,

$$\int d\theta_1 d\theta_2 \exp(-\theta_1 A_{12}\theta_2) = A_{12}. \quad (2.59)$$

More generally, for $2n$ Grassmann variables,

$$\int d\theta_1 \cdots d\theta_{2n} \exp\left(-\frac{1}{2}\theta_i A_{ij}\theta_j\right) = \sqrt{\det A}, \quad (2.60)$$

where A_{ij} is antisymmetric:

$$A_{ij} = -A_{ji}. \quad (2.61)$$

Remark This differs from the ordinary bosonic Gaussian integral, for which one obtains

$$(\det A)^{-1/2}.$$

For Grassmann variables, the determinant appears in the numerator rather than the denominator.

Complex Grassmann Variables

Starting from two real Grassmann variables θ_1, θ_2 , we may define a complex Grassmann variable by

$$\theta = \frac{1}{\sqrt{2}} (\theta_1 + i\theta_2), \quad \theta^* = \frac{1}{\sqrt{2}} (\theta_1 - i\theta_2). \quad (2.62)$$

Complex conjugation reverses the order of Grassmann variables:

$$(\theta\eta)^* = \eta^*\theta^*. \quad (2.63)$$

A general function of θ, θ^* takes the form

$$f(\theta, \theta^*) = c_0 + c_1\theta + \bar{c}_1\theta^* + c_{11}\theta\theta^*. \quad (2.64)$$

We define the integration convention

$$\boxed{\int d\theta d\theta^* \theta^*\theta = 1.} \quad (2.65)$$

Complex Grassmann Gaussian Integrals

For a single complex Grassmann variable,

$$\int d\theta^* d\theta e^{-\theta^* b \theta} = b. \quad (2.66)$$

More generally, for n complex Grassmann variables,

$$\int \prod_{j=1}^n d\theta_j^* d\theta_j \exp(-\theta_i^* A_{ij} \theta_j) = \det A. \quad (2.67)$$

This should be contrasted with the ordinary complex Gaussian integral, which gives

$$(\det A)^{-1}.$$

Gaussian Integral with Sources

Define Grassmann source variables

$$\eta = (\eta_1, \dots, \eta_n)^T. \quad (2.68)$$

Consider the Gaussian integral

$$I(\eta, \eta^*) = \int \prod_{j=1}^n d\theta_j^* d\theta_j \exp(-\theta^\dagger A \theta + \eta^\dagger \theta + \theta^\dagger \eta). \quad (2.69)$$

Completing the square gives

$$\boxed{I(\eta, \eta^*) = \det A \exp(\eta^\dagger A^{-1} \eta).} \quad (2.70)$$

Generating Functional Identity

The source-dependent Gaussian integral generates correlation functions by differentiation. For example,

$$\frac{1}{I(0,0)} \int \prod_{j=1}^n d\theta_j^* d\theta_j \theta_k \theta_l^* e^{-\theta^\dagger A \theta} = \frac{1}{I(\eta, \eta^*)} \left(\frac{\partial}{\partial \eta_k^*} \right) \left(-\frac{\partial}{\partial \eta_l} \right) I(\eta, \eta^*) \Big|_{\eta=\eta^*=0} = (A^{-1})_{kl}. \quad (2.71)$$

Thus Grassmann Gaussian integrals reproduce the inverse matrix in the same way as ordinary bosonic Gaussian integrals.

This identity will be the key ingredient in deriving the fermion propagator from the fermionic path integral.

2.2 Path Integral for Dirac Fields

We now apply Grassmann integration to construct the path integral for fermionic (Dirac) fields.

Grassmann-valued Dirac Fields

The Dirac field $\psi_\alpha(x)$ has four spinor components:

$$\alpha = 1, 2, 3, 4. \quad (2.72)$$

To correctly describe fermions in the path integral formalism, we require that

$$\boxed{\psi_\alpha(x) \text{ are Grassmann variables.}} \quad (2.73)$$

This means:

- x is an ordinary spacetime coordinate,
- for each x , $\psi_\alpha(x)$ is a Grassmann number.

In particular, they satisfy anti-commutation relations:

$$\boxed{\psi_\alpha(x)\psi_\beta(y) = -\psi_\beta(y)\psi_\alpha(x).} \quad (2.74)$$

Fermionic Path Integral

The fermionic path integral is defined in complete analogy with the bosonic case:

$$\langle \Omega | T \mathcal{O} | \Omega \rangle = \frac{\int \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{O}[\psi, \bar{\psi}] e^{iS[\psi, \bar{\psi}]}}{\int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{iS[\psi, \bar{\psi}]}}, \quad (2.75)$$

The only difference from the scalar field case is:

The integration variables are now Grassmann-valued fields.

Measure The measure is defined as

$$\mathcal{D}\psi = \prod_{\alpha=1}^4 \prod_x d\psi_\alpha(x), \quad (2.76)$$

and similarly for $\mathcal{D}\bar{\psi}$.

Generating Functional

To compute correlation functions, we introduce Grassmann sources $\eta, \bar{\eta}$:

$$Z[\eta, \bar{\eta}] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[iS[\psi, \bar{\psi}] + i \int d^4x (\bar{\eta}\psi + \bar{\psi}\eta) \right]. \quad (2.77)$$

Functional derivatives with respect to $\eta, \bar{\eta}$ generate insertions of $\psi, \bar{\psi}$.

Free Dirac Theory

For the free Dirac field, the action is quadratic:

$$S_0 = -i \int d^4x \bar{\psi}(x) (\gamma^\mu \partial_\mu - m) \psi(x). \quad (2.78)$$

It is convenient to rewrite this in matrix form:

$$S_0 = -i \int d^4x d^4y \bar{\psi}_\alpha(x) A_{\alpha\beta}(x-y) \psi_\beta(y), \quad (2.79)$$

where

$$A_{\alpha\beta}(x-y) = -\delta^{(4)}(x-y) (\not{\partial}_y - m)_{\alpha\beta}. \quad (2.80)$$

Thus the path integral is Gaussian in Grassmann variables.

Evaluation of the Gaussian Integral

Using the Grassmann Gaussian formula, we obtain:

$$Z^{(0)}[0, 0] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{iS_0} = \det A. \quad (2.81)$$

Here the determinant is taken over:

- spinor indices,
- spacetime (functional) indices.

Generating Functional Result

The sources $\eta, \bar{\eta}$ are Grassmann-valued fields with the same structure as ψ :

$$\eta_\alpha(x), \quad \bar{\eta}_\alpha(x), \quad \alpha = 1, \dots, 4. \quad (2.82)$$

They are independent Grassmann variables. Including sources, the Gaussian integral gives:

$$Z^{(0)}[\eta, \bar{\eta}] = \det A \exp \left[i \int d^4x d^4y \bar{\eta}_\alpha(x) D_{\alpha\beta}^{(0)}(x-y) \eta_\beta(y) \right], \quad (2.83)$$

where

$$D_{\alpha\beta}^{(0)}(x-y) = A_{\alpha\beta}^{-1}(x-y). \quad (2.84)$$

Remark

- If $\eta = \bar{\eta} = 0$, then $Z^{(0)}$ is a c-number.
- If sources are present, $Z^{(0)}$ is a function of Grassmann variables.

Obtaining Correlation Functions

Correlation functions are obtained by functional derivatives:

$$\langle 0 | T \{ \psi(x_1) \cdots \bar{\psi}(y_n) \} | 0 \rangle = \frac{1}{Z^{(0)}[0, 0]} \frac{\delta}{\delta \bar{\eta}(x_1)} \cdots \frac{\delta}{\delta \eta(y_n)} Z^{(0)}[\eta, \bar{\eta}] \Big|_{\eta=\bar{\eta}=0}. \quad (2.85)$$

Important Rule

- Derivative w.r.t. $\bar{\eta}$ brings down ψ
- Derivative w.r.t. η brings down $\bar{\psi}$

This follows from the source term:

$$\bar{\eta}\psi + \bar{\psi}\eta.$$

Because the exponent is quadratic,

$$Z_0[\eta, \bar{\eta}] \sim \exp(\bar{\eta} D \eta),$$

every differentiation pairs $\bar{\eta}$ with η , producing propagators. Thus:

Correlation functions are given by sums over all pairwise contractions.

Each contraction gives:

$$\langle 0 | T \{ \psi_\alpha(x) \bar{\psi}_\beta(y) \} | 0 \rangle = D_{\alpha\beta}^{(0)}(x-y). \quad (2.86)$$

Fermionic Sign Because Grassmann variables anti-commute:

Every exchange of fermionic fields introduces a minus sign.

Example: Four-point function Consider

$$\langle 0|T\{\psi(x_1)\bar{\psi}(x_2)\bar{\psi}(x_3)\psi(x_4)\}|0\rangle. \quad (2.87)$$

There are two contractions:

- (x_1, x_2) and (x_4, x_3)
- (x_1, x_3) and (x_4, x_2)

The result is

$$\boxed{-D^{(0)}(x_1 - x_2)D^{(0)}(x_4 - x_3) + D^{(0)}(x_1 - x_3)D^{(0)}(x_4 - x_2)} \quad (2.88)$$

The minus sign comes from reordering fermionic fields to form adjacent contractions.

Interacting Theory

For interacting theories, everything proceeds as in the scalar case:

$$\frac{\langle \Omega|T\mathcal{O}|\Omega\rangle}{\langle \Omega|\Omega\rangle} = \frac{\langle 0|T(\mathcal{O}e^{iS_{\text{int}}})|0\rangle}{\langle 0|e^{iS_{\text{int}}}|0\rangle}. \quad (2.89)$$

Key Points

- Same perturbation theory structure as scalar fields
- Vacuum bubble diagrams cancel
- Only difference: fermionic minus signs

2.3 Feynman Rules for Fermions: Differences from Scalar Fields

We now summarize the main differences between fermionic and scalar Feynman rules.

Propagator and Arrow Structure

For fermions, the propagator carries spinor indices and is represented by a line with an arrow indicating **charge flow**.

- Arrow flows from $\bar{\psi}$ to ψ
- This is a **charge arrow**, not necessarily the momentum direction

In coordinate space:

$$\begin{array}{c} \alpha \\ x_1 \end{array} \xleftarrow{\hspace{1.5cm}} \begin{array}{c} \beta \\ x_2 \end{array} = D_{\alpha\beta}^{(0)}(x_1 - x_2) = \langle 0|T\psi_\alpha(x_1)\bar{\psi}_\beta(x_2)|0\rangle \quad (2.90)$$

In momentum space:

$$\begin{array}{c} \alpha \\ \xleftarrow{\hspace{1.5cm}} \end{array} \begin{array}{c} \xrightarrow{k} \hspace{0.5cm} \beta \\ \xleftarrow{\hspace{1.5cm}} \end{array} = - \left(\frac{1}{i\not{k} - m + i\epsilon} \right)_{\alpha\beta} \quad (2.91)$$

Momentum vs Arrow

- Momentum direction is **arbitrary**
- Arrow direction is **fixed** (charge flow)

The arrow on a fermion line represents the flow of charge (fermion number), not necessarily the direction of momentum.

External Points

External insertions depend on whether the operator is ψ or $\bar{\psi}$.

- $\bar{\psi}(x)$: arrow **leaves** the point

$$\begin{array}{ccc} x & \xrightarrow{\quad} & \\ \alpha & & \end{array} \quad \begin{array}{ccc} & k & \\ \alpha & \xrightarrow{\quad} & \end{array} \quad (\text{represents } \bar{\psi}(x)) \quad (2.92)$$

- $\psi(x)$: arrow **enters** the point

$$\begin{array}{ccc} x & \xleftarrow{\quad} & \\ \alpha & & \end{array} \quad \begin{array}{ccc} & k & \\ \alpha & \xleftarrow{\quad} & \end{array} \quad (\text{represents } \psi(x)) \quad (2.93)$$

Spinor Structure

Unlike scalar fields:

- propagators are matrices $D_{\alpha\beta}$
- fields carry spinor indices
- indices must be contracted along fermion lines

Spinor indices are contracted following the direction of the arrows.

Fermionic Signs

Because fermions anti-commute:

Each exchange of fermion operators introduces a minus sign.

(2.94)

This affects:

- Wick contractions
- multi-fermion diagrams

2.4 Scattering Amplitudes and LSZ rules

We now turn to scattering amplitudes. Unlike Green functions, scattering amplitudes describe transitions between asymptotic particle states:

$$\langle k_1, s_1; k_2, \bar{s}_2 \cdots | p_1, r_1; p_2, \bar{r}_2, \cdots \rangle. \quad (2.95)$$

Here, we specify the initial and final momenta and polarizations

- p_i, k_i are momenta
- s_i and r_i are spin (polarization) indices
- $\bar{s}_i$ and $\bar{r}_i$ denote an antiparticle

Key idea (LSZ):

Scattering amplitudes are obtained from Green functions by removing external propagators and attaching appropriate spinor wavefunctions.

Therefore, we need to assign polarization vectors for each initial and final particles.

External Line Rules

Each external fermion line is replaced by a spinor:

Initial states:

- particle:

$$r_1 \xrightarrow{p_1} u_{r_1}(p_1) \quad (2.96)$$

- antiparticle:

$$\bar{r}_2 \xleftarrow{p_2} \bar{v}_{r_2}(p_2) \quad (2.97)$$

Final states:

- particle:

$$s_1 \xleftarrow{k_1} \bar{u}_{s_1}(k_1) \quad (2.98)$$

- antiparticle:

$$\bar{s}_2 \xleftarrow{k_2} v_{s_2}(k_2) \quad (2.99)$$

Origin of external spinors In canonical quantization, a one-particle state is created by

$$|p_1, r_1\rangle \propto \left(a_{\vec{p}_1}^{(r_1)} \right)^\dagger |0\rangle.$$

To understand how this arises from the field, recall the mode expansion of the Dirac field:

$$\psi(x) = \sum_r \int \frac{d^3p}{(2\pi)^3} \left[a_{p,r} u^r(p) e^{ip \cdot x} + b_{p,r}^\dagger v^r(p) e^{-ip \cdot x} \right], \quad (2.100)$$

$$\bar{\psi}(x) = \sum_r \int \frac{d^3p}{(2\pi)^3} \left[a_{p,r}^\dagger \bar{u}^r(p) e^{-ip \cdot x} + b_{p,r} \bar{v}^r(p) e^{ip \cdot x} \right]. \quad (2.101)$$

The key observation is that the creation operator $a_{p,r}^\dagger$ appears in $\bar{\psi}(x)$, not in $\psi(x)$. Therefore, a one-particle state can be extracted from $\bar{\psi}(x)$ by projecting onto the corresponding momentum and spin:

$$|p_1, r_1\rangle \propto \int d^3x e^{-ip_1 \cdot x} \bar{\psi}(x) \gamma^0 u_{r_1}(\vec{p}_1) |0\rangle.$$

This shows that, when relating Green functions to particle states, $\bar{\psi}$ must be contracted with the spinor $u_r(\vec{p})$ in order to isolate the creation operator.

As a result, in the LSZ reduction formula, external fermion lines are replaced by spinors:

- $u_r(p)$ for incoming particles,
- $\bar{u}_r(p)$ for outgoing particles,
- $v_r(p), \bar{v}_r(p)$ for antiparticles.

Rule for Spinor Contractions

Follow each fermion line **against the arrow direction**, starting from $\bar{u}$ or $\bar{v}$, and multiply matrices along the path.

This ensures:

- row spinor $(\bar{u}, \bar{v})$ appears first
- column spinor (u, v) appears last
- result is a scalar

2.5 Example: Yukawa Theory and Fermion Scattering (1930s)

A historically important example is **Yukawa theory**, proposed by **Hideki Yukawa** in the 1930s to explain nuclear forces.

Lagrangian We consider a scalar field ϕ and a Dirac field ψ :

$$\mathcal{L} = \bar{\psi}(\not{\partial} - m)\psi + \frac{1}{2}(\partial_\mu \phi)^2 + \frac{1}{2}m_\phi^2 \phi^2 - g \phi \bar{\psi}\psi. \quad (2.102)$$

This interaction is Lorentz invariant.

Feynman rules:

- scalar propagator:

$$\text{---}\overset{k}{\longrightarrow}\text{---} \quad \phi = \frac{-i}{k^2 + m_\phi^2 - i\epsilon}$$

- fermion propagator:

$$\text{---}\overset{k}{\longleftarrow}\text{---} \quad \psi = \frac{1}{i\not{k} - m + i\epsilon}$$

- vertex:

$$\begin{array}{c} \text{---} \\ | \\ \swarrow \quad \searrow \\ \text{---} \quad \text{---} \end{array} = -ig$$

Scattering Process

We consider:

$$\psi(p_1, s_1) + \psi(p_2, s_2) \rightarrow \psi(p'_1, s'_1) + \psi(p'_2, s'_2). \quad (2.103)$$

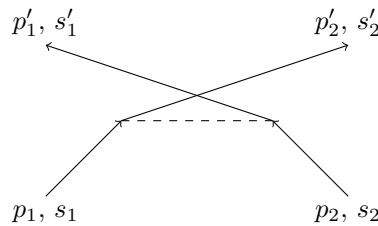
Tree-Level Diagrams

There are two diagrams:

(1) Direct channel



(2) Exchange channel



Scattering Amplitude

Direct diagram

$$\mathcal{M}_1 = (-ig)^2 \bar{u}_{s'_1}(p'_1) u_{s_1}(p_1) \frac{-i}{(p'_1 - p_1)^2 + m_\phi^2 - i\epsilon} \bar{u}_{s'_2}(p'_2) u_{s_2}(p_2). \quad (2.104)$$

Exchange diagram

$$\mathcal{M}_2 = (-ig)^2 \bar{u}_{s'_1}(p'_1) u_{s_2}(p_2) \frac{-i}{(p'_2 - p_1)^2 + m_\phi^2 - i\epsilon} \bar{u}_{s'_2}(p'_2) u_{s_1}(p_1). \quad (2.105)$$

Fermionic Minus Sign

The total amplitude is:

$$\boxed{\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2} \quad (2.106)$$

Origin of the minus sign

- The two diagrams differ by exchanging identical fermions
- Fermionic operators anti-commute
- Therefore, exchanging them introduces a minus sign

Key Rule for Spinor Contractions

Start from $\bar{u}$ or $\bar{v}$ and follow the fermion line **backwards along the arrow**.

This ensures:

- correct ordering of matrices
- proper contraction of spinor indices

Physical Remark

Yukawa used this interaction to explain nuclear forces by postulating a scalar particle exchange. From the strength of the interaction, he predicted the mass of the exchanged particle, later identified as the pion.

3 The Quantization of Maxwell Field

Having completed the discussion of fermions, we now turn to the final ingredient needed for quantum electrodynamics (QED): the quantization of the Maxwell field, i.e. the photon. We begin by reviewing the classical Maxwell theory.

3.1 Classical Theory of Maxwell Field

Lagrangian The Lagrangian density is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A_\mu j^\mu, \quad (3.1)$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (3.2)$$

and j^μ is the electromagnetic current.

Equations of motion Varying the action gives Maxwell's equations:

$$\partial_\mu F^{\mu\nu} = -j^\nu. \quad (3.3)$$

Taking ∂_ν on both sides:

$$\partial_\nu \partial_\mu F^{\mu\nu} = -\partial_\nu j^\nu = 0, \quad (3.4)$$

since $F^{\mu\nu}$ is antisymmetric. Therefore,

$$\boxed{\partial_\mu j^\mu = 0}, \quad (3.5)$$

i.e. the current must be conserved.

Consistency condition This shows that current conservation is required for the consistency of Maxwell's equations.

Gauge Symmetry

The Maxwell theory is invariant under the transformation:

$$A_\mu(x) \rightarrow A'_\mu(x) = A_\mu(x) + \partial_\mu \lambda(x), \quad (3.6)$$

where $\lambda(x)$ is an arbitrary function of spacetime.

Invariance of the theory

- The field strength is invariant:

$$F_{\mu\nu} \rightarrow F_{\mu\nu}, \quad (3.7)$$

since derivatives cancel.

- The equation of motion is invariant because it depends only on $F_{\mu\nu}$.
- The action is invariant:

$$A_\mu j^\mu \rightarrow (A_\mu + \partial_\mu \lambda) j^\mu. \quad (3.8)$$

The extra term becomes

$$\int d^4x (\partial_\mu \lambda) j^\mu = - \int d^4x \lambda \partial_\mu j^\mu, \quad (3.9)$$

which vanishes using current conservation.

Local vs global symmetry

- Global symmetry:
 - transformation parameter is constant
 - corresponds to a physical symmetry
- Gauge (local) symmetry:
 - transformation parameter $\lambda(x)$ depends on spacetime
 - does **not** correspond to a physical symmetry

Physical meaning of gauge symmetry Gauge symmetry implies that the theory contains **redundant degrees of freedom**. Different configurations of A_μ related by

$$A_\mu \sim A_\mu + \partial_\mu \lambda$$

describe the **same physical state**.

Gauge orbits (geometric picture) Consider the space of all configurations of A_μ .

- Starting from a given A_μ , applying all possible $\lambda(x)$ generates a set of configurations.
- This set is called a **gauge orbit**.
- All points on the same orbit represent the same physical state.

Thus, the true physical configuration space is not the full space of A_μ , but the space of gauge orbits.

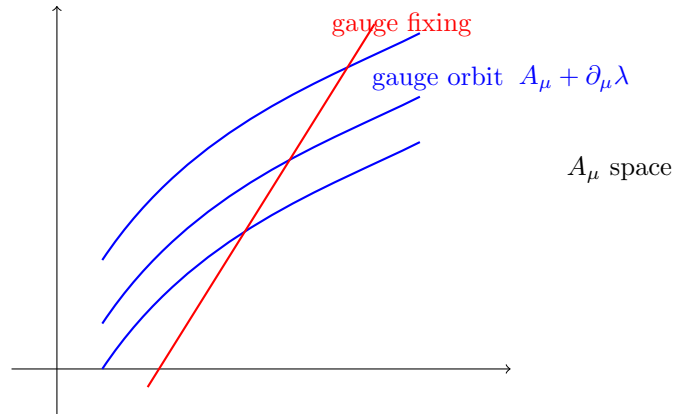


Figure 1: Gauge orbits and gauge fixing. Points along each blue curve are related by gauge transformations and represent the same physical state. The red line selects one representative from each orbit. Physics only depends on the intersections.

Remarks:

Gauge symmetry is not a symmetry between different physical states, but a redundancy in our description of the same physical state.

3.2 Coulomb Gauge

To understand the physical content of Maxwell theory, we count its dynamical degrees of freedom. We often solve Maxwell equation by fixing a gauge.

Decomposition of the gauge field We write

$$A^\mu = (\phi, \vec{A}), \quad j^\mu = (\rho, \vec{j}). \quad (3.10)$$

Coulomb gauge The Coulomb gauge is defined by

$$\nabla \cdot \vec{A} = 0. \quad (3.11)$$

Equations of motion In this gauge, Maxwell equations become:

- Time component:

$$\nabla^2 \phi = -\rho \quad (3.12)$$

- Spatial component:

$$\partial_\mu \partial^\mu \vec{A} = \vec{j} - \nabla(\partial_t \phi) \quad (3.13)$$

Non-dynamical field The equation for ϕ contains no time derivatives:

$$\nabla^2 \phi = -\rho. \quad (3.14)$$

Thus:

- ϕ is not dynamical
- it is determined instantaneously by the charge density ρ

Momentum space interpretation In momentum space:

$$\nabla \cdot \vec{A} = 0 \quad \implies \quad \vec{k} \cdot \vec{A} = 0. \quad (3.15)$$

This means:

- $\vec{A}$ has no component along $\vec{k}$
- only transverse components remain

We decompose:

$$A_i = \vec{A}_i^L + \vec{A}_i^T, \quad (3.16)$$

where:

$$A_i^L \parallel k_i, \quad (3.17)$$

$$A_i^T \cdot k_i = 0. \quad (3.18)$$

Coulomb gauge implies:

$$A_i^L = 0, \quad \vec{A} = \vec{A}_T. \quad (3.19)$$

Physical Degrees of Freedom We now explain why the Maxwell field has only two physical degrees of freedom.

- The gauge field has four components:

$$A_\mu = (\phi, \vec{A}) \Rightarrow 4 \text{ components.} \quad (3.20)$$

- In the Coulomb gauge, the scalar potential $\phi = A_0$ is not dynamical. Its equation of motion contains no time derivatives, so it does not describe an independent propagating degree of freedom. In vacuum, it can be set to zero:

$$\phi = 0. \quad (3.21)$$

Thus only the spatial components $\vec{A}$ remain:

$$\vec{A} = (A_1, A_2, A_3). \quad (3.22)$$

- The Coulomb gauge condition imposes

$$\nabla \cdot \vec{A} = 0. \quad (3.23)$$

In momentum space, this becomes

$$\vec{k} \cdot \vec{A} = 0. \quad (3.24)$$

This constraint removes one component of $\vec{A}$: the component parallel to the momentum $\vec{k}$ (the longitudinal mode).

- Therefore, out of the three spatial components, only two independent components remain:

$$3 - 1 = 2. \quad (3.25)$$

Maxwell theory has 2 physical degrees of freedom

(3.26)

These correspond to the two transverse polarizations of the electromagnetic wave, i.e. the two independent directions perpendicular to the direction of propagation.

Limitations of Coulomb Gauge Although the Coulomb gauge is intuitive, it has important drawbacks:

- **Not Lorentz covariant:**

$$\nabla \cdot \vec{A} = 0$$

is not invariant under Lorentz transformations.

- **Instantaneous interaction:**

$$\nabla^2 \phi = -\rho$$

contains no time derivative.

- ϕ is determined instantaneously by ρ
- suggests “action at a distance”

- **Causality not manifest:** This instantaneous behavior is an artifact of gauge choice.

Important remark Although the Coulomb gauge is not manifestly Lorentz covariant and suggests instantaneous interactions, this does not imply any violation of causality or relativity. The scalar potential ϕ is not a physical observable and its instantaneous behavior is a gauge artifact. Physical quantities are gauge-invariant combinations such as

$$F_{\mu\nu}, \quad \vec{E}, \quad \vec{B}, \quad (3.27)$$

which transform consistently under Lorentz transformations and obey causal propagation.

Thus, while Lorentz covariance is not manifest in the Coulomb gauge, the underlying physics remains fully Lorentz invariant.

3.3 Lorenz Gauge

To maintain Lorentz covariance, we introduce the Lorenz gauge:

$$\partial_\mu A^\mu = 0. \quad (3.28)$$

This condition is Lorentz invariant.

Equations of motion In Lorenz gauge, Maxwell equations simplify to:

$$\partial_\mu \partial^\mu A^\nu = j^\nu. \quad (3.29)$$

Interpretation This equation resembles four decoupled massless scalar fields:

$$\square \phi = \text{source}. \quad (3.30)$$

Thus, each component A^μ behaves like a scalar field with a source. However, they are not independent due to the gauge condition.

Residual Gauge Symmetry

The Lorenz gauge does not completely fix the gauge. Consider:

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda. \quad (3.31)$$

Then:

$$\partial_\mu A^\mu \rightarrow \partial_\mu A^\mu + \partial_\mu \partial^\mu \lambda. \quad (3.32)$$

Thus, the Lorenz gauge is preserved if:

$$\partial_\mu \partial^\mu \lambda = 0. \quad (3.33)$$

Key observation

- Before gauge fixing:
 - $\lambda(x)$ is arbitrary (off-shell)
- After Lorenz gauge:
 - $\lambda(x)$ must satisfy $\partial_\mu \partial^\mu \lambda = 0$
 - residual gauge freedom remains

Lorenz gauge fixes the gauge only partially

(3.34)

Gauge fixing reduces redundancy but may not eliminate it completely.

Gauge Fixing at the Action Level Instead of imposing the Lorenz gauge condition only after deriving the equations of motion, one may fix the gauge directly at the level of the Lagrangian by adding a gauge-fixing term. In the convention used here, we take

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - A_\mu j^\mu - \frac{\xi}{2} (\partial_\mu A^\mu)^2, \quad (3.35)$$

where ξ is a constant gauge parameter. This procedure modifies the action and therefore breaks the full gauge redundancy

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda \quad (3.36)$$

for arbitrary λ . The point of doing so is that the resulting equations of motion are non-degenerate and admit a well-defined propagator, which is especially important for covariant quantization.

Varying the action with respect to A_μ gives

$$\partial_\nu F^{\nu\mu} + \xi \partial^\mu (\partial_\nu A^\nu) = j^\mu. \quad (3.37)$$

Using

$$\partial_\nu F^{\nu\mu} = \partial^2 A^\mu - \partial^\mu \partial_\nu A^\nu, \quad (3.38)$$

we obtain the modified field equation

$$\partial^2 A^\mu + (\xi - 1) \partial^\mu \partial_\nu A^\nu = j^\mu. \quad (3.39)$$

Now take the divergence of this equation:

$$\partial_\mu \partial^2 A^\mu + (\xi - 1) \partial_\mu \partial^\mu (\partial_\nu A^\nu) = \partial_\mu j^\mu. \quad (3.40)$$

Since $\partial_\mu \partial^2 A^\mu = \partial^2 (\partial_\mu A^\mu)$, this becomes

$$\xi \partial^2 (\partial_\mu A^\mu) = \partial_\mu j^\mu. \quad (3.41)$$

For a conserved current,

$$\partial_\mu j^\mu = 0, \quad (3.42)$$

and therefore

$$\partial^2 (\partial_\mu A^\mu) = 0. \quad (3.43)$$

Thus, the divergence $\partial_\mu A^\mu$ satisfies the homogeneous wave equation. In general this equation admits nontrivial solutions, so the gauge-fixing term does *not* force $\partial_\mu A^\mu = 0$ identically by itself. However, if one chooses initial or boundary conditions such that

$$\partial_\mu A^\mu = 0 \quad \text{and} \quad \partial_0 (\partial_\mu A^\mu) = 0 \quad (3.44)$$

on an initial time slice, then the unique solution is

$$\partial_\mu A^\mu = 0 \quad (3.45)$$

for all time. In this sense, the Lorenz condition is preserved dynamically once it is imposed consistently on the initial data.

Gauge fixing at the action level means modifying the Lagrangian so that the equations of motion are well-defined and compatible with the Lorenz condition; with suitable initial or boundary conditions, the condition $\partial_\mu A^\mu = 0$ is then preserved by the dynamics.

Why this approach is useful

- Classically, it gives equations of motion compatible with the Lorenz condition.
- Quantum mechanically, it is essential in covariant quantization and in the path integral formulation.
- It renders the kinetic operator invertible and hence gives a well-defined propagator.

For the special choice $\xi = 1$, the field equation simplifies to

$$\partial^2 A^\mu = j^\mu. \quad (3.46)$$

In this case the Lagrangian can be written, up to a total derivative, as

$$\mathcal{L} = -\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu - j_\mu A^\mu. \quad (3.47)$$

This is the particularly convenient covariant gauge in the present convention.

Interpretation For the special choice $\xi = 1$, the gauge-fixed equations of motion take the simple form

$$\partial^2 A^\mu = j^\mu. \quad (3.48)$$

At first sight, this resembles the equation for four independent massless scalar fields, one for each component of A^μ . However, this interpretation is not correct: the electromagnetic field still describes only two physical propagating degrees of freedom.

The reason is that the four components of A^μ are not all independent. First, they are constrained by the Lorenz condition

$$\partial_\mu A^\mu = 0. \quad (3.49)$$

This removes one degree of freedom. Second, even after imposing the Lorenz condition, there remains a residual gauge freedom

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda, \quad \partial^2 \lambda = 0, \quad (3.50)$$

which preserves the Lorenz gauge. This residual gauge symmetry removes one more degree of freedom. Thus the number of physical propagating modes is

$$4 - 1 - 1 = 2. \quad (3.51)$$

Equivalently, for a plane-wave solution

$$A^\mu(x) = \epsilon^\mu(k) e^{-ik \cdot x}, \quad k^2 = 0, \quad (3.52)$$

the Lorenz condition implies

$$k_\mu \epsilon^\mu = 0. \quad (3.53)$$

This gives one condition on the polarization vector ϵ^μ . In addition, the residual gauge transformation

$$\epsilon^\mu \rightarrow \epsilon^\mu + i k^\mu \alpha \quad (3.54)$$

(with $\lambda = \alpha e^{-ik \cdot x}$ and $k^2 = 0$) allows one further component of ϵ^μ to be removed. Therefore only two independent transverse polarizations remain, corresponding to the two physical photon helicity states. Hence, although the equation $\partial^2 A^\mu = j^\mu$ looks like four decoupled wave equations, the Lorenz condition together with residual gauge invariance ensures that the theory still contains only the two physical degrees of freedom of the electromagnetic field.

3.4 Quantization of the Maxwell Field

We now quantize the free Maxwell field, setting

$$j^\mu = 0. \quad (3.55)$$

The Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \quad (3.56)$$

Subtlety: constrained structure

Canonical quantization is not completely straightforward because not all components of A_μ are dynamical. In particular, the Lagrangian contains no time derivative of A_0 , so the canonical momentum conjugate to A_0 vanishes:

$$\pi^0 \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 A_0)} = 0. \quad (3.57)$$

This is a primary constraint rather than an ordinary equation of motion.

For the spatial components, the canonical momenta are

$$\pi^i \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 A_i)} = -F^{0i} = E^i. \quad (3.58)$$

Thus the momentum conjugate to A_i is the electric field.

Because the theory has constraints, a fully systematic treatment requires the formalism for constrained systems. Instead of developing that machinery here, we proceed by fixing the gauge and quantizing the resulting unconstrained theory. This allows us to isolate the physical degrees of freedom more directly.

3.4.1 Quantization in Coulomb gauge

We begin with quantization in Coulomb gauge, which is conceptually simple. In the absence of sources, Maxwell's equations reduce to

$$\nabla^2 \phi = 0, \quad (3.59)$$

$$\partial_\mu \partial^\mu A_i = -\partial_i(\partial_t \phi), \quad (3.60)$$

together with the gauge condition

$$\nabla \cdot \vec{A} = 0. \quad (3.61)$$

If we impose the boundary condition that $\phi \rightarrow 0$ at spatial infinity, then the Laplace equation

$$\nabla^2 \phi = 0 \quad (3.62)$$

admits only the trivial solution,

$$\phi = 0. \quad (3.63)$$

Thus the scalar potential is fixed completely and carries no independent dynamical degrees of freedom.

With $\phi = 0$, the equation for the spatial components becomes

$$\partial_\mu \partial^\mu A_i = 0. \quad (3.64)$$

Therefore each component A_i satisfies the free massless wave equation. However, the three components are not independent, since they must also satisfy the Coulomb gauge constraint

$$\partial_i A_i = \nabla \cdot \vec{A} = 0. \quad (3.65)$$

This transversality condition removes the longitudinal component of $\vec{A}$, leaving only the two transverse physical polarization states of the electromagnetic field.

Transverse operator field We now promote the spatial vector potential A_i to an operator field. In Coulomb gauge, the condition

$$\partial_i A_i = 0 \quad (3.66)$$

must hold, so only the transverse part of A_i is dynamical. Therefore the quantized photon field contains only the two physical transverse polarization states. At first sight, one might try to impose the usual equal-time canonical commutation relation

$$[A_i(t, \vec{x}), \pi_j(t, \vec{x}')] = i \delta_i^j \delta^{(3)}(\vec{x} - \vec{x}'). \quad (3.67)$$

Equivalently, if one identifies the canonical momentum with the electric field (up to convention-dependent signs), one might write

$$[A_i(t, \vec{x}), E_j(t, \vec{x}')] = i \delta_{ij} \delta^{(3)}(\vec{x} - \vec{x}'). \quad (3.68)$$

However, this cannot be correct in Coulomb gauge, because A_i is constrained to be transverse:

$$\partial_i A_i = 0. \quad (3.69)$$

Indeed, if we apply ∂_i to the naive commutator, we find

$$[\partial_i A_i(t, \vec{x}), E_j(t, \vec{x}')] = i \partial_j \delta^{(3)}(\vec{x} - \vec{x}'), \quad (3.70)$$

which is not zero. But since $\partial_i A_i = 0$ identically, the left-hand side must vanish. Hence the naive canonical commutation relation is incompatible with the Coulomb gauge constraint.

Origin of the problem The problem is that the naive commutator treats the three spatial components of A_i as independent. In Coulomb gauge, however, the longitudinal component is not an independent degree of freedom. Only the transverse part should be quantized.

Transverse fields We therefore decompose the fields into transverse and longitudinal parts, and keep only the transverse components:

$$A_i \rightarrow A_i^T, \quad \pi_i \rightarrow \pi_i^T, \quad (3.71)$$

with

$$\partial_i A_i^T = 0, \quad \partial_i \pi_i^T = 0. \quad (3.72)$$

In the source-free theory and in Coulomb gauge, we have $A_0 = \phi = 0$, so

$$E_i = -\partial_0 A_i \quad (3.73)$$

(up to sign conventions for E_i and π_i). Thus the canonical momentum is proportional to the time derivative of the transverse field.

Correct commutation relation The equal-time commutation relation must be imposed only on the transverse components:

$$[A_i^T(t, \vec{x}), \pi_j^T(t, \vec{x}')] = i \delta_{ij}^T \delta^{(3)}(\vec{x} - \vec{x}'), \quad (3.74)$$

where δ_{ij}^T is the transverse delta function. Equivalently,

$$\delta_{ij}^T(\vec{x} - \vec{x}') = P_{ij}^T \delta^{(3)}(\vec{x} - \vec{x}'), \quad (3.75)$$

with P_{ij}^T the transverse projection operator. Thus one may write

$$[A_i^T(t, \vec{x}), \pi_j^T(t, \vec{x}')] = i P_{ij}^T \delta^{(3)}(\vec{x} - \vec{x}'). \quad (3.76)$$

Transverse projector The transverse projector extracts the divergence-free part of a vector field:

$$A_i^T = P_{ij}^T A_j. \quad (3.77)$$

In coordinate space, it is formally given by

$$P_{ij}^T = \delta_{ij} - \frac{\partial_i \partial_j}{\nabla^2}. \quad (3.78)$$

This expression is nonlocal because of the inverse Laplacian, so it is often clearer in momentum space, where

$$P_{ij}^T(\vec{k}) = \delta_{ij} - \frac{k_i k_j}{\vec{k}^2}. \quad (3.79)$$

It is easy to verify that this indeed projects orthogonally to $\vec{k}$:

$$k_i P_{ij}^T(\vec{k}) = 0. \quad (3.80)$$

Physical interpretation The operator P_{ij}^T removes the longitudinal component and retains only the transverse part of the field. Therefore the commutation relations are imposed only on the physical degrees of freedom. In this way, Coulomb gauge quantization describes directly the two physical photon polarizations.

Mode expansion

Since A_i^T satisfies the free wave equation

$$\partial_\mu \partial^\mu A_i^T = 0, \quad (3.81)$$

we can expand it in plane waves. Because A_i^T has two independent components, we introduce polarization vectors:

$$\epsilon_i^{(r)}(\vec{k}), \quad r = 1, 2. \quad (3.82)$$

which are a basis of transverse vectors. They satisfy the orthogonal relation:

$$\vec{k} \cdot \vec{\epsilon}^{(r)}(\vec{k}) = 0, \quad (3.83)$$

$$\vec{\epsilon}^{(r)}(\vec{k}) \cdot \vec{\epsilon}^{(s)}(\vec{k}) = \delta^{rs}. \quad (3.84)$$

They also satisfy the completeness relation:

$$\sum_{r=1}^2 \epsilon_i^{(r)}(\vec{k}) \epsilon_j^{(r)}(\vec{k}) = \delta_{ij} - \frac{k_i k_j}{\vec{k}^2} = P_{ij}^T(\vec{k}). \quad (3.85)$$

Field expansion We can now write the general solution:

$$A_i^T(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \sum_{r=1}^2 \epsilon_i^{(r)}(\vec{k}) \left[a_{\vec{k}}^{(r)} e^{ik \cdot x} + \left(a_{\vec{k}}^{(r)} \right)^\dagger e^{-ik \cdot x} \right], \quad (3.86)$$

with

$$\omega_k = |\vec{k}|. \quad (3.87)$$

Creation and annihilation operators The only difference from the scalar case is that now we have two independent polarization vectors. Substituting this expansion into the canonical commutation relations, we find that the creation and annihilation operators satisfy

$$[a_{\vec{k}}^{(r)}, (a_{\vec{k}'}^{(s)})^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}') \delta^{rs}, \quad (3.88)$$

and all other commutators vanish. Here $r, s = 1, 2$ label the two transverse polarizations.

Hilbert space and photon states We construct the Hilbert space in the usual way. The vacuum is defined by

$$a_{\vec{k}}^{(r)} |0\rangle = 0, \quad \forall \vec{k}, r = 1, 2. \quad (3.89)$$

Single-particle states are created by

$$(a_{\vec{k}}^{(r)})^\dagger |0\rangle = |\vec{k}, \vec{\epsilon}^{(r)}\rangle. \quad (3.90)$$

These describe one-photon states with momentum $\vec{k}$ and polarization vector $\vec{\epsilon}^{(r)}$.

Summary of Coulomb gauge quantization The Coulomb gauge is conceptually simple:

- The scalar potential ϕ is non-dynamical and can be set to zero.
- The longitudinal component is removed by imposing

$$\partial_i A_i = 0$$

as an operator equation.

- Only the transverse components remain and are quantized.
- The resulting theory is equivalent to two free massless scalar fields.

The drawback is that Lorentz invariance is not manifest.

3.4.2 Quantization in Lorenz Gauge

We now turn to the Lorenz gauge, which preserves manifest Lorentz covariance but requires more care in the quantization procedure. Rather than solving the constraint explicitly, we begin from the gauge-fixed Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{\xi}{2}(\partial_\mu A^\mu)^2. \quad (3.91)$$

For the special choice $\xi = 1$, this simplifies to

$$\mathcal{L} = -\frac{1}{2}\partial_\mu A_\nu \partial^\mu A^\nu. \quad (3.92)$$

In this form, the theory looks like four decoupled massless scalar fields, one for each Lorentz component A^μ . The equation of motion is simply

$$\partial_\mu \partial^\mu A^\nu = 0. \quad (3.93)$$

Taking ∂_ν of this equation gives

$$\partial_\mu \partial^\mu (\partial_\nu A^\nu) = 0. \quad (3.94)$$

Thus $\partial_\mu A^\mu$ itself satisfies the massless wave equation. If we impose boundary conditions such that $\partial_\mu A^\mu = 0$ initially, then the equations of motion preserve the Lorenz gauge condition

$$\partial_\mu A^\mu = 0. \quad (3.95)$$

Canonical quantization Since the Lagrangian has the form of four free massless scalars, we may proceed formally as in the scalar case. The canonical momentum conjugate to A_ν is

$$\pi^\nu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\nu)} = \partial_0 A^\nu. \quad (3.96)$$

The equal-time canonical commutation relation is therefore

$$[A_\mu(t, \vec{x}), \pi^\nu(t, \vec{x}')] = i \delta_\mu^\nu \delta^{(3)}(\vec{x} - \vec{x}'). \quad (3.97)$$

Equivalently,

$$[A_\mu(t, \vec{x}), \partial_0 A^\nu(t, \vec{x}')] = i \delta_\mu^\nu \delta^{(3)}(\vec{x} - \vec{x}'). \quad (3.98)$$

Mode expansion As in the scalar field case, we expand the operator field in plane-wave modes. Since A^μ is a Lorentz vector, it is convenient to introduce a basis of polarization vectors $\epsilon_\mu^{(\alpha)}(\vec{k})$, where

$$\alpha = 0, 1, 2, 3. \quad (3.99)$$

These four polarization vectors form a complete basis for Lorentz vectors at fixed momentum $\vec{k}$. We then write

$$A_\mu(x) = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \sum_{\alpha=0}^3 \left[\epsilon_\mu^{(\alpha)}(\vec{k}) a_{\vec{k}}^{(\alpha)} e^{ik \cdot x} + \left(\epsilon_\mu^{(\alpha)}(\vec{k}) \right)^* \left(a_{\vec{k}}^{(\alpha)} \right)^\dagger e^{-ik \cdot x} \right], \quad (3.100)$$

where

$$k^0 = \omega_{\vec{k}} = |\vec{k}|. \quad (3.101)$$

Choice of polarization vectors A convenient choice is as follows:

- $\epsilon_\mu^{(0)}$: timelike polarization, taken along the time direction;
- $\epsilon_\mu^{(3)}$: longitudinal polarization, taken parallel to the spatial momentum $\vec{k}$;
- $\epsilon_\mu^{(1)}$ and $\epsilon_\mu^{(2)}$: transverse polarizations, orthogonal to $\vec{k}$ and to each other.

These polarization vectors are chosen to satisfy the orthonormality relation

$$\epsilon_\mu^{(\alpha)}(\vec{k}) \epsilon_\nu^{(\beta)}(\vec{k}) \eta^{\mu\nu} = \delta^{\alpha\beta}, \quad (3.102)$$

and the completeness relation

$$\sum_{\alpha, \beta=0}^3 \epsilon_\mu^{(\alpha)}(\vec{k}) \epsilon_\nu^{(\beta)}(\vec{k}) \eta_{\alpha\beta} = \eta_{\mu\nu}. \quad (3.103)$$

Thus they form a complete orthonormal basis of polarization states. For example, if $\vec{k}$ points in the z -direction, one may choose

$$\epsilon_{(0)}^\mu = (1, 0, 0, 0), \quad (3.104)$$

$$\epsilon_{(1)}^\mu = (0, 1, 0, 0), \quad (3.105)$$

$$\epsilon_{(2)}^\mu = (0, 0, 1, 0), \quad (3.106)$$

$$\epsilon_{(3)}^\mu = (0, 0, 0, 1). \quad (3.107)$$

More generally, $\epsilon_{(3)}^\mu$ is aligned with $\vec{k}$, while $\epsilon_{(1,2)}^\mu$ span the plane transverse to $\vec{k}$.

Oscillator commutation relations Substituting the mode expansion into the canonical commutation relations, one finds

$$[a_{\vec{k}}^{(\alpha)}, (a_{\vec{k}'}^{(\beta)})^\dagger] = \eta^{\alpha\beta} (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'), \quad (3.108)$$

with all other commutators vanishing:

$$[a_{\vec{k}}^{(\alpha)}, a_{\vec{k}'}^{(\beta)}] = 0, \quad [(a_{\vec{k}}^{(\alpha)})^\dagger, (a_{\vec{k}'}^{(\beta)})^\dagger] = 0. \quad (3.109)$$

We define the vacuum by

$$a_\alpha(\vec{k}) |0\rangle = 0 \quad \text{for all } \alpha, \vec{k}, \quad (3.110)$$

and construct the Hilbert space by acting with arbitrary creation operators:

$$\mathcal{H}_{\text{big}} = \left\{ a_{\alpha_1}^\dagger(\vec{k}_1) \cdots a_{\alpha_n}^\dagger(\vec{k}_n) |0\rangle \right\}. \quad (3.111)$$

We write $\mathcal{H}_{\text{big}}$ because this space is larger than the physical Hilbert space.

First problem: too many degrees of freedom At this stage, the theory appears to contain four massless modes, corresponding to the four possible polarizations. But this cannot be correct: Maxwell theory has only two physical propagating degrees of freedom, namely the two transverse photon polarizations. Thus the present construction clearly contains two extra, unphysical degrees of freedom.

Second problem: negative norm states There is an even more serious problem. Because the oscillator algebra involves the Minkowski metric $\eta_{\alpha\beta}$, the timelike mode has the wrong sign:

$$[a_{\vec{k}}^{(0)}, (a_{\vec{k}'}^{(0)})^\dagger] = -(2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}'). \quad (3.112)$$

The negative sign in the timelike commutator means that the one-particle state created by the timelike mode has negative norm. Indeed, for

$$|\vec{k}, \epsilon^{(0)}\rangle \propto (a_{\vec{k}}^{(0)})^\dagger |0\rangle, \quad (3.113)$$

we find

$$\langle \vec{k}', \epsilon^{(0)} | \vec{k}, \epsilon^{(0)} \rangle \propto \langle 0 | a_{\vec{k}'}^{(0)} (a_{\vec{k}}^{(0)})^\dagger | 0 \rangle = -(2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}') < 0. \quad (3.114)$$

Thus the state $|\vec{k}, \epsilon^{(0)}\rangle$ has negative norm.

This is not an accident; rather, it is another signal that the covariant quantization above contains unphysical degrees of freedom. Since physical states must have nonnegative norm, the timelike polarization cannot correspond to a genuine physical photon state.

Why this does not yet contradict Maxwell theory At this stage, however, there is no real inconsistency. What we have quantized so far is the gauge-fixed vector field with four components, not yet the physical Maxwell theory. To recover Maxwell theory, two additional steps are required:

1. impose the Lorenz condition

$$\partial_\mu A^\mu = 0, \quad (3.115)$$

in the appropriate quantum-mechanical sense;

2. remove the residual gauge freedom that still remains even after imposing the Lorenz gauge.

Only after these two steps do we obtain the physical Hilbert space. In that physical Hilbert space, the unphysical timelike and longitudinal modes are removed, and only the two transverse photon polarizations remain.

Comparison with Coulomb gauge It is useful to compare this with the Coulomb gauge. In Coulomb gauge, the condition

$$\partial_i A_i = 0 \quad (3.116)$$

is imposed already at the classical level and solved explicitly. One then quantizes only the transverse field. As a result, the transversality condition becomes an operator statement, and the quantum theory is built directly from the two physical transverse modes.

However, in Lorenz gauge, the situation is different. The equations of motion imply only that

$$\square(\partial_\mu A^\mu) = 0. \quad (3.117)$$

Therefore, to conclude that $\partial_\mu A^\mu = 0$, one must impose an additional boundary condition. Classically, this is harmless. Quantum mechanically, however, this means that the Lorenz condition should not be imposed as a strict operator identity.

Why the Lorenz condition cannot be imposed as an operator equation One might try to demand

$$\partial_\mu A^\mu = 0 \quad (3.118)$$

as an operator equation. But this is too strong and is incompatible with the canonical commutation relations. In other words, the full operator field A^μ cannot satisfy the Lorenz condition identically as an operator relation on the whole Hilbert space $\mathcal{H}_{\text{big}}$. The reason is analogous to what happened in Coulomb gauge: constraints reduce the allowed degrees of freedom, so they cannot be imposed naively together with unconstrained canonical commutators.

Condition on states rather than operators Classically, the Lorenz condition is enforced by restricting the allowed solutions through boundary conditions. In quantum theory, restrictions on allowed classical configurations are translated into restrictions on the allowed states. Therefore, in Lorenz gauge, the correct strategy is not to impose

$$\partial_\mu A^\mu = 0 \quad (3.119)$$

as an operator identity, but instead to impose a suitable condition on physical states. Schematically, one requires that physical states be annihilated by the appropriate part of $\partial_\mu A^\mu$. However, even this statement must be treated with care. It is not sufficient simply to require

$$\partial_\mu A^\mu |\psi\rangle = 0, \quad (3.120)$$

since that condition is still too strong. The correct prescription is more subtle and leads to the Gupta–Bleuler condition.

Structure of the Hilbert space The logic is therefore the following. First, we quantize covariantly and obtain the large space $\mathcal{H}_{\text{big}}$, which contains:

- positive-norm states,
- negative-norm states,
- and, after imposing the first physical-state condition, also zero-norm states.

The first step removes the negative-norm states. But this does not yet fully solve the problem, because zero-norm states remain. These null states are associated with the residual gauge freedom left over after imposing the Lorenz condition. After quotienting out these zero-norm states, one obtains the true physical Hilbert space. This final physical Hilbert space contains only two propagating degrees of freedom, exactly as expected for the photon.

Therefore, in Lorenz gauge, covariant quantization gives a large Hilbert space $\mathcal{H}_{\text{big}}$ containing four polarization states. This space includes unphysical negative-norm and zero-norm states. To recover the physical Maxwell theory, one must impose further conditions, leading eventually to a physical Hilbert space with only the two transverse photon polarizations. We now turn to the path integral formulation of Maxwell theory.

3.5 Path integral for Maxwell theory

We now discuss the path integral quantization of Maxwell theory. Since Maxwell theory is free, one might expect the path integral to be straightforward. Indeed, formally it looks like a Gaussian integral. However, the gauge redundancy introduces an important subtlety.

Generating functional before gauge fixing Let us begin with the original Maxwell action, without gauge fixing:

$$S[A] = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right). \quad (3.121)$$

We introduce an external source $J^\mu(x)$ and define the generating functional

$$Z[J] = \int \mathcal{D}A_\mu \exp \left\{ iS[A] + i \int d^4x J_\mu A^\mu \right\}. \quad (3.122)$$

Here J^μ is merely an auxiliary source used to generate correlation functions; it is not the physical electromagnetic current.

Since the action is quadratic in A_μ , this appears to be a Gaussian functional integral. By integrating by parts, we can write the quadratic form as

$$S[A] = -\frac{1}{2} \int d^4x d^4y A_\mu(x) K_0^{\mu\nu}(x, y) A_\nu(y), \quad (3.123)$$

where

$$K_0^{\mu\nu}(x, y) = (\partial^2 \eta^{\mu\nu} - \partial^\mu \partial^\nu) \delta^{(4)}(x - y). \quad (3.124)$$

Here the derivatives act on x . If $K_0^{\mu\nu}$ had a well-defined inverse, we would formally obtain

$$Z[J] \stackrel{?}{=} \mathcal{N} \exp \left[i \int d^4x d^4y J_\mu(x) (K_0^{-1})^{\mu\nu}(x, y) J_\nu(y) \right], \quad (3.125)$$

just as for an ordinary Gaussian integral.

The problem At this point, the expression looks deceptively simple. But this cannot be the whole story. In canonical quantization, gauge redundancy required special care, so the same issue must appear here as well.

The problem is that the functional integral

$$\int \mathcal{D}A_\mu \quad (3.126)$$

integrates over all gauge-equivalent configurations. If two vector potentials differ by a gauge transformation,

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda, \quad (3.127)$$

they represent the same physical configuration. Thus the path integral overcounts physically equivalent field configurations.

This conceptual problem appears mathematically because the quadratic kernel $K_0^{\mu\nu}(x, y)$ is not invertible. To see this most clearly, go to momentum space. Then

$$K_0^{\mu\nu}(k) = -k^2 \eta^{\mu\nu} + k^\mu k^\nu, \quad (3.128)$$

where the overall sign depends on conventions and is not important here. The crucial point is that k_ν is a zero eigenvector:

$$K_0^{\mu\nu}(k) k_\nu = (-k^2 \eta^{\mu\nu} + k^\mu k^\nu) k_\nu = -k^2 k^\mu + k^\mu k^2 = 0. \quad (3.129)$$

Thus $K_0^{\mu\nu}(k)$ has a zero eigenvalue, so it cannot be inverted.

Equivalently, in position space, we have $k_\nu \rightarrow \partial_\nu$, one may write

$$K_0^{\mu\nu} \partial_\nu \lambda = 0 \quad (3.130)$$

for any scalar function $\lambda(x)$. These zero modes are precisely the pure gauge directions.

Therefore,

$$\boxed{K_0^{\mu\nu} \text{ has zero modes and hence no inverse.}} \quad (3.131)$$

This is the mathematical manifestation of gauge redundancy in the path integral.

Geometric meaning Geometrically, the space of all vector potentials A_μ contains entire gauge orbits:

$$A_\mu \sim A_\mu + \partial_\mu \lambda. \quad (3.132)$$

Points lying on the same orbit represent the same physical configuration. The path integral over all A_μ therefore integrates repeatedly over the same physical state. The zero modes of $K_0^{\mu\nu}$ are tangent to these gauge orbits. This is why the quadratic form is degenerate on the full configuration space. To obtain a nondegenerate quadratic form, we must restrict the integral to a slice that intersects each gauge orbit once. In other words, we must fix the gauge. On such a gauge-fixed slice, the pure gauge directions are no longer allowed variations, so the zero modes disappear and the kernel becomes invertible.

Why this is special to gauge symmetry This issue is specific to local symmetries. A gauge symmetry is generated by an arbitrary function $\lambda(x)$, so it identifies infinitely many field configurations along continuous directions in configuration space. These directions are redundant, not physical. By contrast, a global symmetry does not introduce such local redundancy. It relates one physical configuration to another, but it does not mean that the variables contain extra unphysical directions that must be quotiented out point by point in spacetime. Hence global symmetries do not lead to this same obstruction in defining the Gaussian path integral.

Faddeev–Popov gauge fixing The naive path integral

$$Z[J] = \int \mathcal{D}A_\mu \exp \left\{ iS_{\text{Maxwell}}[A] + i \int d^4x J_\mu A^\mu \right\} \quad (3.133)$$

is ill-defined, since the Maxwell action is invariant under gauge transformations

$$A_\mu \rightarrow A_\mu + \partial_\mu \alpha, \quad (3.134)$$

and therefore integrates repeatedly over physically equivalent field configurations. To remove this overcounting, we insert the identity

$$1 = \Delta_{\text{FP}}[A] \int \mathcal{D}\alpha \delta[\partial_\mu A^\mu \alpha], \quad (3.135)$$

where $A_\mu^\alpha = A_\mu + \partial_\mu \alpha$, and $\Delta_{\text{FP}}[A]$ is the Faddeev–Popov determinant. For Maxwell theory,

$$\partial_\mu A^\mu \alpha = \partial_\mu A^\mu + \partial^2 \lambda, \quad (3.136)$$

so that

$$\Delta_{\text{FP}}[A] = \det(\partial^2), \quad (3.137)$$

which is independent of A_μ . Hence in the abelian theory this determinant contributes only an overall normalization factor, which we absorb into $Z[J]$.

After factoring out the infinite gauge volume $\int \mathcal{D}\alpha$, we are left with the gauge-fixed expression

$$Z[J] \propto \int \mathcal{D}A_\mu \delta[\partial_\mu A^\mu] \exp \left\{ iS_{\text{Maxwell}}[A] + i \int d^4x J_\mu A^\mu \right\}. \quad (3.138)$$

Gauge fixing in the path integral We now fix the gauge. In the path integral, this is done by inserting a delta functional that restricts the integration to field configurations satisfying the chosen gauge condition. For the Lorenz gauge,

$$\partial_\mu A^\mu = 0, \quad (3.139)$$

we insert

$$\delta[\partial_\mu A^\mu] \quad (3.140)$$

into the functional integral. This should be understood as a product of ordinary delta functions at each spacetime point,

$$\delta[\partial_\mu A^\mu] \equiv \prod_x \delta(\partial_\mu A^\mu(x)). \quad (3.141)$$

Thus the generating functional becomes

$$Z[J] = \int \mathcal{D}A_\mu \delta[\partial_\mu A^\mu] \exp \left\{ iS_{\text{Maxwell}}[A] + i \int d^4x J_\mu A^\mu \right\}. \quad (3.142)$$

This restricts the integration to a gauge slice, i.e. to field configurations satisfying the Lorenz gauge condition.

A remark on the Jacobian Strictly speaking, what we would really like to do is to restrict the integration directly to gauge-fixed configurations,

$$A_\mu = A_\mu^{\text{fixed}}, \quad (3.143)$$

that is, to integrate only over a cross-section of the gauge orbits. Formally one might think of inserting something like

$$\delta[A_\mu - A_\mu^{\text{fixed}}]. \quad (3.144)$$

However, in practice one usually does not know how to solve the gauge condition explicitly for A_μ^{fixed} . Instead one imposes the simpler condition $\partial_\mu A^\mu = 0$. Passing from a delta functional of the field variables themselves

to a delta functional of the gauge-fixing condition generally introduces a Jacobian, just as in the familiar one-dimensional formula

$$\delta(f(x)) = \sum_{x_i: f(x_i)=0} \frac{\delta(x-x_i)}{|f'(x_i)|}. \quad (3.145)$$

In the present abelian case, the gauge-fixing condition $\partial_\mu A^\mu$ is linear in A_μ . Therefore the corresponding Jacobian is independent of A_μ , so it contributes only an overall constant to the path integral. Since such overall constants are absorbed into the normalization, we may ignore it here.

A useful generalization of the gauge condition Even after inserting

$$\delta[\partial_\mu A^\mu], \quad (3.146)$$

the path integral is not yet convenient to evaluate directly. The standard trick is to generalize the gauge condition slightly. Instead of imposing

$$\partial_\mu A^\mu = 0, \quad (3.147)$$

we impose

$$\partial_\mu A^\mu = B(x), \quad (3.148)$$

where $B(x)$ is an arbitrary function. Geometrically, choosing different functions $B(x)$ simply corresponds to choosing different gauge slices. Since the final physics should not depend on which slice we choose, we may replace

$$\delta[\partial_\mu A^\mu] \longrightarrow \delta[\partial_\mu A^\mu - B]. \quad (3.149)$$

Thus we consider a B -dependent generating functional

$$Z[J; B] = \int \mathcal{D}A_\mu \delta[\partial_\mu A^\mu - B] \exp \left\{ iS_{\text{Maxwell}}[A] + i \int d^4x J_\mu A^\mu \right\}. \quad (3.150)$$

Because the choice of $B(x)$ is arbitrary, $Z[J; B]$ differs from the original expression only by an irrelevant normalization factor.

Averaging over B Since $B(x)$ is arbitrary, we may average over it with any convenient weight. A particularly useful choice is a Gaussian weight:

$$Z[J] \propto \int \mathcal{D}B \exp \left[-\frac{i\xi}{2} \int d^4x B(x)^2 \right] Z[J; B], \quad (3.151)$$

where ξ is an arbitrary gauge parameter. Substituting the expression for $Z[J; B]$, we obtain

$$Z[J] \propto \int \mathcal{D}B \mathcal{D}A_\mu \delta[\partial_\mu A^\mu - B] \exp \left\{ iS_{\text{Maxwell}}[A] + i \int d^4x J_\mu A^\mu - \frac{i\xi}{2} \int d^4x B^2 \right\}. \quad (3.152)$$

The delta functional now makes the B -integration trivial: it simply sets

$$B(x) = \partial_\mu A^\mu(x). \quad (3.153)$$

Therefore,

$$Z[J] \propto \int \mathcal{D}A_\mu \exp \left\{ iS_\xi[A] + i \int d^4x J_\mu A^\mu \right\}, \quad (3.154)$$

where the gauge-fixed action is

$$S_\xi[A] = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{\xi}{2} (\partial_\mu A^\mu)^2 \right]. \quad (3.155)$$

Thus the path integral derivation reproduces precisely the covariant gauge-fixed action introduced earlier.

Why this works The point of introducing the auxiliary field $B(x)$ and averaging over it with a Gaussian weight is that, after integrating out B , the gauge-fixing condition becomes a quadratic term in the action. This converts the path integral into a Gaussian integral with an invertible quadratic kernel. For Maxwell theory this may seem somewhat elaborate, since the Jacobian associated with the gauge-fixing condition is field-independent and can be absorbed into normalization. However, this procedure becomes essential in non-abelian gauge theories, where the Faddeev–Popov determinant is nontrivial.

The gauge-fixed quadratic kernel We now write the gauge-fixed action in quadratic form:

$$S_\xi[A] = \frac{1}{2} \int d^4x d^4y A_\mu(x) K_\xi^{\mu\nu}(x-y) A_\nu(y). \quad (3.156)$$

The corresponding differential operator is

$$K_\xi^{\mu\nu} = \partial^2 \eta^{\mu\nu} - (1 - \xi) \partial^\mu \partial^\nu. \quad (3.157)$$

Unlike the original Maxwell kernel, this operator is now invertible for generic ξ . In momentum space,

$$K_\xi^{\mu\nu}(k) = -k^2 \eta^{\mu\nu} + (1 - \xi) k^\mu k^\nu, \quad (3.158)$$

again up to sign conventions associated with Fourier transform and metric signature.

Evaluating the Gaussian integral Since the action is now quadratic, the path integral is Gaussian and can be evaluated exactly:

$$Z[J] = \mathcal{N} \exp \left[i \int d^4x d^4y J^\mu(x) (K_\xi^{-1})_{\mu\nu}(x-y) J^\nu(y) \right], \quad (3.159)$$

where $\mathcal{N}$ is an irrelevant normalization constant. The inverse kernel $(K_\xi^{-1})^{\mu\nu}$ is the Feynman propagator for the gauge field:

$$(K_\xi^{-1})_{\mu\nu}(x-y) = S_{\mu\nu}^{(F)} = \langle 0 | T A_\mu(x) A_\nu(y) | 0 \rangle \quad (3.160)$$

Photon propagator in momentum space In momentum space, the propagator is

$$S_{\mu\nu}^F(k) = \frac{-i}{k^2 - i\epsilon} \left(\eta^{\mu\nu} + (\xi - 1) \frac{k^\mu k^\nu}{k^2} \right). \quad (3.161)$$

Equivalently, in terms of the transverse and longitudinal projectors,

$$P_T^{\mu\nu} = \eta^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}, \quad P_L^{\mu\nu} = \frac{k^\mu k^\nu}{k^2}, \quad (3.162)$$

we may write

$$S_{\mu\nu}^F(k) = \frac{-i}{k^2 - i\epsilon} (P_T^{\mu\nu} + \xi P_L^{\mu\nu}). \quad (3.163)$$

Remarks

1. For $\xi = 1$ (Feynman gauge), the propagator simplifies to

$$S_{\mu\nu}^F(k) = \frac{-i\eta^{\mu\nu}}{k^2 - i\epsilon}. \quad (3.164)$$

This is often the most convenient choice for perturbative calculations.

2. If one formally sets $\xi = 0$, the propagator becomes proportional to the transverse projector:

$$S_{\mu\nu}^F(k) = \frac{-i}{k^2 - i\epsilon} P_T^{\mu\nu}. \quad (3.165)$$

This corresponds to Landau gauge. Notice that the original Maxwell kernel with no gauge fixing was not invertible; the correct procedure is to first gauge-fix, invert the kernel, and only then consider the limit $\xi \rightarrow 0$.

3. The parameter ξ does *not* represent residual gauge freedom. Rather, it labels different choices of gauge fixing. Physical observables are independent of ξ , even though intermediate expressions such as propagators may depend on it.

Wick contractions Once the propagator is known, Wick's theorem for the gauge field follows exactly as in the scalar and fermionic cases. In perturbation theory, every contraction of two gauge fields contributes a factor of the photon propagator

$$\overbrace{A_\mu(x) A_\nu(y)} = S_{\mu\nu}^F(x-y). \quad (3.166)$$

Thus the Feynman rules for theories involving photons are obtained immediately from the same path-integral machinery used earlier for scalars and fermions.

Interacting theories and Feynman rules At this point, the generalization to interacting theories is immediate. Consider a theory containing a gauge field A_μ , a Dirac field ψ , and possibly a scalar field ϕ . Then for any such interacting theory we may compute vacuum correlation functions by writing the full action as

$$S = S_0 + S_I, \quad (3.167)$$

where S_0 is the free part and S_I is the interaction part. As before, correlation functions in the interacting vacuum may be rewritten in terms of the free theory:

$$\langle \Omega | T \{ \mathcal{O} \} | \Omega \rangle = \frac{\langle 0 | T \{ \mathcal{O} e^{iS_I} \} | 0 \rangle}{\langle 0 | T e^{iS_I} | 0 \rangle}. \quad (3.168)$$

Here $|\Omega\rangle$ denotes the interacting vacuum, while $|0\rangle$ denotes the free vacuum. Therefore, exactly as in the scalar and fermionic cases, we may expand e^{iS_I} perturbatively and then apply Wick's theorem. This generates the perturbation series in terms of Feynman diagrams. The Feynman rules are then read off from:

- the free propagators, obtained from the quadratic part S_0 ,
- the interaction vertices, obtained from the interaction part S_I .

In particular, for theories involving the Maxwell field, each internal gauge-field line contributes the propagator $S_{\mu\nu}^F$, while the remaining propagators and vertices are determined in the same way from the matter fields and their couplings.

Conclusion We have now completed the path-integral quantization of the free Maxwell field. The essential steps were:

1. identify the gauge redundancy and the resulting non-invertibility of the original kernel;
2. fix the gauge by inserting a delta functional;
3. average over gauge slices using an auxiliary field $B(x)$;
4. obtain the covariant gauge-fixed action

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2; \quad (3.169)$$

5. invert the quadratic kernel to obtain the photon propagator.

With this, all the ingredients are in place to study interacting theories involving photons, in particular quantum electrodynamics.

4 QED from Maxwell field and Dirac

4.1 Step-by-step construction of QED

We have discussed the Maxwell action. The Maxwell Lagrangian density is

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu.$$

and J^μ should be a **conserved current** because for the consistency of the Maxwell equation,

$$\partial_\mu J^\mu = 0.$$

So the question is: what provides this J^μ ?

Introduce Dirac fermions and their conserved current Now introduce some other fields: **Dirac fermions**, with a fermionic field ψ . There is a conserved current because for Dirac fermions there is a **global** symmetry:

$$U(1): \quad \psi \rightarrow e^{i\alpha}\psi, \quad \alpha = \text{constant}.$$

This leads to a **Noether current** J^μ , which is conserved.

Couple fermions to the Maxwell field by identifying the currents If we want to couple the fermions to the Maxwell fields, then it is natural to identify the Maxwell current with the fermion Noether current:

$$J_{\text{Maxwell}}^\mu \equiv J_{\text{fermion}}^\mu.$$

Thus, the QED Lagrangian is obtained:

- Maxwell part: the free photon
- Dirac part: the free fermion
- plus coupling between them through $J^\mu A_\mu$

The interaction term is:

$$-e\bar{\psi}\gamma^\mu\psi A_\mu.$$

The role of e (coupling constant and unit of charge) The constant e can play two roles:

1. e as a **coupling constant**: it determines the coupling strength between ψ and A .
2. e as a **unit of charge**: since the Noether current is conserved, you can multiply it by an arbitrary number and it remains conserved. The unit of this current is given by e .

And of course e is just our **standard electric charge**. If ψ is the electron field, then this gives the usual interaction between electron and photon.

Rewrite QED in terms of a covariant derivative We can also slightly rewrite this Lagrangian in a somewhat different way. QED can be written as

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - i\bar{\psi}(\gamma^\mu D_\mu - m)\psi.$$

where we introduce a new derivative called D_μ :

$$D_\mu\psi = \partial_\mu\psi - ieA_\mu\psi.$$

which is called covariant derivative and highlights a new property. We know the Maxwell theory has a **gauge symmetry**. However, in the original Dirac theory we only had a **global** symmetry: the parameter α must be a constant. When we **combine together** and use the new Lagrangian, the new Lagrangian has a **generalized gauge symmetry**.

Generalized gauge transformation of A_μ and ψ The QED Lagrangian is invariant under:

$$A_\mu \rightarrow A'_\mu = A_\mu + \partial_\mu\lambda(x),$$

and simultaneously ψ transforms by a **local** phase:

$$\psi \rightarrow \psi' = e^{ie\lambda(x)}\psi.$$

Here e appears in the local phase, as stated.

Proof: $\mathcal{L}_{\text{QED}}$ is invariant under the above transformation

Step 1: show how $D_\mu\psi$ transforms Compute $D'_\mu\psi'$:

$$D'_\mu\psi' \equiv (\partial_\mu - ieA'_\mu)\psi'.$$

Substitute $A'_\mu = A_\mu + \partial_\mu\lambda$ and $\psi' = e^{ie\lambda}\psi$:

$$\begin{aligned} D'_\mu\psi' &= (\partial_\mu - ie(A_\mu + \partial_\mu\lambda))(e^{ie\lambda}\psi) \\ &= \partial_\mu(e^{ie\lambda}\psi) - ieA_\mu e^{ie\lambda}\psi - ie(\partial_\mu\lambda)e^{ie\lambda}\psi. \end{aligned}$$

Now expand the derivative:

$$\partial_\mu(e^{ie\lambda}\psi) = (\partial_\mu e^{ie\lambda})\psi + e^{ie\lambda}\partial_\mu\psi = ie(\partial_\mu\lambda)e^{ie\lambda}\psi + e^{ie\lambda}\partial_\mu\psi.$$

Plug this back in:

$$\begin{aligned}
D'_\mu \psi' &= [ie(\partial_\mu \lambda)e^{ie\lambda}\psi + e^{ie\lambda}\partial_\mu \psi] - ieA_\mu e^{ie\lambda}\psi - ie(\partial_\mu \lambda)e^{ie\lambda}\psi \\
&= e^{ie\lambda}\partial_\mu \psi - ieA_\mu e^{ie\lambda}\psi \\
&= e^{ie\lambda}(\partial_\mu \psi - ieA_\mu \psi) \\
&= e^{ie\lambda}D_\mu \psi.
\end{aligned}$$

So we have the simple covariant transformation rule:

$$D'_\mu \psi' = e^{ie\lambda(x)} D_\mu \psi.$$

Step 2: check invariance of the fermion kinetic term Consider

$$i\bar{\psi}'\gamma^\mu D'_\mu \psi'.$$

Using $\bar{\psi}' = e^{-ie\lambda}\bar{\psi}$ and $D'_\mu \psi' = e^{ie\lambda}D_\mu \psi$:

$$\begin{aligned}
i\bar{\psi}'\gamma^\mu D'_\mu \psi' &= i(e^{-ie\lambda}\bar{\psi})\gamma^\mu(e^{ie\lambda}D_\mu \psi) \\
&= i\bar{\psi}\gamma^\mu D_\mu \psi.
\end{aligned}$$

The phase factors cancel.

Step 3: check invariance of the mass term The mass term is $-m\bar{\psi}\psi$. Under the transformation:

$$\bar{\psi}'\psi' = (e^{-ie\lambda}\bar{\psi})(e^{ie\lambda}\psi) = \bar{\psi}\psi.$$

So

$$-m\bar{\psi}'\psi' = -m\bar{\psi}\psi.$$

Step 4: check invariance of the Maxwell term We use

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Under $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$,

$$\begin{aligned}
F'_{\mu\nu} &= \partial_\mu(A_\nu + \partial_\nu \lambda) - \partial_\nu(A_\mu + \partial_\mu \lambda) \\
&= (\partial_\mu A_\nu - \partial_\nu A_\mu) + (\partial_\mu \partial_\nu \lambda - \partial_\nu \partial_\mu \lambda) \\
&= F_{\mu\nu},
\end{aligned}$$

since $\partial_\mu \partial_\nu \lambda = \partial_\nu \partial_\mu \lambda$. Therefore

$$-\frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}.$$

Conclusion: explicit invariance of the full Lagrangian density All terms are invariant:

$$\mathcal{L}_{\text{QED}}(A', \psi') = \mathcal{L}_{\text{QED}}(A, \psi).$$

Hence $\mathcal{L}_{\text{QED}}$ is gauge invariant under

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda(x), \quad \psi \rightarrow e^{ie\lambda(x)}\psi.$$

Turning the logic around: gauge symmetry determines the interaction Let us first review the logic from the fermionic example. We started with Maxwell theory, in which the gauge field A_μ couples to some current J^μ :

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + A_\mu J^\mu.$$

On the other hand, for the free Dirac field ψ , there is a conserved current

$$J^\mu = \bar{\psi}\gamma^\mu\psi.$$

So the natural thing to do is to combine the two theories by replacing the current in Maxwell theory by the fermionic current:

$$J^\mu \rightarrow \bar{\psi}\gamma^\mu\psi.$$

What is nontrivial is that, once we combine them, the resulting theory has a *generalized gauge symmetry*. Namely, besides the usual gauge transformation of A_μ ,

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \lambda(x),$$

we can also transform the fermion field locally:

$$\psi(x) \rightarrow e^{-ie\lambda(x)}\psi(x), \quad \bar{\psi}(x) \rightarrow \bar{\psi}(x)e^{ie\lambda(x)}.$$

Thus the full interacting theory is invariant under this enlarged local symmetry.

We can now *turn the logic around*. Instead of first writing down the interaction and then discovering the symmetry, we may proceed as follows:

- Start with the gauge symmetry of Maxwell theory,

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda(x).$$

- Try to generalize this symmetry so that it also acts on the matter field ψ .
- Require that the full theory of A_μ and ψ be invariant under this local symmetry.

Then this requirement forces the interaction between A_μ and ψ to take the form

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu.$$

So although this is logically equivalent to what we already did, it is a much more powerful way to think:

Imposing local gauge symmetry can be used to deduce the interaction.

In this sense, gauge symmetry is not merely a redundancy of description; it is also a dynamical principle that constrains the form of the allowed couplings.

Remark on gauge fixing Extending the gauge symmetry to include the fermion field does not introduce any essentially new gauge-fixing issue. We are not doing gauge fixing at this stage; we are simply enlarging the local symmetry so that it acts consistently on both A_μ and ψ . The gauge-fixing discussion for the Maxwell field remains as before.

4.2 Next example: coupling a charged scalar to the photon

Let us now consider the analogous problem for a charged scalar field. Take a complex scalar field ϕ , with Lagrangian of the form

$$\mathcal{L}_{\text{scalar}} = -\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi - V(\phi^* \phi),$$

This theory has a global $U(1)$ symmetry

$$\phi \rightarrow e^{i\alpha} \phi, \quad \alpha = \text{constant},$$

and hence a conserved Noether current

$$J^\mu = -i(\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi).$$

By analogy with the fermion case, one might try to couple this current to the Maxwell field and identify the current in Maxwell theory with the scalar Noether current. That is the natural first guess.

Constructing scalar QED by imposing gauge invariance The way out is to *impose* that the theory be invariant under the local gauge transformation

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \lambda(x), \quad \phi(x) \rightarrow e^{ie\lambda(x)}\phi(x).$$

To write down a theory invariant under this transformation, we can borrow the key lesson from the fermionic case. There we learned that the ordinary derivative must be replaced by a derivative that transforms covariantly.

The important point is that this construction does not depend on whether the field is a fermion or a boson, nor on whether it is a scalar or a spinor. So for the scalar field we introduce

$$D_\mu \phi \equiv \partial_\mu \phi - ieA_\mu \phi.$$

Similarly,

$$(D_\mu \phi)^* = \partial_\mu \phi^* + ie A_\mu \phi^*.$$

Under the gauge transformation above, $D_\mu \phi$ transforms in the same way as ϕ itself:

$$D_\mu \phi \rightarrow e^{ie\lambda(x)} D_\mu \phi.$$

Thus $D_\mu \phi$ transforms nicely under the gauge symmetry. Because of this, we can immediately write down a gauge-invariant Lagrangian for scalar QED:

$$\mathcal{L}_{\text{sQED}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - (D_\mu \phi)^* (D^\mu \phi) - m^2 \phi^* \phi - V(\phi^* \phi).$$

The mass term and any potential $V(\phi^* \phi)$ are automatically gauge invariant, since they do not involve derivatives.

The derivative D_μ is called the *covariant derivative*, because it transforms covariantly under the gauge transformation. Mathematically, this object has a deep relation to differential geometry, where A_μ plays the role of a *connection*. We will not pursue that viewpoint here, but it is worth keeping in mind.

Comparison with the naive scalar coupling Now let us compare this gauge-invariant scalar QED Lagrangian with the naive theory

$$\tilde{\mathcal{L}}_{\text{sQED}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi - V(\phi \phi^*) + e A_\mu J^\mu.$$

If we expand the covariant-derivative term, we find

$$(D_\mu \phi)^* (D^\mu \phi) = \partial_\mu \phi^* \partial^\mu \phi + ie A^\mu (\phi^* \partial_\mu \phi - (\partial_\mu \phi^*) \phi) + e^2 A_\mu A^\mu \phi^* \phi.$$

Equivalently,

$$(D_\mu \phi)^* (D^\mu \phi) = \partial_\mu \phi^* \partial^\mu \phi + e A_\mu J^\mu + e^2 A_\mu A^\mu \phi^* \phi,$$

where

$$J^\mu = -i (\phi^* \partial^\mu \phi - (\partial^\mu \phi^*) \phi).$$

So the gauge-invariant scalar QED Lagrangian is almost the same as the naive one, except for one extra term:

$$e^2 A_\mu A^\mu \phi^* \phi.$$

This is the crucial difference from the fermion case.

- For fermions, the kinetic term contains only one derivative, so replacing ∂_μ by D_μ generates only the cubic interaction

$$\bar{\psi} \gamma^\mu \psi A_\mu.$$

- For scalars, the kinetic term contains two derivatives, so replacing each ∂_μ by D_μ produces not only the cubic interaction $A_\mu J^\mu$, but also a quartic interaction.

Indeed, scalar QED contains both

$$A \phi^* \phi \quad \text{and} \quad A^2 \phi^* \phi$$

interactions. In words:

- the naive current coupling gives only a cubic interaction involving two scalar fields and one photon;
- gauge invariance requires an additional quartic interaction involving two scalar fields and two photons.

Once this extra term is included, the full theory is gauge invariant and therefore consistent.

4.3 Feynman rules for fermionic QED

Let us now write down the Feynman rules. We begin with the fermionic case.

Propagators The propagators are the same as before, because propagators only depend on the quadratic part of the theory; they do not care about the interactions. So we have:

- a wavy line for the photon field A_μ ;
- a solid line for the fermion field ψ .

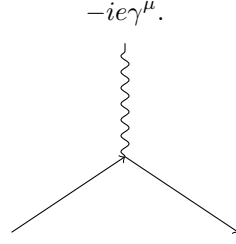
$$\begin{array}{ccc} \text{~~~~~} & & \text{—————} \\ \text{~~~~~} & & \text{~~~~~} \\ A_\mu & & \psi \end{array}$$

These correspond to the photon propagator and the fermion propagator, respectively.

Vertices Now consider the interaction term

$$-eA_\mu\bar{\psi}\gamma^\mu\psi.$$

This gives a vertex with two fermion lines and one photon line. Structurally, this is very similar to the Yukawa coupling discussed before, except that now the bosonic line is a photon line. The corresponding vertex factor is



The origin of the factors is as follows:

- the minus sign comes from the minus sign in the interaction term;
- the factor of i comes from the usual factor in passing from the action to the Feynman rule;
- the fields $\bar{\psi}$, ψ , and A_μ are represented by the attached lines;
- what remains is γ^μ .

So the fermion-photon vertex is

$$\boxed{-ie\gamma^\mu}.$$

One should pay attention that γ^μ is a matrix in spinor space. Therefore, when combining vertices and propagators, one must keep track of the spinor indices carefully.

External photon legs Now, if we consider scattering amplitudes, then we must also specify the external legs. For photons, one needs to include polarization vectors for external photon states. If one is computing a Green's function, this is not necessary: one simply keeps the external propagator. But for a scattering amplitude, the external photon states require polarization vectors.

For an initial-state $|\vec{k}, \alpha\rangle$ photon with momentum k and polarization label α , the external leg contributes a polarization vector

$$\alpha \overset{k}{\sim} \epsilon_\mu^{(\alpha)}(k).$$

For a final-state $\langle \vec{k}, \alpha |$ photon with momentum k and polarization label α , the corresponding factor is the complex conjugate

$$\alpha \overset{k}{\sim} \left(\epsilon_\mu^{(\alpha)}(k) \right)^*.$$

So for external photon legs in scattering amplitudes, one must include the appropriate polarization vector. The same applies for external fermion legs as well: one includes the corresponding external fermion wavefunctions.

4.4 Feynman rules for scalar QED

Let us now turn to scalar QED. The discussion is very similar in spirit to fermionic QED, but there is one important difference: the scalar field carries electric charge, so the arrow on the scalar line has a physical meaning. It is not merely a momentum arrow; rather, it indicates the flow of charge.

We represent the scalar propagator by a dashed line with an arrow:

$$\begin{array}{c} \text{-----} \rightarrow \\ \phi \end{array}$$

Here the arrow indicates the charge flow. One should distinguish this from the momentum label, which can be added separately when needed. The interaction terms in scalar QED come from expanding the covariant derivatives in

$$(D_\mu\phi)^*(D^\mu\phi), \quad D_\mu = \partial_\mu + ieA_\mu.$$

Expanding this gives

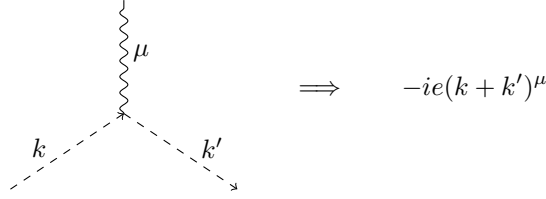
$$(D_\mu\phi)^*(D^\mu\phi) = \partial_\mu\phi^*\partial^\mu\phi + ieA_\mu(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi) + e^2A_\mu A^\mu\phi^*\phi.$$

So there are two interaction vertices:

- a three-point vertex involving one photon and two scalars;
- a four-point vertex involving two photons and two scalars.

The three-point vertex

The current coupling gives the vertex with two scalar lines and one photon line. Schematically:



Here k and k' are the scalar momenta, assigned according to the convention used in the diagram. The reason this factor appears is that the interaction contains derivatives acting on the scalar fields, and each derivative brings down a factor of momentum. Since there are two such terms, the result is proportional to the sum of the two scalar momenta.

Thus the scalar-scalar-photon vertex is

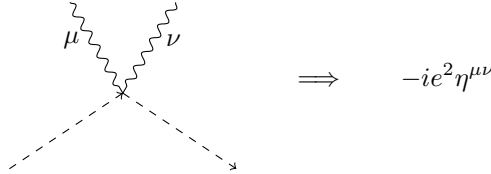
$$\boxed{-ie(k + k')^\mu}.$$

The four-point vertex

There is also the interaction term

$$e^2 A_\mu A^\mu \phi^* \phi,$$

which gives a contact vertex involving two photons and two scalars:



So the scalar-scalar-photon-photon vertex is

$$\boxed{-ie^2 \eta^{\mu\nu}}.$$

5 Scattering and Elementary Processes in QED

5.1 Cross Sections and Decay Rates

After constructing QED and its Feynman rules, the next question is how to connect a scattering amplitude with an experimental observable. A Feynman diagram gives an amplitude $\mathcal{M}$, but experiments count particles: how many particles are produced in a detector, in which direction, and with what energy. The cross section and the decay rate are the bridge between these two descriptions.

Cross sections in nonrelativistic scattering

In ordinary nonrelativistic scattering one considers an incident beam, a target, and a detector placed at an angle (θ, ϕ) . The detector subtends a small solid angle $d\Omega$. The directly measured quantity is the number of scattered particles entering that detector per unit time,

$$\left(\frac{dN}{dt} \right)_{\text{out}}.$$

This number is proportional both to the intensity of the incoming beam and to the size of the detector. Therefore it is not by itself an intrinsic property of the interaction.

The incident flux is the number of incoming particles per unit time per unit area,

$$\left(\frac{dN}{dt dA} \right)_{\text{in}}.$$

The differential cross section is defined by removing this trivial beam factor:

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\left(\frac{dN}{dt d\Omega}\right)_{\text{out}}}{\left(\frac{dN}{dt dA}\right)_{\text{in}}}}. \quad (5.1)$$

It has the dimension of area and may be interpreted as the effective area presented by the target to the incoming beam. The total cross section is obtained by integrating over all directions,

$$\sigma_{\text{tot}} = \int d\Omega \frac{d\sigma}{d\Omega}. \quad (5.2)$$

Relativistic transition probability

In relativistic quantum field theory, the initial state may contain two incoming particles and the final state may contain any number of outgoing particles,

$$1 + 2 \rightarrow 3 + 4 + \cdots + n.$$

Let the initial state be denoted by α and the final state by β . The scattering amplitude is defined by

$$\langle \beta, +\infty | \alpha, -\infty \rangle = (2\pi)^4 \delta^{(4)}(p_\alpha - p_\beta) \mathcal{M}_{\alpha \rightarrow \beta}. \quad (5.3)$$

Here p_α and p_β are the total four-momenta of the initial and final states. The delta function expresses energy-momentum conservation.

Using relativistic normalization,

$$\langle p' | p \rangle = (2\pi)^3 2p^0 \delta^{(3)}(\mathbf{p} - \mathbf{p}'), \quad (5.4)$$

one has, for the same momentum,

$$\langle p | p \rangle = 2p^0 V, \quad (5.5)$$

where $V = (2\pi)^3 \delta^{(3)}(0)$ is the normalization volume. Squaring the amplitude gives

$$\left[\delta^{(4)}(p_\alpha - p_\beta) \right]^2 = \delta^{(4)}(p_\alpha - p_\beta) \delta^{(4)}(0),$$

and

$$(2\pi)^4 \delta^{(4)}(0) = VT. \quad (5.6)$$

The factor T is the total time interval of the experiment. Dividing by T therefore gives the transition probability per unit time.

For a process with two incoming particles of momenta p_1, p_2 and n outgoing particles of momenta $k_1, \dots, k_n$, the final momenta are measured with finite resolution. We must multiply by the density of final states,

$$\prod_{j=1}^n \frac{V d^3 k_j}{(2\pi)^3}.$$

All final-state volume factors cancel. The remaining result can be written as

$$d\dot{P}_{\alpha \rightarrow \beta} = \frac{|\mathcal{M}_{\alpha \rightarrow \beta}|^2}{4E_1 E_2 V} d\mu_\beta, \quad (5.7)$$

where the Lorentz-invariant phase-space measure is

$$\boxed{d\mu_\beta = (2\pi)^4 \delta^{(4)} \left(p_1 + p_2 - \sum_{j=1}^n k_j \right) \prod_{j=1}^n \frac{d^3 k_j}{(2\pi)^3 2k_j^0}}. \quad (5.8)$$

The factor $d^3 k / (2k^0)$ is Lorentz invariant for an on-shell particle, and the four-dimensional delta function is also Lorentz invariant.

Invariant flux factor

In the rest frame of particle 2, the incident flux of particle 1 is

$$\left(\frac{dN_1}{dt dA} \right)_{\text{in}} = \frac{v_1}{V},$$

since there is one incoming particle in the normalization volume V . Dividing the transition rate by this flux gives

$$d\sigma_{\alpha \rightarrow \beta} = \frac{|\mathcal{M}_{\alpha \rightarrow \beta}|^2}{4E_1 E_2 v_1} d\mu_\beta$$

in that special frame.

To write this result covariantly, we need a Lorentz-invariant quantity that reduces to $E_1 E_2 v_1$ in the rest frame of particle 2. With our metric convention, $p_i^2 = -m_i^2$. Define

$$\Sigma \equiv \sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}. \quad (5.9)$$

In the rest frame of particle 2, $p_2 = (m_2, \mathbf{0})$, so

$$\Sigma = m_2 |\mathbf{p}_1| = E_1 E_2 v_1.$$

Therefore the general $2 \rightarrow n$ cross section is

$$d\sigma_{\alpha \rightarrow \beta} = \frac{|\mathcal{M}_{\alpha \rightarrow \beta}|^2}{4\Sigma} d\mu_\beta. \quad (5.10)$$

It is often useful to introduce

$$s \equiv -(p_1 + p_2)^2. \quad (5.11)$$

Then

$$p_1 \cdot p_2 = -\frac{1}{2}(s - m_1^2 - m_2^2),$$

and

$$\Sigma = \frac{1}{2} \sqrt{\lambda(s, m_1^2, m_2^2)}, \quad (5.12)$$

where the Kallen function is

$$\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2ac - 2bc. \quad (5.13)$$

In the center-of-mass frame,

$$|\mathbf{p}_{\text{cm}}| = \frac{\Sigma}{\sqrt{s}} = \frac{\sqrt{\lambda(s, m_1^2, m_2^2)}}{2\sqrt{s}}, \quad (5.14)$$

and hence

$$d\sigma_{\alpha \rightarrow \beta} = \frac{|\mathcal{M}_{\alpha \rightarrow \beta}|^2}{4|\mathbf{p}_{\text{cm}}|\sqrt{s}} d\mu_\beta. \quad (5.15)$$

$2 \rightarrow 2$ scattering

For a $2 \rightarrow 2$ process,

$$p_1 + p_2 \rightarrow k_1 + k_2,$$

we define the Mandelstam variables

$$s = -(p_1 + p_2)^2 = -(k_1 + k_2)^2, \quad (5.16)$$

$$t = -(p_1 - k_1)^2 = -(p_2 - k_2)^2, \quad (5.17)$$

$$u = -(p_1 - k_2)^2 = -(p_2 - k_1)^2. \quad (5.18)$$

They satisfy

$$s + t + u = m_1^2 + m_2^2 + m_1'^2 + m_2'^2. \quad (5.19)$$

The two-body phase-space measure in the center-of-mass frame is

$$d\mu_2 = (2\pi)^4 \delta^{(4)}(p_1 + p_2 - k_1 - k_2) \frac{d^3 k_1}{(2\pi)^3 2E_1'} \frac{d^3 k_2}{(2\pi)^3 2E_2'}. \quad (5.20)$$

Using the spatial delta function to set $\mathbf{k}_2 = -\mathbf{k}_1$, and using the remaining energy delta function to fix $|\mathbf{k}_1|$, one obtains

$$d\mu_2 = \frac{|\mathbf{k}_{\text{cm}}|}{16\pi^2\sqrt{s}} d\Omega_{\text{cm}}. \quad (5.21)$$

Therefore

$$\frac{d\sigma}{d\Omega_{\text{cm}}} = \frac{|\mathcal{M}|^2}{64\pi^2 s} \frac{|\mathbf{k}_{\text{cm}}|}{|\mathbf{p}_{\text{cm}}|}. \quad (5.22)$$

This is the main formula that we will use for elementary QED processes.

Decay rate

If the initial state contains only one particle,

$$p \rightarrow k_1 + \cdots + k_n,$$

then the process is a decay rather than scattering. There is no incoming flux to divide by. Repeating the same normalization argument, but using

$$\langle p|p \rangle = 2E_p V,$$

gives

$$d\Gamma_{p \rightarrow \beta} = \frac{|\mathcal{M}_{p \rightarrow \beta}|^2}{2E_p} d\mu_\beta, \quad (5.23)$$

where now

$$d\mu_\beta = (2\pi)^4 \delta^{(4)}\left(p - \sum_{j=1}^n k_j\right) \prod_{j=1}^n \frac{d^3 k_j}{(2\pi)^3 2k_j^0}. \quad (5.24)$$

The total decay rate is

$$\Gamma_{\text{tot}} = \sum_{\beta} \int d\Gamma_{p \rightarrow \beta}, \quad (5.25)$$

and the lifetime is

$$\tau = \frac{1}{\Gamma_{\text{tot}}}. \quad (5.26)$$

The decay rate is not Lorentz invariant. Since the formula contains $1/(2E_p)$, a moving particle with $E_p = \gamma m$ has

$$\Gamma_{\text{lab}} = \frac{1}{\gamma} \Gamma_{\text{rest}}, \quad \tau_{\text{lab}} = \gamma \tau_{\text{rest}}, \quad (5.27)$$

which is the usual time-dilation effect.

5.2 Elementary Processes in QED

We now apply the cross-section formula to concrete QED processes. The common structure is simple:

1. write the amplitude $\mathcal{M}$ from the Feynman rules;
2. square the amplitude;
3. sum over final spin or polarization states and average over initial ones;
4. insert the result into the appropriate phase-space formula.

Spin sums and unpolarized scattering

The amplitude depends not only on momenta but also on external spin and polarization labels. For spin- $\frac{1}{2}$ fermions these labels appear in

$$u_r(p), \quad v_r(p),$$

and for photons they appear in polarization vectors

$$\epsilon_\mu^{(\lambda)}(p).$$

Most beams are unpolarized, and final-state polarizations are often not measured. Therefore the experimentally relevant squared amplitude is obtained by summing over final labels and averaging over initial labels. For two incoming spin- $\frac{1}{2}$ particles,

$$\overline{|\mathcal{M}|^2} = \frac{1}{4} \sum_{\text{initial spins}} \sum_{\text{final spins}} |\mathcal{M}|^2. \quad (5.28)$$

The useful spin-sum identities in our conventions are

$$\sum_r u_r(p) \bar{u}_r(p) = i(\not{p} + m), \quad \sum_r v_r(p) \bar{v}_r(p) = -i(-\not{p} + m). \quad (5.29)$$

Together with trace identities for gamma matrices, these turn spin sums into traces.

$e^- e^+ \rightarrow b \bar{b}$

Consider

$$e^-(p_1, r_1) + e^+(p_2, \bar{r}_2) \rightarrow b(k_1, s_1) + \bar{b}(k_2, \bar{s}_2),$$

where b is another charged fermion, such as a muon or a quark. Let the electron mass be m , and let the final fermion mass be m' . At leading order the process is mediated by a virtual photon:

$$e^- + e^+ \rightarrow \gamma^* \rightarrow b + \bar{b}.$$

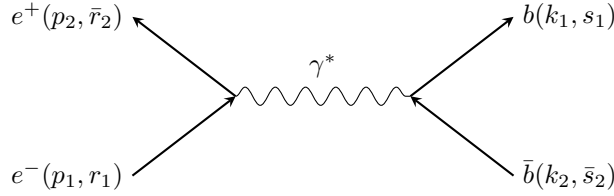


Figure 2: Leading QED diagram for $e^- e^+ \rightarrow b \bar{b}$.

Following the fermion lines gives

$$\mathcal{M} = \bar{v}_{r_2}(p_2)(-ie\gamma^\mu)u_{r_1}(p_1) \left(\frac{-i\eta_{\mu\nu}}{q^2 - i\epsilon} \right) \bar{u}_{s_1}(k_1)(-ie\gamma^\nu)v_{s_2}(k_2), \quad (5.30)$$

where

$$q = p_1 + p_2 = k_1 + k_2, \quad q^2 = -s.$$

Up to an irrelevant overall phase,

$$\mathcal{M} = -\frac{ie^2}{s} [\bar{v}_{r_2}(p_2)\gamma^\mu u_{r_1}(p_1)] [\bar{u}_{s_1}(k_1)\gamma_\mu v_{s_2}(k_2)]. \quad (5.31)$$

Spin-averaged squared amplitude

For the unpolarized process,

$$\overline{|\mathcal{M}|^2} = \frac{1}{4} \sum_{r_1, r_2, s_1, s_2} |\mathcal{M}|^2.$$

Using the spin sums and trace identities,

$$\overline{|\mathcal{M}|^2} = \frac{e^4}{4s^2} \text{tr} [(-i\not{p}_2 + m)\gamma^\mu(i\not{p}_1 + m)\gamma^\nu] \text{tr} [(i\not{k}_1 + m')\gamma_\mu(-i\not{k}_2 + m')\gamma_\nu]. \quad (5.32)$$

The final result is

$$\overline{|\mathcal{M}|^2} = \frac{8e^4}{s^2} [(k_1 \cdot p_1)(k_2 \cdot p_2) + (p_1 \cdot k_2)(p_2 \cdot k_1) - m^2(k_1 \cdot k_2) - m'^2(p_1 \cdot p_2) + 2m^2 m'^2]. \quad (5.33)$$

In terms of the Mandelstam variables

$$s = -(p_1 + p_2)^2, \quad t = -(p_1 - k_1)^2, \quad u = -(p_1 - k_2)^2,$$

with

$$s + t + u = 2m^2 + 2m'^2,$$

this becomes

$$\overline{|\mathcal{M}|^2} = \frac{2e^4}{s^2} \left[(t - m^2 - m'^2)^2 + (u - m^2 - m'^2)^2 + 2s(m^2 + m'^2) \right]. \quad (5.34)$$

Center-of-mass result

In the center-of-mass frame,

$$p_1 = (E, \mathbf{p}), \quad p_2 = (E, -\mathbf{p}),$$

$$k_1 = (E, \mathbf{k}), \quad k_2 = (E, -\mathbf{k}), \quad E = \frac{\sqrt{s}}{2}.$$

The magnitudes of the initial and final momenta are

$$|\mathbf{p}_{\text{cm}}|^2 = E^2 - m^2, \quad |\mathbf{k}_{\text{cm}}|^2 = E^2 - m'^2.$$

Let θ be the angle between $\mathbf{p}$ and $\mathbf{k}$. Then

$$\overline{|\mathcal{M}|^2} = e^4 \left[1 + \frac{m^2 + m'^2}{E^2} + \frac{|\mathbf{p}_{\text{cm}}|^2 |\mathbf{k}_{\text{cm}}|^2}{E^4} \cos^2 \theta \right]. \quad (5.35)$$

Using the $2 \rightarrow 2$ formula,

$$\frac{d\sigma}{d\Omega_{\text{cm}}} = \frac{\overline{|\mathcal{M}|^2} |\mathbf{k}_{\text{cm}}|}{64\pi^2 s |\mathbf{p}_{\text{cm}}|}. \quad (5.36)$$

Writing

$$\alpha = \frac{e^2}{4\pi},$$

the total cross section is

$$\sigma(e^- e^+ \rightarrow b\bar{b}) = \frac{4\pi\alpha^2}{3s} \frac{\sqrt{1 - \frac{m'^2}{E^2}}}{\sqrt{1 - \frac{m^2}{E^2}}} \left(1 + \frac{m^2}{2E^2} \right) \left(1 + \frac{m'^2}{2E^2} \right). \quad (5.37)$$

The numerator square root describes the final-state threshold. If $E < m'$, the final pair cannot be produced.

$$e^- e^+ \rightarrow \mu^- \mu^+$$

For muon-pair production, $b = \mu^-$. Since $m_\mu \simeq 207m_e$, the electron mass can usually be neglected once the muon threshold is open. Near threshold,

$$|\mathbf{k}_{\text{cm}}| \ll m_\mu, \quad E \simeq m_\mu,$$

and

$$\sigma(e^- e^+ \rightarrow \mu^- \mu^+) = \frac{\pi\alpha^2}{2m_\mu^2} \frac{|\mathbf{k}_{\text{cm}}|}{m_\mu} + \dots. \quad (5.38)$$

The cross section vanishes at threshold because the final-state phase space closes.

In the ultra-relativistic regime,

$$E \gg m_e, \quad E \gg m_\mu,$$

both masses may be neglected. Then

$$\frac{d\sigma}{d\Omega_{\text{cm}}} = \frac{\alpha^2}{4s} (1 + \cos^2 \theta), \quad (5.39)$$

and

$$\sigma(e^- e^+ \rightarrow \mu^- \mu^+) = \frac{4\pi\alpha^2}{3s}. \quad (5.40)$$

The $1/s$ scaling follows directly from dimensional analysis: in the high-energy massless limit, s is the only available scale and a cross section has mass dimension -2 .

Inclusive hadron production and quark colors

Now consider

$$e^- + e^+ \rightarrow \text{hadrons}.$$

At short distances the leading process is

$$e^- + e^+ \rightarrow q + \bar{q}.$$

Although quarks are confined and do not appear as isolated final particles, an inclusive sum over all hadronic final states can be approximated by summing over the short-distance quark production channels:

$$\sigma(e^-e^+ \rightarrow \text{hadrons}) \simeq \sum_{\text{allowed quarks } q_i} \sigma(e^-e^+ \rightarrow q_i \bar{q}_i).$$

For a quark of electric charge $Q_i e$, the final vertex contributes a factor Q_i , so the cross section is multiplied by Q_i^2 . Each quark flavor also has N_c color states. Hence

$$\sigma(e^-e^+ \rightarrow \text{hadrons}) = N_c \sum_{\text{allowed } i} Q_i^2 \sigma(e^-e^+ \rightarrow \mu^- \mu^+). \quad (5.41)$$

Define

$$R \equiv \frac{\sigma(e^-e^+ \rightarrow \text{hadrons})}{\sigma(e^-e^+ \rightarrow \mu^- \mu^+)} = N_c \sum_{\text{allowed } i} Q_i^2. \quad (5.42)$$

For the light quark charges,

$$Q_u = Q_c = \frac{2}{3}, \quad Q_d = Q_s = Q_b = -\frac{1}{3}.$$

As $\sqrt{s}$ increases, more flavors become kinematically accessible. Roughly,

$$\begin{aligned} \sqrt{s} < 3 \text{ GeV} : \quad R &= N_c \frac{2}{3}, \\ 3 \text{ GeV} < \sqrt{s} < 9.5 \text{ GeV} : \quad R &= N_c \frac{10}{9}, \\ \sqrt{s} > 9.5 \text{ GeV} : \quad R &= N_c \frac{11}{9}. \end{aligned}$$

Experimentally these plateaus are compatible with

$$N_c = 3. \quad (5.43)$$

Thus the inclusive $e^-e^+ \rightarrow \text{hadrons}$ cross section gives direct evidence that quarks come in three colors.

Crossing symmetry

Compare the annihilation process

$$A : \quad e^-(p_1) + e^+(p_2) \rightarrow \mu^-(k_1) + \mu^+(k_2)$$

with elastic electron-muon scattering,

$$B : \quad e^-(p_1) + \mu^-(p_2) \rightarrow \mu^-(k_1) + e^-(k_2).$$

The second process is obtained from the first by viewing the same diagram in a different channel. The incoming positron in process A becomes an outgoing electron in process B , and the outgoing antimuon becomes an incoming muon.

At the level of momenta, moving a particle from the initial state to the final state, or from the final state to the initial state, reverses the sign of its momentum:

$$p_2 \rightarrow -k_2, \quad k_2 \rightarrow -p_2.$$

For scalar external particles this would imply the simple relation

$$\mathcal{M}_A(p_1, -k_2; k_1, -p_2) = \mathcal{M}_B(p_1, p_2; k_1, k_2).$$

For fermions, crossing also changes the type of external spinor wavefunction:

incoming particle	$u_r(p)$
outgoing particle	$\bar{u}_s(k)$
incoming antiparticle	$\bar{v}_r(p)$
outgoing antiparticle	$v_s(k)$

Therefore a polarized amplitude is not obtained by a completely trivial replacement. However, after summing over spins, the relation becomes simple. For example,

$$\sum_r v_r(-k) \bar{v}_r(-k) = -i(\not{k} + m) = -\sum_s u_s(k) \bar{u}_s(k).$$

In the example above two fermion lines are crossed, so the two minus signs cancel:

$$\boxed{\sum_{\text{spins}} |\mathcal{M}_A(p_1, -k_2; k_1, -p_2)|^2 = \sum_{\text{spins}} |\mathcal{M}_B(p_1, p_2; k_1, k_2)|^2.} \quad (5.44)$$

This is the basic content of crossing symmetry. It is not a symmetry relating two physical processes at the same kinematic point; rather, it is a relation between amplitudes after analytic continuation of the external momenta.

Compton scattering

Another elementary QED process is Compton scattering,

$$e^-(p_1, r_1) + \gamma(p_2, \alpha_1) \rightarrow e^-(k_1, s_1) + \gamma(k_2, \alpha_2).$$

Compared with $e^-e^+ \rightarrow \mu^-\mu^+$, this process introduces two new ingredients: the internal propagator is a fermion propagator, and the external photon states carry polarization vectors.

At tree level there are two diagrams. Along the electron line, the electron can first absorb the incoming photon and then emit the outgoing photon, or it can first emit the outgoing photon and then absorb the incoming photon. Define

$$q_1 = p_1 + p_2 = k_1 + k_2, \quad q_2 = p_1 - k_2 = k_1 - p_2.$$

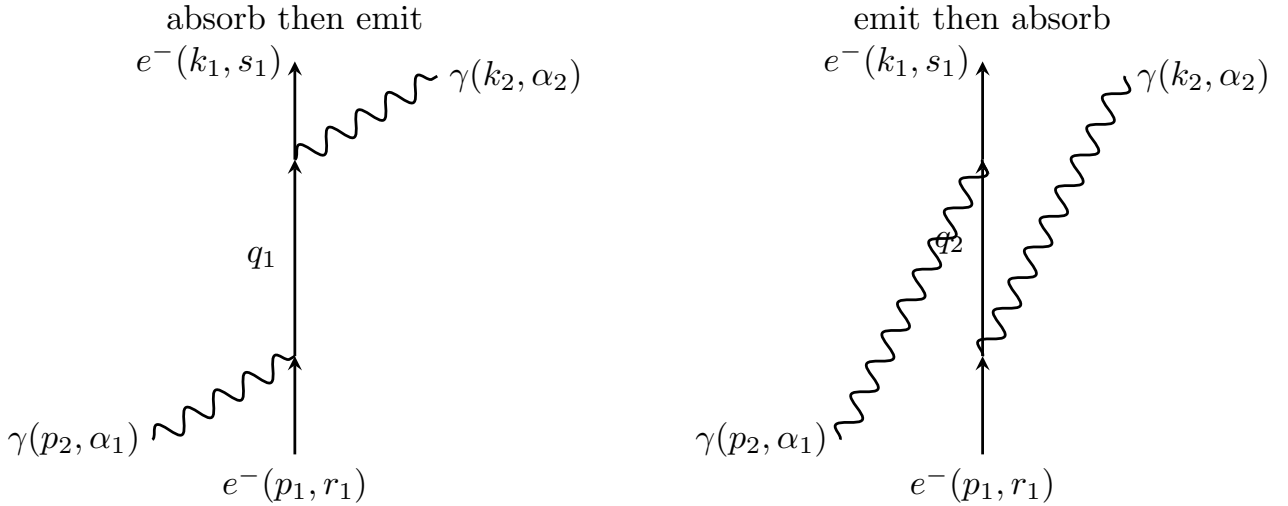


Figure 3: The two tree-level diagrams for Compton scattering.

The fermion propagator is

$$S_F(q) = -\frac{1}{i\not{q} - m + i\epsilon} = \frac{i\not{q} + m}{q^2 + m^2 - i\epsilon}. \quad (5.45)$$

The amplitude is

$$\mathcal{M} = \epsilon_\mu^{(\alpha_2)*}(k_2) \epsilon_\nu^{(\alpha_1)}(p_2) \bar{u}_{s_1}(k_1) [(-ie\gamma^\mu)S_F(q_1)(-ie\gamma^\nu) + (-ie\gamma^\nu)S_F(q_2)(-ie\gamma^\mu)] u_{r_1}(p_1). \quad (5.46)$$

The order of the matrices is important because the propagator and gamma matrices act on spinor indices.

Photon polarization sums and the Ward identity

For unpolarized Compton scattering,

$$\overline{|\mathcal{M}|^2} = \frac{1}{4} \sum_{r_1, s_1, \alpha_1, \alpha_2} |\mathcal{M}|^2.$$

The fermion spin sums are treated by traces. For photons, it is convenient to replace the sum over the two physical transverse polarizations by a covariant sum,

$$\sum_{\alpha=1,2} \epsilon_\mu^{(\alpha)*}(k) \epsilon_\nu^{(\alpha)}(k) \longrightarrow \eta_{\mu\nu}, \quad (5.47)$$

provided the full amplitude is gauge invariant.

The reason is the Ward identity. If an amplitude with one external photon has the form

$$\mathcal{M} = \epsilon_\mu^{(\alpha)}(k) M^\mu,$$

then the shift

$$\epsilon_\mu(k) \rightarrow \epsilon_\mu(k) + c k_\mu$$

cannot change a physical amplitude. Hence

$$\boxed{k_\mu M^\mu = 0.} \quad (5.48)$$

This identity makes the unphysical time-like and longitudinal polarization contributions cancel. In Compton scattering,

$$\mathcal{M} = \epsilon_\mu^{(\alpha_2)*}(k_2) \epsilon_\nu^{(\alpha_1)}(p_2) T^{\mu\nu},$$

and the Ward identity for each external photon allows the replacement

$$\sum_{\alpha_1, \alpha_2} |\mathcal{M}|^2 = \eta_{\mu\lambda} \eta_{\nu\rho} T^{\mu\nu} T^{\lambda\rho*}, \quad (5.49)$$

with the remaining spin sums handled by trace technology.

Compton formula and Klein–Nishina cross section

It is convenient to work in the rest frame of the incoming electron:

$$p_1 = (m, \mathbf{0}), \quad p_2 = (\omega, 0, 0, \omega).$$

Let the outgoing photon have energy ω' and direction $\mathbf{n}$, with

$$\mathbf{n} \cdot \hat{\mathbf{z}} = \cos \theta, \quad k_2 = (\omega', \omega' \mathbf{n}).$$

The outgoing electron momentum is fixed by conservation,

$$k_1 = p_1 + p_2 - k_2.$$

Imposing $k_1^2 = -m^2$ gives the Compton shift formula

$$\boxed{\omega' = \frac{\omega}{1 + \frac{\omega}{m}(1 - \cos \theta)}}. \quad (5.50)$$

The unpolarized differential cross section is the Klein–Nishina formula,

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2m^2} \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \theta \right]}. \quad (5.51)$$

Equivalently,

$$\boxed{\frac{d\sigma}{d \cos \theta} = \frac{\pi \alpha^2}{m^2} \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \theta \right]}. \quad (5.52)$$

For soft incoming photons,

$$\omega \ll m,$$

we have $\omega' \simeq \omega$, and the Klein–Nishina formula reduces to the Thomson result

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2m^2} (1 + \cos^2 \theta)}. \quad (5.53)$$

For finite ω/m , the outgoing photon generally has $\omega' < \omega$ unless $\theta = 0$; the missing energy becomes recoil energy of the electron. This frequency shift is the characteristic quantum feature of Compton scattering.

A Angular momentum and spin of the Dirac field

The Dirac action is invariant under Lorentz transformations. According to Noether's theorem, this symmetry leads to a conserved current associated with Lorentz transformations. The corresponding conserved charges are the generators of spacetime rotations and boosts.

For an infinitesimal Lorentz transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + \omega^\mu{}_\nu x^\nu, \quad (\text{A.1})$$

with antisymmetric parameters $\omega_{\mu\nu} = -\omega_{\nu\mu}$, the Dirac field transforms as

$$\psi(x) \rightarrow \left(1 - \frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu}\right) \psi(x), \quad (\text{A.2})$$

where

$$\Sigma^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu] \quad (\text{A.3})$$

are the generators of Lorentz transformations acting on spinors.

The conserved Noether current associated with Lorentz symmetry is

$$M^{\mu\nu\rho} = x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu} - \bar{\psi} \gamma^\mu \Sigma^{\nu\rho} \psi, \quad (\text{A.4})$$

where $T^{\mu\nu}$ is the energy-momentum tensor. The corresponding conserved charges are

$$J^{\nu\rho} = \int d^3x M^{0\nu\rho}. \quad (\text{A.5})$$

For spatial indices i, j , these charges generate rotations and therefore represent the angular momentum operator

$$J^{ij} = \int d^3x (x^i T^{0j} - x^j T^{0i} - \bar{\psi} \gamma^0 \Sigma^{ij} \psi). \quad (\text{A.6})$$

The first two terms correspond to the orbital angular momentum, while the last term represents the intrinsic spin of the Dirac field.

The spin operator is obtained from

$$S^k = \frac{1}{2} \epsilon^{kij} \Sigma^{ij}. \quad (\text{A.7})$$

Using the Dirac matrices one finds

$$\vec{S} = \frac{1}{2} \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix}, \quad (\text{A.8})$$

where $\vec{\sigma}$ are the Pauli matrices.

Since the eigenvalues of σ^3 are ± 1 , the eigenvalues of the spin operator S_z are

$$S_z = \pm \frac{1}{2}. \quad (\text{A.9})$$

Thus the excitations of the Dirac field carry spin- $\frac{1}{2}$, which explains why each particle has two independent spin polarizations.

B Momentum operator from Noether theorem

In this appendix we derive the expression for the momentum operator of the Dirac field from Noether's theorem.

Spatial translations

Consider an infinitesimal spacetime translation

$$x^\mu \rightarrow x'^\mu = x^\mu + \epsilon^\mu,$$

where ϵ^μ is constant. Under this transformation, the Dirac field transforms as

$$\psi(x) \rightarrow \psi'(x) = \psi(x - \epsilon),$$

so that, to first order,

$$\delta\psi(x) = \psi'(x) - \psi(x) = -\epsilon^\mu \partial_\mu \psi(x). \quad (\text{B.1})$$

We focus on spatial translations, $\epsilon^\mu = (0, \vec{\epsilon})$, so that

$$\delta\psi(x) = -\epsilon^i \partial_i \psi(x). \quad (\text{B.2})$$

Dirac Lagrangian and canonical momentum

The Dirac Lagrangian is

$$\mathcal{L} = -i\bar{\psi}(\gamma^\mu \partial_\mu - m)\psi. \quad (\text{B.3})$$

The derivative of the Lagrangian with respect to $\partial_\mu \psi$ is

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} = -i\bar{\psi}\gamma^\mu. \quad (\text{B.4})$$

Noether current and energy-momentum tensor

For spacetime translations, the Noether current is

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \delta\psi + \mathcal{L} \delta x^\mu. \quad (\text{B.5})$$

Since $\delta x^\mu = \epsilon^\mu$ and $\delta\psi = -\epsilon^\nu \partial_\nu \psi$, we obtain

$$j^\mu = -i\bar{\psi}\gamma^\mu(-\epsilon^\nu \partial_\nu \psi) + \mathcal{L}\epsilon^\mu = \epsilon_\nu T^{\mu\nu}, \quad (\text{B.6})$$

where the canonical energy-momentum tensor is

$$T^{\mu\nu} = i\bar{\psi}\gamma^\mu \partial^\nu \psi + \eta^{\mu\nu} \mathcal{L}. \quad (\text{B.7})$$

Momentum operator

The conserved charges associated with translations are

$$P^\nu = \int d^3x T^{0\nu}. \quad (\text{B.8})$$

For spatial components, since $\eta^{0i} = 0$ and $(\gamma^0)^2 = -1$,

$$T^{0i} = i\bar{\psi}\gamma^0 \partial^i \psi = -i\psi^\dagger \partial^i \psi. \quad (\text{B.9})$$

Therefore,

$$P^i = \int d^3x \psi^\dagger (-i\partial^i) \psi. \quad (\text{B.10})$$

Interpretation

This expression has the same form as the quantum mechanical momentum operator $-i\nabla$, acting on the field ψ , integrated over all space. It can be interpreted as the total momentum obtained by summing the contribution of each field degree of freedom.

C Dimensional Analysis in Natural Units

In quantum field theory, it is standard to work in **natural units**

$$\hbar = c = 1. \quad (\text{C.1})$$

In these units, all physical quantities can be expressed in powers of a single unit, which we take to be **energy** (or equivalently mass).

Basic relations

We first recall two fundamental relations:

(1) Relativity From special relativity,

$$E = mc^2. \quad (\text{C.2})$$

Setting $c = 1$, we obtain

$$E \sim m. \quad (\text{C.3})$$

(2) **Quantum mechanics** From the Planck relation,

$$E = \hbar\omega, \quad (\text{C.4})$$

and since $\omega \sim 1/\text{time}$, setting $\hbar = 1$ gives

$$\boxed{E \sim \frac{1}{\text{time}}}. \quad (\text{C.5})$$

(3) **Relation between time and length** From relativity,

$$c = \frac{\text{length}}{\text{time}}. \quad (\text{C.6})$$

Setting $c = 1$ implies

$$\boxed{\text{time} \sim \text{length}}. \quad (\text{C.7})$$

Consequences

Combining the above results, we find

$$\boxed{E \sim m \sim \frac{1}{\text{length}} \sim \frac{1}{\text{time}}}. \quad (\text{C.8})$$

Thus, in natural units, all quantities can be expressed in terms of a single dimension (e.g. mass or energy).

Dimension assignments

It is convenient to define the **mass dimension** $[\cdot]$ of a quantity. From the above relations, we obtain

$$[\text{energy}] = [\text{mass}] = 1, \quad [\text{length}] = [\text{time}] = -1. \quad (\text{C.9})$$

In particular,

- Spacetime coordinate:

$$[x^\mu] = -1 \quad (\text{C.10})$$

- Derivative:

$$[\partial_\mu] = 1 \quad (\text{C.11})$$

- Momentum:

$$[p_\mu] = 1 \quad (\text{C.12})$$

Dimension of the action

The action S appears in the path integral as

$$e^{iS}, \quad (\text{C.13})$$

and must therefore be dimensionless. Hence,

$$\boxed{[S] = 0}. \quad (\text{C.14})$$

Since

$$S = \int d^4x \mathcal{L}, \quad (\text{C.15})$$

and $[d^4x] = -4$, we conclude

$$\boxed{[\mathcal{L}] = 4}. \quad (\text{C.16})$$